Linear Algebra

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Lecture 1: The Geometry of Linear Equations

The fundamental problem of linear algebra is to solve a system of linear equations. Start with the normal case of n equations in n unknowns, and look at it in three ways: the row picture (one equation at a time), the column picture (one column at a time), and the matrix form.

1.1 Two equations, two unknowns

Take the system

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y &= 3 \end{aligned} \quad \Longleftrightarrow \quad \underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 \\ 3 \end{bmatrix}}_{\mathbf{b}}.$$

The coefficient matrix A is a rectangular array of numbers, here two rows and two columns. The vector $\mathbf{x}$ holds the unknowns and $\mathbf{b}$ holds the right-hand sides, so the system is $A\mathbf{x} = \mathbf{b}$.

Row picture. Each equation is a line in the xy-plane. The first line, $2x - y = 0$, passes through the origin and $(1, 2)$. The second, $-x + 2y = 3$, does not pass through the origin; it contains $(-3, 0)$ and $(-1, 1)$. The solution is the point on both lines, $(x, y) = (1, 2)$. Indeed $2 \cdot 1 - 2 = 0$ and $-1 + 2 \cdot 2 = 3$.

Column picture. Write the system as one vector equation in the columns of A:

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

Given vectors $\mathbf{c}, \mathbf{d}$ and scalars x, y, the sum $x\mathbf{c} + y\mathbf{d}$ is a linear combination of $\mathbf{c}$ and $\mathbf{d}$; it is the most fundamental operation of the course. The equation asks for the combination of the columns that produces $\mathbf{b}$. With $x = 1$ and $y = 2$, one copy of $(2, -1)$ followed by two copies of $(-1, 2)$, placed head to tail, ends at $(0, 3)$. The combinations of these two columns, over all x and y, fill the whole plane, so every right-hand side can be reached.

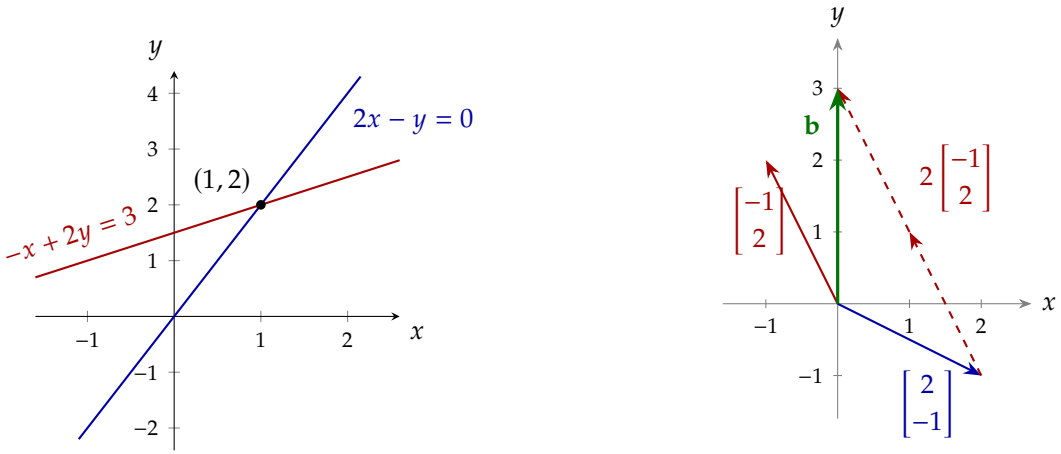


Figure 1: Left: the row picture, two lines meeting at the solution $(1, 2)$. Right: the column picture, one copy of column 1 plus two copies of column 2 (dashed) equals $\mathbf{b} = (0, 3)$.

1.2 Three equations, three unknowns

Consider

$$\begin{aligned} 2x - y \quad &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned} \qquad A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Row picture. Each equation is now a plane in three-dimensional space. For instance, the second plane misses the origin and contains $(1, 0, 0)$, $(0, 0, 1)$, and $(0, -\frac{1}{2}, 0)$. Two of the planes meet in a line, and when nothing is special (no parallel planes), the third plane cuts that line in a single point, the solution. Three planes are already hard to draw, and in higher dimensions the row picture becomes harder still to see.

Column picture. The same system asks for a combination of three vectors in $\mathbb{R}^3$:

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}.$$

Here $\mathbf{b}$ is the third column itself, so $(x, y, z) = (0, 0, 1)$: the three planes of the row picture meet at that point. Keeping A and changing the right-hand side to $\mathbf{b} = (1, 1, -3)$, which is column 1 plus column 2, gives the solution $(x, y, z) = (1, 1, 0)$. In the row picture these are three new planes; in the column picture the columns are the same and only the target changes. Elimination, the systematic method that finds $\mathbf{x}$ for any system, comes in Lecture 3.

1.3 Can every right-hand side be reached?

For a given A, can $A\mathbf{x} = \mathbf{b}$ be solved for every $\mathbf{b}$? In the language of combinations: do the linear combinations of the columns fill the whole space? These are the same question.

For the 3×3 matrix above the answer is yes; A is NONSINGULAR, or INVERTIBLE. The answer is no when the three columns lie in a common plane through the origin, for example when column 3 is the sum of columns 1 and 2. Then every combination lies in that plane, so only right-hand sides in the plane are reachable, and most $\mathbf{b}$ are not. Such a matrix is SINGULAR and its columns are LINEARLY DEPENDENT: their combinations lie on a point, a line, or a plane and do not fill $\mathbb{R}^3$.

The same thinking extends to nine equations in nine unknowns: nine columns in $\mathbb{R}^9$ and their combinations. A random 9×9 matrix is invertible, and its columns' combinations fill $\mathbb{R}^9$. If instead the ninth column equals the eighth, it contributes nothing new, and the combinations fill only an eight-dimensional "plane" inside $\mathbb{R}^9$.

1.4 Multiplying a matrix by a vector

There are two ways to compute $A\mathbf{x}$. By columns, the entries of $\mathbf{x}$ are the coefficients of a combination of the columns of A:

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

By rows, each entry is the dot product of a row of A with $\mathbf{x}$:

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 5 \cdot 2 \\ 1 \cdot 1 + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}.$$

The column view is the one to keep: *$A\mathbf{x}$ is a combination of the columns of A.* In particular, if the right-hand side were $\mathbf{b} = (12, 7)$, the solution would be $\mathbf{x} = (1, 2)$.

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Lecture 2: An Overview of Key Ideas

Linear algebra progresses from vectors to matrices to subspaces. Two 3×3 examples, which differ only in their third column, bring out most of the key ideas: combinations, the matrix as a transform, inverses, dependence, bases, and subspaces.

2.1 Vectors and their combinations

What do you do with vectors? Take combinations: multiply by scalars and add. Given $\mathbf{u}, \mathbf{v}, \mathbf{w}$, a LINEAR COMBINATION is $x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w} = \mathbf{b}$. Take the vectors in $\mathbb{R}^3$

$$\mathbf{u} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \qquad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

All multiples of $\mathbf{u}$ fill a line through the origin, and all multiples of $\mathbf{v}$ fill another line. Since $\mathbf{u}$ and $\mathbf{v}$ point in different directions, all combinations $x_1\mathbf{u} + x_2\mathbf{v}$ fill a plane through the origin, not all of $\mathbb{R}^3$. Taking all combinations of some vectors produces a *subspace*.

2.2 The matrix as a difference transform

Put the vectors in the columns of a matrix. Multiplying by $\mathbf{x}$ takes the combination of the columns:

$$A\mathbf{x} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix} = x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w}.$$

Computing row by row (dot products) gives the same entries. The calculation is the same as before, but the perspective has changed: instead of numbers multiplying vectors, the matrix multiplies the numbers in $\mathbf{x}$. This A is a (first) DIFFERENCE MATRIX, since the entries of $A\mathbf{x}$ are differences of the entries of $\mathbf{x}$. For example, the differences of the squares are the odd numbers:

$$A \begin{bmatrix} 1 \\ 4 \\ 9 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}.$$

2.3 Solving $A\mathbf{x} = \mathbf{b}$: the inverse as a sum matrix

The deeper question goes backward: given $\mathbf{b}$, which $\mathbf{x}$ gives $A\mathbf{x} = \mathbf{b}$? The equations are

$$x_1 = b_1, \qquad x_2 - x_1 = b_2, \qquad x_3 - x_2 = b_3.$$

The matrix is lower triangular, so solve from the top down, substituting each answer into the next equation:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_1 + b_2 \\ b_1 + b_2 + b_3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}}_{S} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

The columns of S are read off as the coefficients of b_1, b_2, b_3. The solution exists for every $\mathbf{b}$ and is unique, and it is $\mathbf{x} = A^{-1}\mathbf{b}$ with $A^{-1} = S$. Think of A as a transform $\mathbf{x} \mapsto \mathbf{b}$ with an inverse transform $\mathbf{b} \mapsto \mathbf{x}$. The inverse of the difference matrix is a SUM MATRIX; for instance $S(1, 3, 5) = (1, 4, 9)$ undoes the earlier example. This mirrors calculus, where the fundamental theorem says that integration (sums) inverts differentiation (differences). In particular, $A\mathbf{x} = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$.

2.4 A circular difference matrix

Keep $\mathbf{u}$ and $\mathbf{v}$ but replace $\mathbf{w}$ by $\mathbf{w}^* = (-1, 0, 1)$:

$$C\mathbf{x} = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{bmatrix}.$$

The differences are now circular, and the difficulty with C shows up in several ways.

- *Nonzero solutions of* $C\mathbf{x} = \mathbf{0}$. The differences of a constant vector are zero, so $C\mathbf{x} = \mathbf{0}$ for every $\mathbf{x} = (c, c, c)$, a whole line of solutions through $(1, 1, 1)$. Then C has no inverse: if C^{-1} existed, multiplying $C\mathbf{x} = \mathbf{0}$ by it would give $\mathbf{x} = \mathbf{0}$. Once a nonzero vector is sent to zero, nothing can bring it back.

- *A condition on* $\mathbf{b}$. Adding the three equations $x_1 - x_3 = b_1$, $x_2 - x_1 = b_2$, $x_3 - x_2 = b_3$ gives

  $$0 = b_1 + b_2 + b_3.$$

  So $C\mathbf{x} = \mathbf{b}$ has no solution unless the components of $\mathbf{b}$ add to zero. When they do, there are solutions: setting $x_3 = 0$ gives $x_1 = b_1$, $x_2 = b_1 + b_2$, and the third equation $x_3 - x_2 = b_3$ holds because $b_1 + b_2 + b_3 = 0$. For springs and masses arranged in a cycle, this condition says the forces must balance.

- *Geometry.* The three columns of C lie in one plane, since $\mathbf{w}^* = -(\mathbf{u} + \mathbf{v})$. Combinations of

$\mathbf{u}, \mathbf{v}, \mathbf{w}^*$ add nothing beyond the plane of $\mathbf{u}$ and $\mathbf{v}$, and every $\mathbf{b}$ off that plane is unreachable.

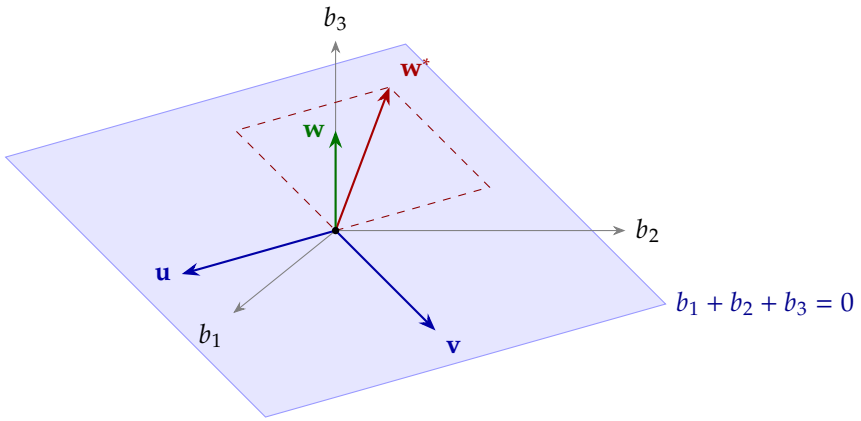


Figure 2: The columns $\mathbf{u}, \mathbf{v}, \mathbf{w}^*$ of C lie in the plane $b_1 + b_2 + b_3 = 0$ (shaded patch, drawn schematically); the dashed parallelogram shows $\mathbf{w}^* = (-\mathbf{u}) + (-\mathbf{v})$. The column $\mathbf{w}$ of A leaves the plane, so the columns of A reach all of $\mathbb{R}^3$.

2.5 Independence, bases, and subspaces

The columns of C are DEPENDENT: one is a combination of the others. The columns of A are INDEPENDENT: $\mathbf{w}$ is not in the plane of $\mathbf{u}$ and $\mathbf{v}$, and their combinations give all of $\mathbb{R}^3$, because $A\mathbf{x} = \mathbf{b}$ was solved for every $\mathbf{b}$.

Definition 2.1 [BASIS OF $\mathbb{R}^n$]. A BASIS for $\mathbb{R}^n$ is a collection of n independent vectors in $\mathbb{R}^n$. Equivalently, it is a collection of n vectors whose combinations fill $\mathbb{R}^n$; equivalently again, the $n \times n$ matrix with those vectors as its columns is invertible.

So $\mathbf{u}, \mathbf{v}, \mathbf{w}$ form a basis for $\mathbb{R}^3$ and $\mathbf{u}, \mathbf{v}, \mathbf{w}^*$ do not. An invertible matrix is one with a basis in its columns.

The combinations of the columns of C are exactly the vectors $C\mathbf{x}$, since multiplying by $\mathbf{x}$ forms the combination. That set is the plane of vectors $\mathbf{b}$ with $b_1 + b_2 + b_3 = 0$. The same plane appears in three languages: as all combinations of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}^*$ (each of whose components add to zero), as all outputs $C\mathbf{x}$, and as the solvability condition found by adding the equations.

Definition 2.2 [VECTOR SPACE AND SUBSPACE]. A VECTOR SPACE is a collection of vectors closed under linear combinations. A SUBSPACE is a vector space inside another vector space.

A plane through the origin in $\mathbb{R}^3$ is a subspace; a triangle in that plane is not, since $2\mathbf{u}$ can leave it. A single point is a subspace only if it is the origin: multiplying a nonzero vector by 5 leaves the point, but the zero subspace $\{\mathbf{0}\}$ is closed. The subspaces of $\mathbb{R}^3$ are the origin alone, the lines through the origin, the planes through the origin, and all of $\mathbb{R}^3$, of dimensions 0, 1, 2, and 3.

2.6 Looking ahead

When you look at a matrix, ask what it is doing. Matrices can be rectangular: seven equations in three unknowns give a 7×3 matrix A, which takes $\mathbb{R}^3$ to $\mathbb{R}^7$ and cannot be inverted. With a rectangular A, the product $A^T A$ keeps appearing. It is square (3×7 times 7×3 is 3×3), it is always symmetric, and it may be invertible. In networks, for example, A carries Kirchhoff's voltage law and A^T his current law.

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Lecture 3: Elimination with Matrices

Elimination is the method every software package uses to solve $A\mathbf{x} = \mathbf{b}$. When A is a good (invertible) matrix it succeeds, and along the way it reveals whether A is good. Back substitution completes the solution. The main point of the lecture is to express each elimination step as multiplication by a matrix.

3.1 Elimination and back substitution

The running example is

$$\begin{aligned} x + 2y + z &= 2 \\ 3x + 8y + z &= 12 \\ 4y + z &= 2 \end{aligned} \qquad A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}.$$

The entry 1 in the upper left is the first pivot. Keep row 1 and subtract 3 times it from row 2; the multiplier 3 is chosen to put a zero in position $(2,1)$. The $(3,1)$ entry is already zero, so that step uses multiplier 0. The second pivot is the 2 now in position $(2,2)$; subtracting 2 times row 2 from row 3 clears position $(3,2)$ and leaves the third pivot 5:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \xrightarrow{(2,1)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \xrightarrow{(3,2)} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = U.$$

The goal of elimination is this upper triangular matrix U, with zeros below the diagonal and the pivots $1, 2, 5$ on it. Looking ahead, the determinant of A is the product of the pivots, here 10.

To carry the right-hand side along, append $\mathbf{b}$ as an extra column, forming the augmented matrix $[\,A \mid \mathbf{b}\,]$, and apply the same row operations to it. Elimination turns $A\mathbf{x} = \mathbf{b}$ into $U\mathbf{x} = \mathbf{c}$:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}\right] = [\,U \mid \mathbf{c}\,].$$

The triangular system $x + 2y + z = 2$, $2y - 2z = 6$, $5z = -10$ is solved in reverse order by back substitution: $z = -2$, then $2y + 4 = 6$ gives $y = 1$, then $x + 2 - 2 = 2$ gives $x = 2$. This also solves the original system.

Algorithm 3.1: Elimination with back substitution

Input: An $n \times n$ matrix A and a vector $\mathbf{b}$.
Output: The solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$, or a report that A is not invertible.

```
for k ← 1 to n do
    if a_kk = 0 then
        if some a_ik ≠ 0 with i > k then
            exchange rows i and k of [ A | b ]
        else
            return "A is not invertible"
    for i ← k + 1 to n do
        ℓ_ik ← a_ik / a_kk                          // multiplier
        subtract ℓ_ik times row k from row i of [ A | b ]
// now [ A | b ] = [ U | c ]
for i ← n down to 1 do
    x_i ← (c_i − Σ_{j>i} u_ij x_j) / u_ii
return x
```

3.2 When elimination fails

Pivots cannot be zero. A zero in the pivot position is a temporary failure if there is a nonzero entry below it: exchange rows and continue. If the 8 in A were a 6, the first step would leave $(0, 0, -2)$ in row 2, a zero in the second pivot position; exchanging rows 2 and 3 brings up the 4 and elimination goes on.

Failure is complete when a zero lands in the pivot position with only zeros below it. If the $(3, 3)$ entry 1 were -4, the same two steps would turn row 3 into $(0, 4, -4) - 2(0, 2, -2) = (0, 0, 0)$, leaving no third pivot. That matrix is not invertible, and $A\mathbf{x} = \mathbf{b}$ has no unique solution.

3.3 Elimination matrices

A matrix times a column is a combination of the columns of the matrix. In parallel, a row times a matrix is a combination of the *rows* of the matrix:

$$\begin{bmatrix} 1 & 2 & 7 \end{bmatrix} A = 1\,(\text{row } 1 \text{ of } A) + 2\,(\text{row } 2 \text{ of } A) + 7\,(\text{row } 3 \text{ of } A).$$

Elimination works with rows, so this is the view it needs. Each row of a product EA is a combination of the rows of A, with coefficients from the corresponding row of E. The IDENTITY MATRIX I, with ones on the diagonal, leaves every row in place.

To subtract 3 times row 1 from row 2 and leave rows 1 and 3 alone, change only row 2 of the identity:

$$E_{21}A = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}.$$

For instance, the $(2,3)$ entry is row 2 of E_{21} times column 3 of A: $-3 \cdot 1 + 1 \cdot 1 + 0 \cdot 1 = -2$. The ELIMINATION MATRIX E_{ij} produces the zero in position (i, j); it is the identity with $-\ell_{ij}$ in position (i, j). The second step uses

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \qquad E_{32}(E_{21}A) = U,$$

with $E_{31} = I$ here since that multiplier was zero.

Matrix multiplication is ASSOCIATIVE: the parentheses can be moved, provided the order of the matrices is kept. So $E_{32}(E_{21}A) = (E_{32}E_{21})A$, and the single matrix $E = E_{32}E_{21}$ carries out all of elimination at once. Many later proofs come down to moving parentheses like this.

3.4 Permutation matrices

A PERMUTATION MATRIX exchanges rows. To build one, perform the exchange on the identity matrix:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}, \qquad P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

To exchange *columns*, multiply on the right:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}.$$

Row operations multiply on the left and column operations on the right. In particular $PA \neq AP$ in general: the COMMUTATIVE law fails for matrices, even though the associative law holds.

3.5 Inverses of elimination steps

The way back from U to A reverses the steps. The step E_{21} subtracted 3 times row 1 from row 2; to undo it, add 3 times row 1 back to row 2:

$$E_{21}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad E_{21}^{-1} E_{21} = I.$$

Every elimination step can be reversed this way. That is why elimination is valid: if $U\mathbf{x} = EA\mathbf{x} = E\mathbf{b} = \mathbf{c}$, then also $A\mathbf{x} = \mathbf{b}$, so the two systems have the same solutions.

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Lecture 4: Multiplication and Inverse Matrices

There are several ways to multiply matrices, all giving the same answer and all useful. The second topic is the inverse of a square matrix: when it exists and how Gauss–Jordan elimination finds it.

4.1 Four ways to multiply

Let A be $m \times n$ and B be $n \times p$. The product is defined only when the number of columns of A equals the number of rows of B, and $C = AB$ is $m \times p$.

Row times column. The entry c_{ij} is the dot product of row i of A with column j of B:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

For example, $c_{34} = a_{31}b_{14} + a_{32}b_{24} + \cdots + a_{3n}b_{n4}$.

Columns. Think of B as p columns side by side. Then

$$A \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & \cdots & A\mathbf{b}_p \end{bmatrix} :$$

column j of C is A times column j of B. Hence the columns of C are combinations of the columns of A, with coefficients taken from B.

Rows. Row i of C is row i of A times B. Hence the rows of C are combinations of the rows of B.

Column times row. A column of A ($m \times 1$) times a row of B ($1 \times p$) is a full $m \times p$ matrix. For example,

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}.$$

Its columns are multiples of $(2, 3, 4)$ and its rows are multiples of $(1, 6)$; the rows $(2, 12)$, $(3, 18)$, $(4, 24)$ all lie on the line through $(1, 6)$. In later language, both its row space and its column space are lines, so this is a minimal kind of matrix. The product AB is the sum of such matrices, column k

of A times row k of B:

$$AB = \sum_{k=1}^{n} \begin{bmatrix} a_{1k} \\ \vdots \\ a_{mk} \end{bmatrix} \begin{bmatrix} b_{k1} & \cdots & b_{kp} \end{bmatrix}.$$

For instance,

$$\begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}.$$

4.2 Block multiplication

If A and B are cut into blocks whose sizes match, the product can be computed block by block, block rows times block columns, exactly as with entries:

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} C_1 & C_2 \\ C_3 & C_4 \end{bmatrix}, \qquad C_1 = A_1B_1 + A_2B_3.$$

For example, 20×20 matrices can be cut into four 10×10 blocks. All five descriptions perform the same multiplications, grouped differently.

4.3 Inverses and singular matrices

Definition 4.1 [Inverse]. A square matrix A is invertible, or nonsingular, if there is a matrix A^{-1} with $A^{-1}A = I = AA^{-1}$. Otherwise A is singular.

For square matrices a left inverse is automatically a right inverse; this is true but not easy to prove. For rectangular matrices there can be a left inverse that is not a right inverse.

Whether a square matrix is invertible is the most important question about it. Consider

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}.$$

It is singular, and this can be seen in several ways. Its determinant is zero. Its columns lie on one line, so the columns of any product AB, being combinations of those columns, also lie on that line; no product AB can equal I, whose column $(1, 0)$ is off the line. Most importantly, there is a nonzero solution of $A\mathbf{x} = \mathbf{0}$:

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

three times column 1 minus column 2 is the zero column.

Proposition 4.2 [Singularity from a null vector]. *If $A\mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \neq \mathbf{0}$, then A has no inverse.*

Proof. If A^{-1} existed, multiplying $A\mathbf{x} = \mathbf{0}$ on the left by it would give $\mathbf{x} = A^{-1}A\mathbf{x} = A^{-1}\mathbf{0} = \mathbf{0}$, a contradiction. Once A sends a nonzero vector to zero, no matrix can bring it back. □

4.4 Gauss–Jordan elimination

Take the invertible matrix A below, whose columns point in different directions, and read $AA^{-1} = I$ column by column:

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \Longleftrightarrow \quad A \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; A \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

In general, A times column j of A^{-1} equals column j of I: finding A^{-1} means solving n systems with the same matrix and different right-hand sides. The Gauss–Jordan idea solves them together by augmenting A with the whole identity. Gauss's elimination reaches upper triangular form; Jordan's idea is to continue upward and clear the entries above the pivots as well (and to divide each row by its pivot, here already 1):

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|cc} 1 & 0 & 7 & -3 \\ 0 & 1 & -2 & 1 \end{array}\right].$$

The first step subtracts 2 times row 1 from row 2; the second subtracts 3 times row 2 from row 1. So $A^{-1} = \left[\begin{smallmatrix} 7 & -3 \\ -2 & 1 \end{smallmatrix}\right]$, and indeed $A^{-1}A$ has entries $7 - 6 = 1$, $21 - 21 = 0$, $-2 + 2 = 0$, $-6 + 7 = 1$.

Proposition 4.3 [Why Gauss–Jordan works]. *If the row operations that reduce $[\,A \mid I\,]$ to $[\,I \mid X\,]$ are collected into one matrix E, then $X = A^{-1}$.*

Proof. Each step multiplies on the left by an elimination matrix, so their product E satisfies, by block multiplication,

$$E\,[\,A \mid I\,] = [\,EA \mid E\,] = [\,I \mid X\,].$$

The left half says $EA = I$, so $E = A^{-1}$. The right half says $X = E$, which is A^{-1}. □

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Lecture 5: Factorization into $A = LU$

Gaussian elimination, written in matrix form, becomes the factorization $A = LU$: the most basic factorization of a matrix. Two facts about products come first, and the lecture ends with the cost of elimination and the permutation matrices needed for row exchanges.

5.1 Inverse and transpose of a product

Proposition 5.1 [Inverse of a product]. *If A and B are invertible and of the same size, then $(AB)^{-1} = B^{-1}A^{-1}$.*

Proof. Move the parentheses: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and on the other side $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$. □

The inverses come in reverse order, as when undoing "socks, then shoes" by removing the shoes first.

The transpose A^T exchanges rows and columns: the entry in row i, column j of A^T is the entry in row j, column i of A. Transposes of products also reverse order, $(AB)^T = B^T A^T$.

Proposition 5.2 [Inverse of a transpose]. *If A is invertible, then so is A^T, and $(A^T)^{-1} = (A^{-1})^T$.*

Proof. Transpose $AA^{-1} = I$ and $A^{-1}A = I$, using $I^T = I$ and the reversed order: $(A^{-1})^T A^T = I$ and $A^T(A^{-1})^T = I$. □

So for a single matrix, transposing and inverting can be done in either order.

5.2 From $EA = U$ to $A = LU$

Assume for now that elimination needs no row exchanges: no zero ever appears in a pivot position. In the 2×2 case one step produces U, and moving its inverse to the other side gives $A = LU$:

$$\underbrace{\begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix}}_{E_{21}} \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}, \qquad \underbrace{\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_{L = E_{21}^{-1}} \underbrace{\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}}_{U}.$$

(Had the 7 been a 4, the second pivot would be zero and A singular.) U is upper triangular with the pivots on its diagonal; L is lower triangular with ones on its diagonal. Dividing each row of U by

its pivot splits off a diagonal matrix D and makes the factorization more balanced, with ones on the diagonals of both triangular factors:

$$\begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix}}_{U}.$$

For a 3×3 matrix, elimination is $E_{32}E_{31}E_{21}A = U$. Inverting the product (in reverse order) puts everything on the right:

$$A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U = LU.$$

The reason to prefer L over E shows up in an example. Take $E_{31} = I$ and

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix}}_{E_{32}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix}}_{E}.$$

The 10 appears because 2 times row 1 was subtracted from row 2, and then 5 times the new row 2 from row 3; altogether 10 times row 1 went into row 3. In the reverse order the multipliers do not interfere:

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{E_{21}^{-1}} \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}}_{E_{32}^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix}}_{L}.$$

L has a 0 where E had the 10.

Theorem 5.3 [$A = LU$]. *If elimination on A needs no row exchanges, then $A = LU$, where U is the upper triangular result of elimination and L is lower triangular with ones on the diagonal and the multiplier ℓ_{ij} in position (i, j) below it.*

Proof. When row i is processed, multiples of rows $1, \ldots, i-1$ are subtracted from it. Each of those rows is already in its final form, a row of U, at the moment it is used. So

$$(\text{row } i \text{ of } U) = (\text{row } i \text{ of } A) - \sum_{j<i} \ell_{ij} \, (\text{row } j \text{ of } U),$$

that is, $(\text{row } i \text{ of } A) = \sum_{j<i} \ell_{ij} (\text{row } j \text{ of } U) + 1 \cdot (\text{row } i \text{ of } U)$. By the row view of multiplication, this says row i of A equals row i of L times U. □

So L costs no extra work: record each multiplier as it is used. In a computer, once row i is finished, the new row of U and the multipliers for row i are stored and the original row of A can be forgotten;

A is overwritten by L and U.

5.3 The cost of elimination

Count one operation as a multiplication together with a subtraction, and assume A is $n \times n$ with no zero entries to exploit. Clearing the first column changes essentially all entries below row 1, about n^2 operations. That leaves an $(n-1) \times (n-1)$ problem, costing about $(n-1)^2$, and so on. The total for the left side is about

$$n^2 + (n-1)^2 + \cdots + 2^2 + 1^2 \approx \int_0^n x^2 \, dx = \tfrac{1}{3} n^3.$$

It is not n^3, because the problems shrink; the factor $\frac{1}{3}$ accounts for that. Doubling n multiplies the cost by about 8.

The right-hand side $\mathbf{b}$ costs much less: forward elimination and back substitution on one extra column take about n^2 operations. So with several right-hand sides, pay $\frac{1}{3} n^3$ once to factor $A = LU$, then about n^2 for each $\mathbf{b}$.

5.4 Permutation matrices

Row exchanges are needed when a zero appears in a pivot position, and they are carried out by PERMUTATION MATRICES: the identity with its rows reordered. The six 3×3 permutation matrices are

$$I, \quad P_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad P_{23} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

The last two move all three rows in a cycle. Doing row exchanges and then more row exchanges is again a reordering, and undoing a reordering is a reordering, so products and inverses stay in this family: the permutation matrices form a GROUP. Each exchange of two rows is its own inverse, the two cycles are inverses of each other, and every permutation matrix satisfies $P^{-1} = P^T$. There are $n!$ permutation matrices of size n: six for $n = 3$ and 24 for $n = 4$.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 5, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 6: Transposes, Permutations, Vector Spaces

This lecture finishes elimination with row exchanges and transposes, then begins linear algebra at a higher level: spaces of vectors, their subspaces, and the column space of a matrix.

6.1 Elimination with row exchanges: $PA = LU$

$A = LU$ assumes no row exchanges. If a zero appears in a pivot position, a nonzero entry from a lower row is exchanged into place, possibly more than once. Collecting all exchanges into one permutation matrix P that puts the rows of A in a good order from the start gives the general form of elimination:

$$PA = LU,$$

valid for every invertible A. For most matrices $P = I$. In practice, software such as MATLAB also exchanges rows to avoid pivots that are merely small, since pivots near zero are numerically inaccurate; algebra requires exchanges only for exact zeros.

A permutation matrix is the identity with its rows reordered (including the trivial reordering). There are $n! = n(n-1)\cdots(2)(1)$ of size n, all invertible, with

$$P^{-1} = P^T, \qquad \text{that is,} \qquad P^T P = I.$$

In $P^T P$, the ones of each row of P meet themselves and nothing else, which produces the ones of the identity.

6.2 Transposes and symmetric matrices

The transpose flips a matrix across its main diagonal, so rows become columns:

$$(A^T)_{ij} = A_{ji}, \qquad \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}.$$

A 3×2 matrix becomes 2×3.

Definition 6.1 [Symmetric matrix]. A matrix A is symmetric if $A^T = A$.

A symmetric matrix is square, and each entry above the diagonal is repeated in the mirror position below it. Symmetric matrices arise constantly from rectangular ones.

Proposition 6.2 [R^TR is symmetric]. *For any matrix R, not necessarily square, R^TR is square and symmetric.*

Proof. Transpose it, reversing the order of the factors: $(R^TR)^T = R^T(R^T)^T = R^TR$, since transposing twice gives R back. □

For example, with $R = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$,

$$R^TR = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 11 & 7 \\ 11 & 13 & 11 \\ 7 & 11 & 17 \end{bmatrix}.$$

The $(1,2)$ and $(2,1)$ entries are the same dot product, $(1,3) \cdot (2,3) = 11$, taken in either order.

6.3 Vector spaces

Vectors can be added and multiplied by numbers (scalars). A space of vectors must allow exactly these operations, that is, all linear combinations, without leaving the space; the operations must also obey eight familiar rules (listed in the text), which are rarely in doubt.

The standard examples:

- $\mathbb{R}^2$, all vectors with two real components, such as $(3,2)$, $(0,0)$, (π, e), pictured as the xy-plane. Vectors add a component at a time.
- $\mathbb{R}^3$, all vectors with three real components. The vector $(3,2,0)$ belongs to $\mathbb{R}^3$, not $\mathbb{R}^2$.
- $\mathbb{R}^n$, all column vectors with n real components.

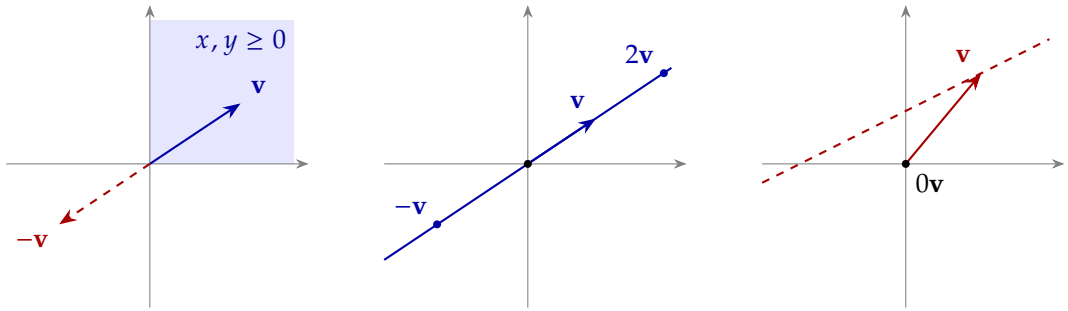
Every vector space contains the zero vector, since $0\mathbf{v}$ must be in it, as must $\mathbf{v} + (-\mathbf{v})$. So the plane with the origin removed is not a vector space.

The quarter-plane of vectors with nonnegative components is not a vector space either. It is closed under addition: $(3,2) + (5,6)$ stays in it. It is not closed under multiplication by all scalars: $-5 \cdot (3,2)$ leaves it.

6.4 Subspaces

Definition 6.3 [Subspace]. A subspace of a vector space is a set of vectors inside it that is itself a vector space: closed under addition and under multiplication by every scalar.

If a subspace of $\mathbb{R}^2$ contains a nonzero vector $\mathbf{v}$, it contains every multiple $c\mathbf{v}$, so it contains the whole line through $\mathbf{v}$ and the origin. That line is a subspace, since sums of vectors on it stay on it. A line in $\mathbb{R}^2$ that misses the origin is not a subspace: multiplying any of its vectors by 0 gives $\mathbf{0}$, which is not on the line. Every subspace contains the zero vector.



not a subspace subspace not a subspace

Figure 3: Sets in $\mathbb{R}^2$. Left: the quarter-plane is closed under addition but not under multiplication by negative numbers. Middle: a line through the origin is a subspace. Right: a line missing the origin is not, since it does not contain $0\mathbf{v} = \mathbf{0}$.

The subspaces of $\mathbb{R}^2$ are

1. all of $\mathbb{R}^2$,
2. any line through $\mathbf{0}$,
3. the zero vector alone, $Z = \{\mathbf{0}\}$.

The set Z satisfies the rules: $\mathbf{0} + \mathbf{0}$ and $17 \cdot \mathbf{0}$ stay in it. A line through the origin in $\mathbb{R}^2$ looks like $\mathbb{R}^1$ but is not $\mathbb{R}^1$, because its vectors have two components. The subspaces of $\mathbb{R}^3$ are all of $\mathbb{R}^3$, any plane through $\mathbf{0}$, any line through $\mathbf{0}$, and Z.

6.5 The column space

Subspaces come out of matrices. Let

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}.$$

Its columns are vectors in $\mathbb{R}^3$. A subspace containing them must also contain their sums, such as $(4, 5, 5)$, their multiples, the zero vector, and every combination.

Definition 6.4 [Column space]. The column space $C(A)$ of a matrix A is the set of all linear combinations of its columns. For an $m \times n$ matrix it is a subspace of $\mathbb{R}^m$.

Here $C(A)$ contains the whole line through $(1, 2, 4)$, the whole line through $(3, 3, 1)$, and everything obtained by adding a vector from one line to a vector from the other. Since the two columns are not on one line, $C(A)$ is a plane through the origin in $\mathbb{R}^3$. Had the columns been on the same line, the column space would be only that line. In the same way, five vectors in $\mathbb{R}^{10}$ produce a subspace of $\mathbb{R}^{10}$, not $\mathbb{R}^5$, since its vectors have ten components; it is some flat piece through the origin whose size depends on the five vectors. The next lecture uses the column space to describe when $A\mathbf{x} = \mathbf{b}$ can be solved.

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Lecture 7: Column Space and Nullspace

Two subspaces attached to a matrix are central: the column space, which decides which right-hand sides $\mathbf{b}$ can be reached, and the nullspace, which collects the solutions of $A\mathbf{x} = \mathbf{0}$.

7.1 Subspaces, unions, and intersections

A vector space is closed under linear combinations: for any $\mathbf{v}$, $\mathbf{w}$ in the space and any scalars c, d, the vector $c\mathbf{v} + d\mathbf{w}$ is in the space. A subspace is a vector space inside another one. In $\mathbb{R}^3$, a plane P through $\mathbf{0}$ and a line L through $\mathbf{0}$ are subspaces; every subspace contains $\mathbf{0}$.

The union $P \cup L$, all vectors in P or L or both, is in general not a subspace: adding a vector of P to a vector of L usually lands outside both. The intersection behaves well. For a line L not in the plane P, $P \cap L = \{\mathbf{0}\}$, which is a subspace, and this holds in general.

Proposition 7.1 [Intersections of subspaces]. *If S and T are subspaces of the same space, then $S \cap T$ is a subspace.*

Proof. Let $\mathbf{v}, \mathbf{w} \in S \cap T$. Both are in the subspace S, so $\mathbf{v} + \mathbf{w} \in S$; both are in T, so $\mathbf{v} + \mathbf{w} \in T$. Hence $\mathbf{v} + \mathbf{w} \in S \cap T$. Likewise $c\mathbf{v}$ lies in S and in T for every scalar c. □

7.2 The column space and solvability

Take the 4×3 example

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}.$$

Its columns are in $\mathbb{R}^4$, so the column space $C(A)$, all combinations of the columns, is a subspace of $\mathbb{R}^4$. Three vectors cannot fill four-dimensional space, so $C(A) \neq \mathbb{R}^4$. Correspondingly, $A\mathbf{x} = \mathbf{b}$, four equations in three unknowns,

$$x_1 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix},$$

is not solvable for every $\mathbf{b}$. It is solvable for some: $\mathbf{b} = \mathbf{0}$ (take $\mathbf{x} = \mathbf{0}$), $\mathbf{b} = (1, 2, 3, 4)$ (take $\mathbf{x} = (1, 0, 0)$), and $\mathbf{b} = (1, 1, 1, 1)$ (take $\mathbf{x} = (0, 1, 0)$). A good way to find such $\mathbf{b}$ is to choose $\mathbf{x}$ first and compute $A\mathbf{x}$.

Theorem 7.2 [Solvability]. *$A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is in the column space $C(A)$.*

Proof. $A\mathbf{x}$ is the combination of the columns of A with coefficients $x_1, \ldots, x_n$, so $C(A)$ consists exactly of all vectors $A\mathbf{x}$. If $\mathbf{b}$ is a combination of the columns, its coefficients give a solution $\mathbf{x}$; if not, no $\mathbf{x}$ works. □

How large is $C(A)$ here? Do all three columns contribute something new? Column 3 is column 1 plus column 2, so dropping it leaves the same column space. Columns 1 and 2 point in different directions, so $C(A)$ is a two-dimensional subspace of $\mathbb{R}^4$: the plane of all combinations of $(1, 2, 3, 4)$ and $(1, 1, 1, 1)$. Column 1 could have been dropped instead; the convention is to keep the earliest columns as long as each adds a new direction. These will be called the *pivot columns*, and column 3 will not be one.

7.3 The nullspace

Definition 7.3 [Nullspace]. The nullspace $N(A)$ of an $m \times n$ matrix A is the set of all solutions $\mathbf{x}$ of $A\mathbf{x} = \mathbf{0}$. It is a subset of $\mathbb{R}^n$.

For the example, $C(A)$ lies in $\mathbb{R}^4$ while $N(A)$ lies in $\mathbb{R}^3$: the number of columns n fixes the number of unknowns. The nullspace contains $\mathbf{x} = \mathbf{0}$, as it does for every matrix. It also contains $(1, 1, -1)$, since column 1 plus column 2 minus column 3 is zero, and every multiple:

$$N(A) = \left\{ c \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} : c \in \mathbb{R} \right\},$$

a line through the origin in $\mathbb{R}^3$. There are no other solutions: substituting column 3 = column 1 + column 2 turns $A\mathbf{x} = \mathbf{0}$ into $(x_1 + x_3)$(column 1) + $(x_2 + x_3)$(column 2) = $\mathbf{0}$, and since columns 1 and 2 are not on one line, $x_1 = x_2 = -x_3$.

Proposition 7.4 [The nullspace is a subspace]. *For every matrix A, $N(A)$ is a subspace of $\mathbb{R}^n$.*

Proof. If $A\mathbf{v} = \mathbf{0}$ and $A\mathbf{w} = \mathbf{0}$, then by the distributive law $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0}$. A scalar moves outside a product, so $A(c\mathbf{v}) = cA\mathbf{v} = \mathbf{0}$. □

7.4 Solutions with $\mathbf{b} \neq \mathbf{0}$

Change the right-hand side to $\mathbf{b} = (1, 2, 3, 4)$, which is in $C(A)$. Solutions include $(1, 0, 0)$ and $(0, -1, 1)$ (minus column 2 plus column 3). The solutions do *not* form a subspace, because $\mathbf{x} = \mathbf{0}$ is not one of them. They form a line that misses the origin: the difference of two solutions solves $A\mathbf{x} = \mathbf{0}$, so every solution differs from $(1, 0, 0)$ by a vector of $N(A)$, and the solutions are

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \qquad c \in \mathbb{R},$$

a line parallel to the nullspace. For instance, $c = -1$ gives $(0, -1, 1)$.

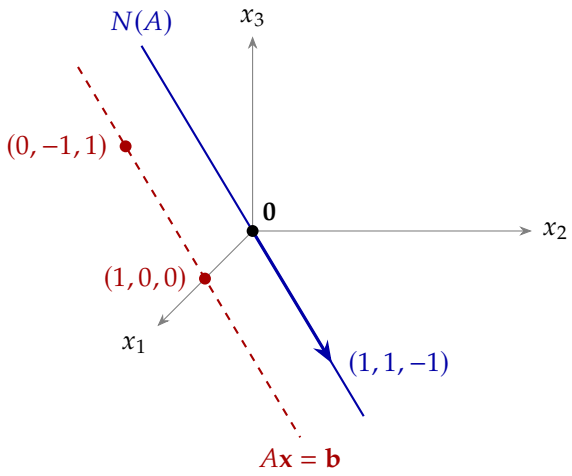


Figure 4: The nullspace $N(A)$, the line through $\mathbf{0}$ in the direction $(1, 1, -1)$, and the parallel line of solutions of $A\mathbf{x} = (1, 2, 3, 4)$ through $(1, 0, 0)$ and $(0, -1, 1)$, which misses the origin (schematic perspective).

The two subspaces illustrate the two natural ways to specify a subspace. The column space is *built up*: start from a few vectors and take all their combinations. The nullspace is *described by equations*: it consists of the vectors satisfying $A\mathbf{x} = \mathbf{0}$, and its members must be found. Both constructions always produce subspaces.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 7, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 8: Solving $A\mathbf{x} = 0$: Pivot Variables, Special Solutions

The nullspace $N(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ of a rectangular $m \times n$ matrix is computed by elimination, continued past columns that have no pivot. The result is a basis of special solutions.

8.1 Elimination on a rectangular matrix

Row operations replace equations by combinations of equations and can be undone, so they do not change the solutions of $A\mathbf{x} = \mathbf{0}$: *elimination preserves the nullspace*. It does change the column space. Since the right side stays $\mathbf{0}$, no augmented column is needed.

Example: *A* 3×4 *matrix.* Column 2 of

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

is twice column 1, and row 3 is row 1 plus row 2. Subtracting 2 and 3 times row 1 from rows 2 and 3 gives

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}.$$

Column 2 has no pivot, and nothing below can be exchanged into its pivot position, so elimination moves on to column 3. Its pivot is the 2 in row 2; subtracting row 2 from row 3 gives

$$U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

U is in ECHELON (staircase) form. A zero row appears because the original row 3 was a combination of rows 1 and 2.

Definition 8.1 [RANK, PIVOT AND FREE COLUMNS**].** The RANK r of A is its number of pivots. Columns containing pivots are PIVOT COLUMNS; the others are FREE COLUMNS. The variables multiplying them are PIVOT VARIABLES and FREE VARIABLES.

In the example $r = 2$, columns 1 and 3 are pivot columns, and x_2, x_4 are free.

8.2 Special solutions

Since $U\mathbf{x} = \mathbf{0}$ has the same solutions as $A\mathbf{x} = \mathbf{0}$, the free variables can be given any values and the pivot variables then follow by back substitution. The equations of the example are

$$x_1 + 2x_2 + 2x_3 + 2x_4 = 0, \qquad 2x_3 + 4x_4 = 0.$$

Setting one free variable to 1 and the others to 0 gives a SPECIAL SOLUTION:

$$x_2 = 1,\ x_4 = 0: \quad \mathbf{s}_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad x_2 = 0,\ x_4 = 1: \quad \mathbf{s}_2 = \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}.$$

Each records a dependency among the columns: $\mathbf{s}_1$ says that $-2(\text{col } 1) + (\text{col } 2) = \mathbf{0}$, and $\mathbf{s}_2$ says that $2(\text{col } 1) - 2(\text{col } 3) + (\text{col } 4) = \mathbf{0}$. The nullspace is the plane of all combinations $\mathbf{x} = c_1\mathbf{s}_1 + c_2\mathbf{s}_2$ in $\mathbb{R}^4$.

Proposition 8.2 [SPECIAL SOLUTIONS SPAN THE NULLSPACE]. *Let A be $m \times n$ with rank r. There are $n - r$ free variables and one special solution for each. Every solution of $A\mathbf{x} = \mathbf{0}$ is a unique combination of the special solutions, so the nullspace consists exactly of their combinations.*

Proof. Each pivot uses one column, so r of the n variables are pivot variables and $n - r$ are free. Let $\mathbf{x} \in N(A)$ have free variables $x_{j_1}, \ldots, x_{j_{n-r}}$, and let $\mathbf{s}_k$ be the special solution with $x_{j_k} = 1$. Then $\mathbf{y} = \mathbf{x} - \sum_k x_{j_k}\mathbf{s}_k$ lies in $N(A)$ and has every free variable equal to 0. Back substitution with zero free variables forces every pivot variable to be 0, so $\mathbf{y} = \mathbf{0}$. The coefficients are unique because they are read off from the free entries of $\mathbf{x}$. □

A matrix with more columns than rows, $n > m$, has $r \le m < n$, hence at least one free variable and a nonzero vector in its nullspace.

Algorithm 8.1: Nullspace by elimination

Input: An $m \times n$ matrix A.
Output: The special solutions, whose combinations form $N(A)$.

Eliminate to echelon form U; when a column has no available pivot, continue with the next column
$r \leftarrow$ number of pivots; mark pivot and free columns
for each free variable x_j **do**
    Set $x_j = 1$ and the other free variables to 0
    Solve $U\mathbf{x} = \mathbf{0}$ for the pivot variables by back substitution
    Record $\mathbf{x}$ as a special solution
return the $n - r$ special solutions

8.3 Reduced row echelon form

Elimination can continue upward to produce zeros above each pivot, and each pivot row can be divided by its pivot. The result is the REDUCED ROW ECHELON FORM $R = \mathrm{rref}(A)$: pivots equal to 1 with zeros above and below them. In the example, subtracting row 2 from row 1 and then dividing row 2 by 2 gives

$$U = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

The pivot rows and pivot columns of R contain an identity matrix. The free columns hold $(2, 0)$ and $(-2, 2)$, and the special solutions contain exactly these numbers with signs reversed, in the pivot positions.

To see why, move the pivot columns of R to the front, which only reorders the unknowns. Then R has the block form

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix},$$

with I the $r \times r$ identity, F of size $r \times (n - r)$, and $m - r$ zero rows that come from dependent rows of A. Writing $\mathbf{x} = (\mathbf{x}_{\text{pivot}}, \mathbf{x}_{\text{free}})$, the system $R\mathbf{x} = \mathbf{0}$ is

$$\mathbf{x}_{\text{pivot}} + F\mathbf{x}_{\text{free}} = \mathbf{0}, \qquad \text{so} \qquad \mathbf{x}_{\text{pivot}} = -F\mathbf{x}_{\text{free}}.$$

Choosing the free part to be each column of the $(n - r) \times (n - r)$ identity gives all special solutions at once, as the columns of the NULLSPACE MATRIX

$$N = \begin{bmatrix} -F \\ I \end{bmatrix}, \qquad RN = -F + F = 0.$$

(In the original column order, the rows of N are permuted back the same way.)

Example: *The transpose.* Take

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix},$$

the transpose of the previous matrix. Column 3 is the sum of columns 1 and 2, so it should be free. Subtracting 2 times row 1 from each lower row gives rows $(0, 0, 0)$, $(0, 2, 2)$, $(0, 4, 4)$. Column 2 has a

zero in the pivot position, so exchange rows 2 and 3 and eliminate:

$$U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

The rank is again 2, with one free column. Setting $x_3 = 1$ gives $x_2 = -1$ and $x_1 = -1$, which is $-F$ above $I = [1]$. So

$$N(A) = \left\{ c \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} : c \in \mathbb{R} \right\},$$

a line in $\mathbb{R}^3$: minus column 1 minus column 2 plus column 3 is zero.

A and A^T both have two pivot columns here. This is no accident: a matrix and its transpose always have the same rank.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 8, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 9: Solving $A\mathbf{x} = \mathbf{b}$: Row Reduced Form R

Elimination decides whether $A\mathbf{x} = \mathbf{b}$ has a solution and, if it does, produces all of them: one particular solution plus the nullspace. The rank alone determines how many solutions there can be.

9.1 Solvability conditions on b

Continue with the matrix of Lecture 8, whose row 3 is row 1 plus row 2. Apply elimination to the augmented matrix $[\,A \;\; \mathbf{b}\,]$, so that each row operation acts on $\mathbf{b}$ as well:

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{array}\right] \to \cdots \to \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array}\right].$$

The last equation reads $0 = b_3 - b_2 - b_1$, so the system is solvable only if $b_3 = b_1 + b_2$. For example, $\mathbf{b} = (1, 5, 6)$ is allowed, while $(1, 5, 17)$ is not.

Proposition 9.1 [Solvability]. *$A\mathbf{x} = \mathbf{b}$ is solvable exactly when $\mathbf{b}$ lies in the column space $C(A)$. Equivalently, whenever a combination of the rows of A gives the zero row, the same combination of the entries of $\mathbf{b}$ gives zero.*

The first statement holds because $A\mathbf{x}$ is the combination of the columns with coefficients x_j. For the second, elimination ends with zero rows exactly where combinations of rows of A vanish, and the corresponding entries of the transformed $\mathbf{b}$ are those same combinations of b_i; the remaining nonzero rows have pivots and can always be solved by back substitution. Both conditions are therefore equivalent to solvability, hence to each other.

9.2 The complete solution

Proposition 9.2 [Particular plus nullspace]. *If $A\mathbf{x}_p = \mathbf{b}$, then the solutions of $A\mathbf{x} = \mathbf{b}$ are exactly*

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n, \qquad \mathbf{x}_n \in N(A).$$

Proof. If $A\mathbf{x}_n = \mathbf{0}$, then $A(\mathbf{x}_p + \mathbf{x}_n) = \mathbf{b} + \mathbf{0} = \mathbf{b}$. Conversely, if $A\mathbf{x} = \mathbf{b}$, then $A(\mathbf{x} - \mathbf{x}_p) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{x} - \mathbf{x}_p \in N(A)$. □

A convenient particular solution sets every free variable to zero and solves for the pivot variables.

Algorithm 9.1: Complete solution of $A\mathbf{x} = \mathbf{b}$

Input: An $m \times n$ matrix A and $\mathbf{b} \in \mathbb{R}^m$.
Output: All solutions, or a report that none exist.

Eliminate on $[\,A \;\; \mathbf{b}\,]$ to echelon form $[\,U \;\; \mathbf{c}\,]$
if some zero row of U has $c_i \neq 0$ **then**
    return no solution
Set all free variables to 0; solve for the pivot variables to get $\mathbf{x}_p$
Find the special solutions $\mathbf{s}_1, \ldots, \mathbf{s}_{n-r}$ of $A\mathbf{x} = \mathbf{0}$
return $\mathbf{x}_p + c_1\mathbf{s}_1 + \cdots + c_{n-r}\mathbf{s}_{n-r}$ for all scalars c_k

Example: *The example with* $\mathbf{b} = (1, 5, 6)$. Elimination turns $\mathbf{b}$ into $(1, 3, 0)$, so the last equation is $0 = 0$. With $x_2 = x_4 = 0$ the remaining equations are

$$x_1 + 2x_3 = 1, \qquad 2x_3 = 3,$$

so $x_3 = 3/2$ and $x_1 = -2$. Adding the special solutions from Lecture 8,

$$\mathbf{x} = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \qquad c_1, c_2 \in \mathbb{R}.$$

There is no free constant in front of $\mathbf{x}_p$: multiplying it by 3 would change the right side to $3\mathbf{b}$. The solution set is not a subspace, since it does not contain $\mathbf{0}$ when $\mathbf{b} \neq \mathbf{0}$. It is the two-dimensional plane $N(A)$ shifted to pass through $\mathbf{x}_p$, as in Figure 5.

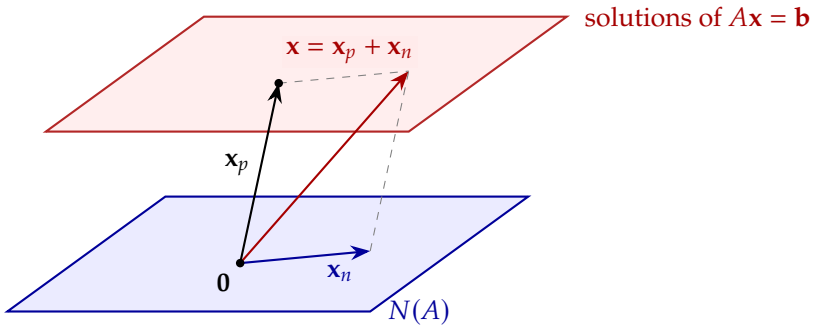


Figure 5: Schematic picture of the solutions of $A\mathbf{x} = \mathbf{b}$ (drawn in three dimensions; in the example they lie in $\mathbb{R}^4$). The nullspace is a plane through $\mathbf{0}$; the solution set is the parallel plane through $\mathbf{x}_p$, and each solution is $\mathbf{x}_p + \mathbf{x}_n$.

9.3 Rank and the number of solutions

For an $m \times n$ matrix of rank r, each row and each column contains at most one pivot, so $r \leq m$ and $r \leq n$. The cases of FULL RANK are the extremes.

Full column rank, $r = n \leq m$. Every column has a pivot, so there are no free variables and $N(A) = \{\mathbf{0}\}$. A solution, if one exists, is just $\mathbf{x}_p$ and is unique: there are 0 or 1 solutions. The reduced form is $R = \begin{bmatrix} I \\ 0 \end{bmatrix}$. For instance,

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix}, \qquad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Four equations in two unknowns are solvable only for $\mathbf{b}$ in the plane $C(A)$. The right side $\mathbf{b} = (4, 3, 7, 6)$, the sum of the two columns, has the single solution $\mathbf{x} = (1, 1)$.

Full row rank, $r = m \leq n$. Every row has a pivot, so elimination produces no zero rows and puts no condition on $\mathbf{b}$: $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b} \in \mathbb{R}^m$. There are $n - r = n - m$ free variables and as many special solutions, and $R = [\, I \;\; F \,]$, possibly with the columns of I and F interleaved. The transpose

$$A = \begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix}$$

has rank 2, pivots in columns 1 and 2, and two free columns.

Full rank, $r = m = n$. A is square and invertible, $R = I$, $N(A) = \{\mathbf{0}\}$, and $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$, for example for $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$.

Rank below both, $r < m$ and $r < n$. Zero rows impose conditions on $\mathbf{b}$, and free variables give a nontrivial nullspace, so there are either no solutions or infinitely many.

With the pivot columns placed first, the four cases are:

	$r = m = n$	$r = n < m$	$r = m < n$	$r < m,\ r < n$
R	I	$\begin{bmatrix} I \\ 0 \end{bmatrix}$	$\begin{bmatrix} I & F \end{bmatrix}$	$\begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
solutions of $A\mathbf{x} = \mathbf{b}$	1	0 or 1	∞	0 or ∞

The rank determines the number of solutions; the entries of A and $\mathbf{b}$ are needed only to find them.

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Lecture 10: Independence, Basis and Dimension

Independence, spanning, basis, and dimension are properties of a collection of vectors or of a space, not of a matrix. Placing vectors in the columns of a matrix turns each question into one about its nullspace and pivots.

10.1 Linear independence

Proposition 10.1 [More unknowns than equations]. *If A is $m \times n$ with $n > m$, then $A\mathbf{x} = \mathbf{0}$ has a nonzero solution.*

Proof. There are at most m pivots, so at least $n - m \geq 1$ free variables. Setting one free variable to 1 gives a nonzero special solution (Lecture 8). □

Definition 10.2 [Independence]. Vectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$ in a vector space are linearly independent if

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n = \mathbf{0} \quad \text{only when} \quad c_1 = c_2 = \cdots = c_n = 0.$$

Otherwise they are dependent: some combination other than the zero combination gives $\mathbf{0}$.

In the plane, $\mathbf{v}_1 = \mathbf{v}$ and $\mathbf{v}_2 = 2\mathbf{v}$ are dependent, since $2\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$. Any collection containing the zero vector is dependent: take coefficient 1 on $\mathbf{0}$ and 0 on the rest. Two vectors are independent when they do not lie on a line through the origin, and three when they do not lie in a plane. Three vectors in $\mathbb{R}^2$ are always dependent: as columns of a 2×3 matrix they give a nonzero null vector by Proposition 10.1.

When $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are the columns of A, a combination $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ is $A\mathbf{c}$. Hence

$$\text{columns independent} \iff N(A) = \{\mathbf{0}\} \iff r = n \iff \text{no free variables},$$

and the columns are dependent exactly when $r < n$. A free column is a combination of the pivot columns before it.

10.2 Spanning, basis, and dimension

Definition 10.3 [Span]. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k$ span a space when the space consists of all their combinations.

The columns of A span $C(A)$. The space spanned by given vectors is the smallest space containing

them, since any space containing the vectors contains all their combinations.

Definition 10.4 [Basis]. A basis for a vector space is a sequence of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_d$ that are independent and span the space.

Spanning means there are enough vectors; independence means there are not too many. Every vector in the space is then a combination of the basis vectors in exactly one way.

Example: *Bases of* $\mathbb{R}^3$. The standard vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ are independent, since $c_1(1,0,0) + c_2(0,1,0) + c_3(0,0,1) = (c_1, c_2, c_3)$ is zero only when every $c_i = 0$, and they span $\mathbb{R}^3$. They are the columns of I, whose nullspace is $\{\mathbf{0}\}$.

The vectors $(1,1,2)$ and $(2,2,5)$ are independent but span only a plane in $\mathbb{R}^3$, so they form a basis for that plane, not for $\mathbb{R}^3$. Adding $(3,3,7)$, their sum, keeps the plane and destroys independence. Adding $(3,3,8)$ fails as well: the matrix with columns $(1,1,2)$, $(2,2,5)$, $(3,3,8)$ has two identical rows, so it is not invertible and its columns are dependent. In general:

Proposition 10.5 [Bases of $\mathbb{R}^n$]. *n vectors in $\mathbb{R}^n$ form a basis exactly when the $n \times n$ matrix with these columns is invertible.*

Bases are far from unique, but every basis of a given space has the same size.

Theorem 10.6 [Invariance of dimension]. *If $\mathbf{v}_1, \ldots, \mathbf{v}_d$ and $\mathbf{w}_1, \ldots, \mathbf{w}_e$ are both bases of the same space, then $d = e$.*

Proof. Suppose $e > d$. Each $\mathbf{w}_j$ is a combination of the $\mathbf{v}$'s, say $\mathbf{w}_j = \sum_i a_{ij}\mathbf{v}_i$. The $d \times e$ matrix (a_{ij}) has more columns than rows, so by Proposition 10.1 some $\mathbf{c} \neq \mathbf{0}$ has $\sum_j a_{ij}c_j = 0$ for every i. Then $\sum_j c_j\mathbf{w}_j = \sum_i \big(\sum_j a_{ij}c_j\big)\mathbf{v}_i = \mathbf{0}$, contradicting the independence of the $\mathbf{w}$'s. By symmetry $e > d$ and $d > e$ are both impossible. □

Definition 10.7 [Dimension]. The dimension of a space is the number of vectors in any basis.

Thus every basis of $\mathbb{R}^n$ has n vectors. If a space has dimension d, then any d independent vectors in it automatically span it: otherwise some vector outside their span could be added, giving $d + 1$ independent vectors, which is impossible.

10.3 Bases for the column space and nullspace

Example: *A rank-2 matrix.* Let

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}.$$

Its four columns span $C(A)$ but are not independent: column 3 is column 1 plus column 2, and column 4 equals column 1. Columns 1 and 2 are independent and are the pivot columns, so they form a basis of $C(A)$, and the rank is 2. Other bases include columns 1 and 3, columns 2 and 3, columns 2 and 4, or any two independent vectors in $C(A)$, such as $(2,2,2)$ and $(7,5,7)$.

The two free variables x_3, x_4 give the special solutions

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

which say $-(\text{col } 1) - (\text{col } 2) + (\text{col } 3) = \mathbf{0}$ and $-(\text{col } 1) + (\text{col } 4) = \mathbf{0}$. They are independent (look at their free entries) and span $N(A)$ by Lecture 8, so they form a basis and $\dim N(A) = 2$.

The same reasoning applies to every matrix.

Theorem 10.8 [Rank and dimension]. *For any $m \times n$ matrix A of rank r,*

$$\operatorname{rank}(A) = \#\,\text{pivot columns} = \dim C(A), \qquad \dim N(A) = \#\,\text{free variables} = n - r.$$

The pivot columns of A form a basis of $C(A)$, and the special solutions form a basis of $N(A)$.

Proof. Elimination preserves the nullspace, so $A\mathbf{x} = \mathbf{0}$ exactly when $R\mathbf{x} = \mathbf{0}$: the columns of A satisfy the same linear relations as the columns of R. The pivot columns of R are distinct columns of an identity matrix, hence independent, and every free column of R is a combination of them. The same is therefore true of the columns of A, so its r pivot columns are independent and span $C(A)$. The statement for $N(A)$ is Lecture 8's result that the $n - r$ special solutions span $N(A)$, together with their independence. □

Words attach to the right objects: a matrix has a rank, and a subspace has a dimension. The rank of A is the dimension of its column space.

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Lecture 11: The Four Fundamental Subspaces

Every $m \times n$ matrix determines four subspaces, two in $\mathbb{R}^n$ and two in $\mathbb{R}^m$. Their bases come from the reduced form R and their dimensions from the rank alone.

11.1 The four subspaces

Definition 11.1 [Fundamental subspaces]. For an $m \times n$ matrix A:

- the column space $C(A)$, all combinations of the columns, is a subspace of $\mathbb{R}^m$;
- the nullspace $N(A)$, all solutions of $A\mathbf{x} = \mathbf{0}$, is a subspace of $\mathbb{R}^n$;
- the row space, all combinations of the rows, is a subspace of $\mathbb{R}^n$. Transposing turns the rows into columns, so it is written $C(A^T)$;
- the left nullspace $N(A^T)$, all solutions of $A^T\mathbf{y} = \mathbf{0}$, is a subspace of $\mathbb{R}^m$.

Transposing $A^T\mathbf{y} = \mathbf{0}$ gives $\mathbf{y}^T A = \mathbf{0}^T$: a row vector multiplying A from the left gives the zero row, which explains the name. Any of the four spaces may be just $\{\mathbf{0}\}$.

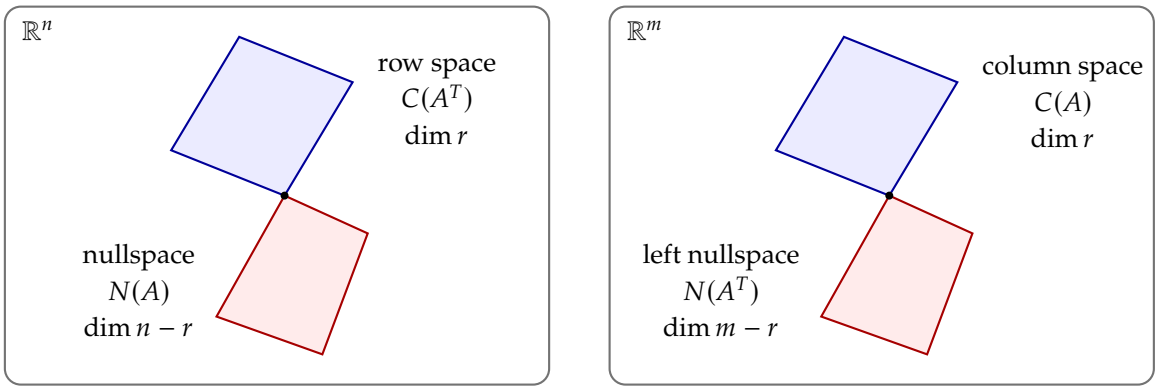


Figure 6: The four fundamental subspaces of an $m \times n$ matrix of rank r and their dimensions. The row space and nullspace lie in $\mathbb{R}^n$; the column space and left nullspace lie in $\mathbb{R}^m$.

11.2 Bases and dimensions

Theorem 11.2 [Dimensions of the four subspaces]. *If A is $m \times n$ with rank r, then*

$$\dim C(A) = r, \qquad \dim C(A^T) = r, \qquad \dim N(A) = n - r, \qquad \dim N(A^T) = m - r.$$

In $\mathbb{R}^n$, $\dim C(A^T) + \dim N(A) = n$; in $\mathbb{R}^m$, $\dim C(A) + \dim N(A^T) = m$.

The statements for $C(A)$ and $N(A)$ are Lecture 10: the r pivot columns of A are a basis of $C(A)$, and the $n - r$ special solutions are a basis of $N(A)$. The other two follow from the row space and the matrix E below. The equality $\dim C(A^T) = \dim C(A)$, row rank equals column rank, is the remarkable one. Applying the nullspace count to the $n \times m$ matrix A^T, which has m columns and rank r, gives $\dim N(A^T) = m - r$.

Row space. Rather than reduce A^T, use the reduced form of A. For

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix},$$

subtract row 1 from rows 2 and 3, multiply the new row 2 by -1, and subtract 2 times it from row 1:

$$A \to \cdots \to R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}.$$

Row operations change the column space: $(1, 1, 1) \in C(A)$ but $(1, 1, 1) \notin C(R)$. They preserve the row space.

Proposition 11.3 [Row space basis]. *A and its reduced form R have the same row space, and the first r rows of R form a basis for it. Hence* $\dim C(A^T) = r$.

Proof. Each row operation replaces rows by combinations of rows, so every row of R is a combination of rows of A. The operations can be reversed, so every row of A is a combination of rows of R. The two row spaces are therefore equal. The zero rows of R contribute nothing, and the r nonzero rows are independent because each has a 1 in a pivot column where the others have 0. □

In the example the row space basis is $(1, 0, 1, 1)$, $(0, 1, 1, 0)$. The rows of A itself need not include a basis in this clean form; the nonzero rows of R are the natural one.

Left nullspace. Row reduction amounts to multiplying A on the left by one $m \times m$ matrix E, the product of the elementary steps. Record it by reducing the augmented matrix as in Gauss–Jordan elimination:

$$\left[\, A_{m\times n} \;\; I_{m\times m} \,\right] \longrightarrow \left[\, R_{m\times n} \;\; E_{m\times m} \,\right], \qquad EA = R.$$

If A is square and invertible, then $R = I$ and $E = A^{-1}$. Applying the steps above to I gives

$$EA = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

Proposition 11.4 [Left nullspace basis]. *The last $m - r$ rows of E form a basis of $N(A^T)$.*

Proof. The last $m - r$ rows of $EA = R$ are zero, so each of the last $m - r$ rows $\mathbf{y}^T$ of E satisfies $\mathbf{y}^T A = \mathbf{0}^T$. These rows are independent because E is invertible, and there are $m - r = \dim N(A^T)$ of them, so they form a basis. □

Here $m - r = 1$, and the basis is $\mathbf{y} = (-1, 0, 1)$: minus row 1 plus row 3 of A is the zero row. The vectors in $N(A^T)$ record the dependencies among the rows of A, just as the vectors in $N(A)$ record the dependencies among its columns.

Example: *A dependence seen through rows.* The vectors $(1, 1, 2)$, $(2, 2, 5)$, $(3, 3, 8)$ of Lecture 10 are the columns of a 3×3 matrix whose first two rows are equal. Its row space therefore has dimension at most 2, so its column space does too, and the three columns cannot be independent.

11.3 A space of matrices

The set M of all 3×3 real matrices is a vector space: matrices can be added and multiplied by scalars, the zero matrix is the additive identity, and the usual rules hold. Matrix multiplication plays no role in this structure. Subspaces of M include

- all upper triangular matrices;
- all symmetric matrices;
- all diagonal matrices D.

A matrix that is both symmetric and upper triangular has zeros below the diagonal and hence above it, so D is the intersection of the first two subspaces. It has dimension 3, with basis

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

These are independent, and every diagonal matrix is a combination of them. Lecture 12 computes the dimensions of the other two subspaces.

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Lecture 12: Matrix Spaces; Rank 1; Small World Graphs

Basis and dimension make sense in any space whose elements can be added and multiplied by scalars: matrices, functions, and solution sets. Rank-one matrices are the simplest matrices and the building blocks of all others.

12.1 Spaces of matrices

Let M be the space of all 3×3 matrices, with its subspaces S (symmetric), U (upper triangular), and $D = S \cap U$ (diagonal) from Lecture 11. A sum of symmetric matrices is symmetric even though a product need not be; only addition and scalar multiplication matter.

Dimension of M. A matrix in M is specified by nine numbers, so M is $\mathbb{R}^9$ written in a square. The nine matrices E_{ij} with a single 1 in position (i, j) and zeros elsewhere form a basis, and $\dim M = 9$.

Dimensions of S, U, and D. A symmetric matrix is fixed by its three diagonal entries and the three entries above the diagonal, so $\dim S = 6$, with basis

$$E_{11},\ E_{22},\ E_{33},\ E_{12} + E_{21},\ E_{13} + E_{31},\ E_{23} + E_{32}.$$

An upper triangular matrix has the same freedom, so $\dim U = 6$, with basis $E_{11}, E_{22}, E_{33}, E_{12}, E_{13}, E_{23}$. This basis of U happens to be a subset of the basis of M, but no basis of S is: of the E_{ij} only E_{11}, E_{22}, E_{33} are symmetric, so a basis for a subspace generally has to be found afresh. The basis E_{11}, E_{22}, E_{33} of D is the intersection of the bases chosen for S and U, and $\dim D = 3$.

Intersection and sum. The intersection of two subspaces is a subspace, but the union $S \cup U$ is not: like two lines in $\mathbb{R}^2$, the union is not closed under addition. Filling it in gives the sum

$$S + U = \{\, \mathbf{s} + \mathbf{u} : \mathbf{s} \in S,\ \mathbf{u} \in U \,\},$$

the smallest subspace containing both. Here $S + U = M$: if L is the strictly lower triangular part of a matrix X, then $L + L^T$ is symmetric and $X - (L + L^T)$ is upper triangular. In general the dimensions of two subspaces satisfy

$$\dim S + \dim U = \dim(S \cap U) + \dim(S + U),$$

which here reads $6 + 6 = 3 + 9$.

12.2 Functions: solutions of a differential equation

The solutions of

$$\frac{d^2y}{dx^2} + y = 0$$

form a vector space: it is the "nullspace" of the operator $y \mapsto y'' + y$, so sums and multiples of solutions are solutions. Among the solutions are $\cos x$, $\sin x$, and e^{ix}, but the last is already the combination $\cos x + i \sin x$. The complete solution is

$$y = c_1 \cos x + c_2 \sin x,$$

with arbitrary (complex) constants. So $\cos x$, $\sin x$ is a basis, like the special solutions of $A\mathbf{x} = \mathbf{0}$, and the solution space has dimension 2, the order of the equation. The pair e^{ix}, e^{-ix} is another basis.

12.3 Subsets that are and are not subspaces

Let M now be the space of 5×17 matrices. The rank-4 matrices in M, even with the zero matrix added, do not form a subspace: the sum of two rank-4 matrices need not have rank 4. In general $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$, and here every rank is at most 5. Likewise the rank-one matrices are not a subspace, since the sum of two of them usually has rank 2.

Example: *A nullspace in* $\mathbb{R}^4$. The vectors $\mathbf{v} = (v_1, v_2, v_3, v_4)$ with $v_1 + v_2 + v_3 + v_4 = 0$ form a subspace S: it contains $\mathbf{0}$ and is closed under addition and scalar multiples. It is the nullspace of the 1×4 matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix},$$

which has rank $r = 1$ with pivot variable v_1 and free variables v_2, v_3, v_4. Hence $\dim S = n - r = 3$, with basis of special solutions

$$\begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

The other three subspaces of A are the row space, the multiples of $(1, 1, 1, 1)$, of dimension 1; the column space $C(A) = \mathbb{R}^1$, of dimension 1; and the left nullspace $N(A^T) = \{\mathbf{0}\}$, since the only combination of the single row giving zero is the zero combination. The zero space has dimension 0 and the empty set as its basis. The counts check: $3 + 1 = 4 = n$ and $1 + 0 = 1 = m$.

12.4 Rank-one matrices

The rank of a matrix is the dimension of its column space, which equals the dimension of its row space. In

$$A = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix},$$

the second row is twice the first and every column is a multiple of $(1, 2)$. The row space has basis $(1, 4, 5)$ and the column space has basis $(1, 2)$, both of dimension 1.

Proposition 12.1 [Rank-one matrices]. *Every rank-one matrix has the form $A = \mathbf{u}\mathbf{v}^T$ for nonzero column vectors $\mathbf{u}$ and $\mathbf{v}$. More generally, a matrix of rank r is a sum of r rank-one matrices.*

Proof. Let $\mathbf{u}_1, \ldots, \mathbf{u}_r$ be a basis of $C(A)$. Column j of A is a combination $\sum_i v_{ij}\mathbf{u}_i$. Collecting the coefficients v_{ij} for fixed i into a vector $\mathbf{v}_i$ gives $A = \sum_{i=1}^{r} \mathbf{u}_i \mathbf{v}_i^T$, since column j of $\mathbf{u}_i\mathbf{v}_i^T$ is $v_{ij}\mathbf{u}_i$. For $r = 1$ this is $A = \mathbf{u}\mathbf{v}^T$. □

For example, a 5×17 matrix of rank 4 is a combination of four rank-one matrices.

12.5 Small world graphs

A graph $G = \{\text{nodes}, \text{edges}\}$ is a collection of nodes joined by edges (Figure 7). In a friendship graph each node is a person, and an edge joins two people who are friends. The distance between two people is the smallest number of friend-to-friend edges joining them. The question of the largest distance in such a graph lies behind the phrases "six degrees of separation" and "it's a small world". The World Wide Web is another graph, with sites as nodes and links as edges. Describing graphs by matrices, as in Lecture 13, makes such questions accessible to linear algebra.

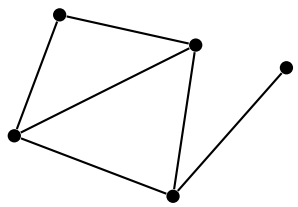


Figure 7: A graph with 5 nodes and 6 edges.

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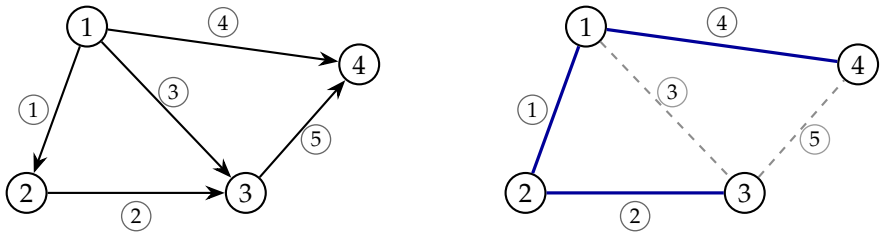
Lecture 13: Graphs, Networks, Incidence Matrices

Matrices from applications carry structure. The incidence matrix of a directed graph is the basic example: its four subspaces describe potentials, currents, loops, and trees in an electrical network.

13.1 The incidence matrix

A graph has n nodes and m edges. To describe currents, give each edge a direction, which fixes the sign of a current along it. The example graph has $n = 4$ nodes and $m = 5$ edges (Figure 8a).

Definition 13.1 [Incidence matrix]. The incidence matrix A of a directed graph has one row for each edge and one column for each node. If edge k runs from node a to node b, row k has -1 in column a, $+1$ in column b, and zeros elsewhere.



(a) directed graph (b) tree of edges 1, 2, 4

Figure 8: (a) The directed graph with nodes 1 to 4 and edges 1 to 5: $1 \to 2$, $2 \to 3$, $1 \to 3$, $1 \to 4$, $3 \to 4$. (b) Edges 1, 2, 4 (solid) connect all four nodes without a loop; adding edge 3 or 5 would close one.

For this graph

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{matrix} \text{edge 1} \\ \text{edge 2} \\ \text{edge 3} \\ \text{edge 4} \\ \text{edge 5} \end{matrix}$$

Every row has exactly two nonzeros, so the incidence matrix of a large graph is sparse: it has $2m$ nonzero entries out of mn.

Edges 1, 2, 3 form a loop through nodes 1, 2, 3, and correspondingly row 3 equals row 1 plus row 2. In general, traversing a loop and adding or subtracting each row according to whether its edge is traversed with or against its arrow cancels every node, giving the zero row: *loops correspond to dependent rows*.

13.2 The nullspace: potentials

Let $\mathbf{x} = (x_1, x_2, x_3, x_4)$ be the POTENTIALS at the nodes. Then

$$A\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix}$$

lists the POTENTIAL DIFFERENCES across the edges. These vanish exactly when all potentials are equal, so

$$N(A) = \{\, c(1,1,1,1) : c \in \mathbb{R} \,\}, \qquad \dim N(A) = 1, \qquad r = n - 1 = 3.$$

Potentials are determined only up to an additive constant; only their differences make current flow. GROUNDING a node, for example setting $x_4 = 0$, removes its column, and the remaining three columns are independent.

13.3 The left nullspace: Kirchhoff's current law

Let $\mathbf{y} = (y_1, \ldots, y_5)$ be the currents on the edges. The equation $A^T\mathbf{y} = \mathbf{0}$ reads

$$\begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

one equation per node:

$$-y_1 - y_3 - y_4 = 0, \qquad y_1 - y_2 = 0, \qquad y_2 + y_3 - y_5 = 0, \qquad y_4 + y_5 = 0.$$

Each says that the net current out of a node is zero. This is KIRCHHOFF'S CURRENT LAW (KCL), a balance or conservation law: charge does not accumulate at nodes.

Since A^T is 4×5 with rank 3, $\dim N(A^T) = m - r = 5 - 3 = 2$. A basis can be read off the graph without elimination: a unit current around any loop satisfies KCL, entering and leaving each node it visits once. Around the loop through nodes 1, 2, 3 and around the loop through nodes 1, 3, 4,

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \qquad \mathbf{y}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 1 \end{bmatrix}.$$

They are independent (only $\mathbf{y}_1$ uses edge 1, only $\mathbf{y}_2$ uses edge 4), so they form a basis. The current around the outer loop, $(1, 1, 0, -1, 1)$, also satisfies KCL, but it equals $\mathbf{y}_1 + \mathbf{y}_2$: the two currents on edge 3 cancel.

13.4 The row space: trees

The row space has dimension $r = 3$, so three rows of A (three columns of A^T) are independent. Rows 1, 2, 3 are not, because edges 1, 2, 3 form a loop. The pivot columns of A^T are columns 1, 2, 4, corresponding to edges 1, 2, 4, which connect all four nodes without forming a loop (Figure 8b). A graph with no loops is a TREE. Independent sets of rows of A correspond to sets of edges without loops, and a basis of the row space corresponds to a tree reaching every node.

13.5 Euler's formula

For a connected graph the nullspace of A is spanned by $(1, \ldots, 1)$, so $r = n - 1$. The number of independent loops is $\dim N(A^T) = m - r = m - (n - 1)$. Rearranging gives

$$\boxed{\text{(number of nodes)} - \text{(number of edges)} + \text{(number of loops)} = 1}$$

for every connected graph, EULER'S FORMULA. Nodes are zero-dimensional, edges one-dimensional, and loops enclose two-dimensional regions. The example has $4 - 5 + 2 = 1$; a graph with 5 nodes, 7 edges, and 3 independent loops has $5 - 7 + 3 = 1$.

13.6 The equations of a network

Three steps connect potentials to currents (Figure 9).

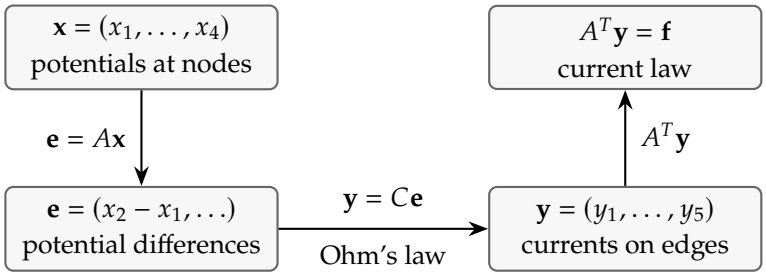


Figure 9: The framework of a network: A takes potentials to potential differences, C takes them to currents, and A^T takes currents to the net current into each node.

The potential differences are $\mathbf{e} = A\mathbf{x}$. OHM'S LAW says that the current on each edge is its conductance times the potential drop, $\mathbf{y} = C\mathbf{e}$, where C holds the conductances. Kirchhoff's current law gives $A^T\mathbf{y} = \mathbf{0}$, or $A^T\mathbf{y} = \mathbf{f}$ when current sources $\mathbf{f}$ send current into or out of the nodes. Combining the

three,

$$\boxed{A^T C A\,\mathbf{x} = \mathbf{f}}$$

This is the basic equilibrium equation of applied mathematics, and the same pattern governs hydraulic networks and structures. The matrix $A^T A$ is always symmetric, since $(A^T A)^T = A^T A$.

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Lecture 14: Orthogonal Vectors and Subspaces

This lecture shows that the four fundamental subspaces of an $m \times n$ matrix A of rank r come in perpendicular pairs: the row space is orthogonal to the nullspace in $\mathbb{R}^n$, and the column space is orthogonal to the left nullspace in $\mathbb{R}^m$. It ends with the matrix A^TA, which will be central for solving $A\mathbf{x} = \mathbf{b}$ when there is no exact solution.

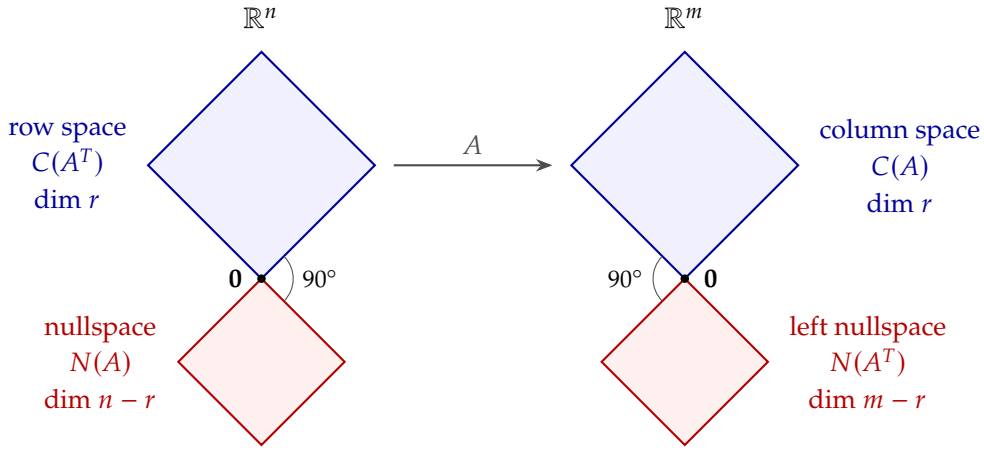


Figure 10: Schematic of the four subspaces. In $\mathbb{R}^n$ the row space and nullspace meet at right angles, and their dimensions r and $n - r$ add to n; in $\mathbb{R}^m$ the same holds for the column space and the left nullspace.

14.1 Orthogonal vectors

Definition 14.1 [Orthogonal vectors]. Vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are orthogonal (perpendicular), written $\mathbf{x} \perp \mathbf{y}$, if

$$\mathbf{x}^T\mathbf{y} = x_1y_1 + x_2y_2 + \cdots + x_ny_n = 0.$$

The length squared of $\mathbf{x}$ is $\|\mathbf{x}\|^2 = \mathbf{x}^T\mathbf{x} = x_1^2 + \cdots + x_n^2$, which is positive unless $\mathbf{x} = \mathbf{0}$.

The dot product test agrees with geometry. Two vectors are perpendicular exactly when $\mathbf{x}$, $\mathbf{y}$, and the hypotenuse $\mathbf{x} + \mathbf{y}$ form a right triangle, that is, when Pythagoras holds:

$$\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 = \|\mathbf{x} + \mathbf{y}\|^2.$$

Expanding the right side,

$$(\mathbf{x} + \mathbf{y})^T(\mathbf{x} + \mathbf{y}) = \mathbf{x}^T\mathbf{x} + \mathbf{x}^T\mathbf{y} + \mathbf{y}^T\mathbf{x} + \mathbf{y}^T\mathbf{y},$$

and $\mathbf{y}^T\mathbf{x} = \mathbf{x}^T\mathbf{y}$ for real vectors. So Pythagoras holds exactly when $2\mathbf{x}^T\mathbf{y} = 0$. For other triangles it fails.

Example: For $\mathbf{x} = (1, 2, 3)$ and $\mathbf{y} = (2, -1, 0)$, $\mathbf{x}^T\mathbf{y} = 2 - 2 + 0 = 0$. Indeed $\|\mathbf{x}\|^2 = 14$, $\|\mathbf{y}\|^2 = 5$, and $\mathbf{x} + \mathbf{y} = (3, 1, 3)$ has $\|\mathbf{x} + \mathbf{y}\|^2 = 19 = 14 + 5$.

The zero vector is orthogonal to every vector, since $\mathbf{0}^T\mathbf{y} = 0$.

14.2 Orthogonal subspaces

Definition 14.2 [Orthogonal subspaces]. A subspace S is orthogonal to a subspace T if every vector in S is orthogonal to every vector in T.

Orthogonal subspaces meet only in $\mathbf{0}$: a nonzero vector in both would have to be orthogonal to itself. So the plane of a wall and the plane of the floor in $\mathbb{R}^3$ are *not* orthogonal, even though they look perpendicular: a vector along the line where they meet lies in both. Meeting only at $\mathbf{0}$ is necessary but not sufficient. In $\mathbb{R}^2$ the subspaces are $\{\mathbf{0}\}$, lines through the origin, and the whole plane:

- $\{\mathbf{0}\}$ is orthogonal to every line through the origin;
- a line through the origin is never orthogonal to the whole plane;
- two lines through the origin are orthogonal exactly when they meet at a right angle.

14.3 The nullspace is perpendicular to the row space

Theorem 14.3 [Orthogonality of the four subspaces]. *For every matrix A, the row space $C(A^T)$ is orthogonal to the nullspace $N(A)$, and the column space $C(A)$ is orthogonal to the left nullspace $N(A^T)$.*

Proof. Let $\mathbf{x} \in N(A)$. Writing $A\mathbf{x} = \mathbf{0}$ row by row,

$$\begin{bmatrix} \text{row } 1 \\ \vdots \\ \text{row } m \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix},$$

so $(\text{row } i)^T\mathbf{x} = 0$ for each i: $\mathbf{x}$ is orthogonal to every row. The row space also contains all combinations of the rows, and

$$\big(c_1 \,\text{row } 1 + \cdots + c_m \,\text{row } m\big)^T \mathbf{x} = c_1 \cdot 0 + \cdots + c_m \cdot 0 = 0.$$

The second statement is the first applied to A^T, whose row space is $C(A)$ and whose nullspace is $N(A^T)$. □

More is true. The dimensions r and $n - r$ of the row space and nullspace add to n, so the two subspaces split $\mathbb{R}^n$ into two perpendicular pieces. In $\mathbb{R}^3$, for instance, two perpendicular lines can never be the row space and nullspace of a matrix, since $1 + 1 \neq 3$.

Definition 14.4 [Orthogonal complement]. The orthogonal complement of a subspace V of $\mathbb{R}^n$ is the subspace of *all* vectors orthogonal to every vector in V.

Theorem 14.5 [Orthogonal complements]. *In $\mathbb{R}^n$, the nullspace is the orthogonal complement of the row space, and the row space is the orthogonal complement of the nullspace. Likewise $N(A^T)$ and $C(A)$ are orthogonal complements in $\mathbb{R}^m$.*

Proof. If $\mathbf{x}$ is orthogonal to the whole row space, it is orthogonal to each row, so $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x} \in N(A)$. Conversely, let V be the set of vectors orthogonal to all of $N(A)$. By Theorem 14.3, $C(A^T) \subseteq V$. Put a basis of $N(A)$ into the rows of an $(n-r) \times n$ matrix B of rank $n-r$. Then $V = N(B)$ has dimension $n-(n-r) = r = \dim C(A^T)$, so $V = C(A^T)$. Apply the same argument to A^T for the second pair. □

Example: Let

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}.$$

Here $m = 2$, $n = 3$, and $r = 1$. The row space is the line through $(1, 2, 5)$; the second row adds nothing. The nullspace is the solution set of the single equation $x_1 + 2x_2 + 5x_3 = 0$, which is the plane through the origin with normal vector $(1, 2, 5)$: exactly the vectors perpendicular to the row space. Its dimension is $n - r = 2$.

Together, these facts form part two of the fundamental theorem of linear algebra. Part one gave the dimensions of the four subspaces; part two says they come in orthogonal complementary pairs; part three will provide orthogonal bases for them.

14.4 The matrix A^TA

The rest of this chapter solves $A\mathbf{x} = \mathbf{b}$ in the best possible way when there is no solution, that is, when $\mathbf{b}$ is not in the column space. This is typical when $m > n$: many measurements (positions of a satellite, repeated readings of a pulse rate) determine a few parameters, and noise in $\mathbf{b}$ makes the equations inconsistent. Discarding equations until the system is square and invertible wastes information; the aim is to use every measurement. Elimination only reports that there is no solution.

The central equation will be

$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b},$$

whose solution $\hat{\mathbf{x}}$ is the best estimate. The matrix A^TA is $n \times n$, so square, and symmetric, since $(A^TA)^T = A^T(A^T)^T = A^TA$. The question is when it is invertible.

Example: For

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix}, \qquad A^TA = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 8 & 30 \end{bmatrix},$$

which is invertible. With three equations and two unknowns, $A\mathbf{x} = \mathbf{b}$ is solvable only when $\mathbf{b}$ lies in the plane $C(A)$, so it usually has no solution. If instead the columns are $(1, 1, 1)$ and $(3, 3, 3)$, then A has rank one and

$$A^T A = \begin{bmatrix} 3 & 9 \\ 9 & 27 \end{bmatrix}$$

is singular. A product involving a rank-one matrix has rank at most one.

Theorem 14.6 [Nullspace and rank of $A^T A$**].** *For every matrix A,*

$$N(A^T A) = N(A) \qquad \textit{and} \qquad \operatorname{rank}(A^T A) = \operatorname{rank}(A).$$

Hence $A^T A$ is invertible exactly when A has independent columns.

Proof. If $A\mathbf{x} = \mathbf{0}$ then $A^T A\mathbf{x} = \mathbf{0}$. Conversely, if $A^T A\mathbf{x} = \mathbf{0}$, then

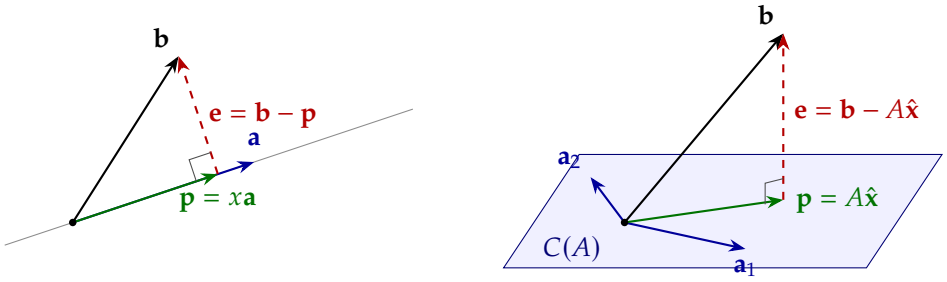
$$0 = \mathbf{x}^T A^T A\mathbf{x} = (A\mathbf{x})^T (A\mathbf{x}) = \|A\mathbf{x}\|^2,$$

so $A\mathbf{x} = \mathbf{0}$. Both matrices have n columns and the same nullspace, so their ranks agree. The square matrix $A^T A$ is invertible exactly when its nullspace is $\{\mathbf{0}\}$, that is, when $N(A) = \{\mathbf{0}\}$, which means the columns of A are independent. □

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 14, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 15: Projections onto Subspaces

Projecting a vector $\mathbf{b}$ onto a subspace finds the closest point $\mathbf{p}$ of that subspace. The key fact is that the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to the subspace. This gives three formulas, for the coefficients $\hat{\mathbf{x}}$, the projection $\mathbf{p}$, and the projection matrix P, first for a line and then for any subspace.



(a) onto a line (b) onto the plane $C(A)$

Figure 11: The closest point $\mathbf{p}$ to $\mathbf{b}$ on a line and in a plane. In both cases the error $\mathbf{e} = \mathbf{b} - \mathbf{p}$ is perpendicular to the subspace. Panel (b) is a perspective sketch.

15.1 Projection onto a line

Let $\mathbf{a} \neq \mathbf{0}$ determine a line through the origin. The projection $\mathbf{p}$ of $\mathbf{b}$ lies on the line, so $\mathbf{p} = x\mathbf{a}$ for some number x, and the error $\mathbf{e} = \mathbf{b} - x\mathbf{a}$ is perpendicular to $\mathbf{a}$:

$$\mathbf{a}^T(\mathbf{b} - x\mathbf{a}) = 0 \quad \Longrightarrow \quad x\,\mathbf{a}^T\mathbf{a} = \mathbf{a}^T\mathbf{b} \quad \Longrightarrow \quad x = \frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}, \qquad \mathbf{p} = \mathbf{a}\,\frac{\mathbf{a}^T\mathbf{b}}{\mathbf{a}^T\mathbf{a}}.$$

Trigonometry would give the same point through $\cos\theta$, with messier formulas. Doubling $\mathbf{b}$ doubles $\mathbf{p}$. Doubling $\mathbf{a}$, or replacing it by $-\mathbf{a}$, does not change $\mathbf{p}$: the line is the same, and the factors cancel in the formula.

Moving the parentheses, $\mathbf{p} = P\mathbf{b}$ for the PROJECTION MATRIX

$$\boxed{P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}}}.$$

The numerator $\mathbf{a}\mathbf{a}^T$ is a column times a row, an $n \times n$ matrix, while the denominator is a number. This P has the following properties.

- Its column space is the line through $\mathbf{a}$, because $P\mathbf{b}$ always lies on that line; so $\operatorname{rank} P = 1$.
- It is symmetric: $P^T = P$, since $(\mathbf{a}\mathbf{a}^T)^T = \mathbf{a}\mathbf{a}^T$.

- $P^2 = P$: projecting a second time leaves a vector on the line where it is.

15.2 Why project?

The system $A\mathbf{x} = \mathbf{b}$ may have no solution, typically when there are more equations than unknowns. The vector $A\mathbf{x}$ always lies in the column space of A, and $\mathbf{b}$ usually does not. So replace $\mathbf{b}$ by the closest vector $\mathbf{p}$ in the column space and solve

$$A\hat{\mathbf{x}} = \mathbf{p}$$

instead. The hat marks the best possible $\hat{\mathbf{x}}$, not an exact solution of the original system.

15.3 Projection onto a subspace

Let $\mathbf{a}_1, \ldots, \mathbf{a}_n$ be independent vectors in $\mathbb{R}^m$, for example a basis $\mathbf{a}_1, \mathbf{a}_2$ of a plane in $\mathbb{R}^3$. They need not be perpendicular. Put them into the columns of A, so that the subspace is $C(A)$. The projection is a combination of the columns,

$$\mathbf{p} = \hat{x}_1\mathbf{a}_1 + \cdots + \hat{x}_n\mathbf{a}_n = A\hat{\mathbf{x}},$$

and the error $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ must be perpendicular to the subspace, that is, to every $\mathbf{a}_i$:

$$\mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0, \ \ldots, \ \mathbf{a}_n^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \qquad \Longleftrightarrow \qquad A^T(\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0}.$$

For a single column this is the equation $\mathbf{a}^T(\mathbf{b} - x\mathbf{a}) = 0$ from before. It says that $\mathbf{e}$ lies in $N(A^T)$, which by Lecture 14 is exactly the set of vectors perpendicular to $C(A)$, as it should be.

Rearranged, these are the NORMAL EQUATIONS

$$\boxed{A^TA\hat{\mathbf{x}} = A^T\mathbf{b}}.$$

Since the columns of A are independent, the $n \times n$ matrix A^TA is invertible (Lecture 14). Instead of dividing by the number $\mathbf{a}^T\mathbf{a}$, we multiply by $(A^TA)^{-1}$:

$$\hat{\mathbf{x}} = (A^TA)^{-1}A^T\mathbf{b}, \qquad \mathbf{p} = A\hat{\mathbf{x}} = A(A^TA)^{-1}A^T\mathbf{b}, \qquad \boxed{P = A(A^TA)^{-1}A^T}.$$

For one column these reduce to the three formulas for a line.

Proposition 15.1 [Projection matrices]. *The matrix $P = A(A^TA)^{-1}A^T$ satisfies $P^T = P$ and $P^2 = P$.*

Proof. The matrix A^TA is symmetric, so its inverse is symmetric too. Transposing reverses the order of the three factors: $P^T = (A^T)^T\big((A^TA)^{-1}\big)^TA^T = A(A^TA)^{-1}A^T = P$. For the square,

$$P^2 = A(A^TA)^{-1}\underbrace{(A^TA)(A^TA)^{-1}}_{I}A^T = A(A^TA)^{-1}A^T = P.$$

Geometrically, the first projection lands in the column space and the second leaves the vector there. □

Note. It is tempting to write $(A^TA)^{-1} = A^{-1}(A^T)^{-1}$, which would make $P = I$. This is not allowed when A is rectangular, since A has no inverse. If A is square and invertible, the simplification is valid: then $C(A)$ is the whole space, $\mathbf{b}$ is already in it, and indeed $P = I$.

15.4 Least squares: fitting a line

The main application is fitting data by a line. Given the points $(t, b) = (1, 1),\ (2, 2),\ (3, 2)$, find the line $b = C + Dt$ closest to them. A line through all three points would satisfy

$$\begin{aligned} C + D &= 1 \\ C + 2D &= 2 \\ C + 3D &= 2 \end{aligned} \quad\Longleftrightarrow\quad \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} C \\ D \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}}_{\mathbf{b}}.$$

No line passes through all three points, so this system has no solution. Instead we solve the normal equations $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$ for the best $\hat{C}$ and $\hat{D}$; Lecture 16 completes the computation.

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Lecture 16: Projection Matrices and Least Squares

The projection matrix $P = A(A^TA)^{-1}A^T$ splits every vector into a part in the column space and a part perpendicular to it. Applied to fitting a line through data, it gives the method of least squares.

16.1 What the projection matrix does

Let the columns of A be independent, so that $P = A(A^TA)^{-1}A^T$ projects onto $C(A)$. Two extreme cases show the formula at work.

- If $\mathbf{b} \in C(A)$, then $\mathbf{b} = A\mathbf{x}$ for some $\mathbf{x}$, and $P\mathbf{b} = A(A^TA)^{-1}(A^TA)\mathbf{x} = A\mathbf{x} = \mathbf{b}$.
- If $\mathbf{b} \perp C(A)$, then $\mathbf{b} \in N(A^T)$, so $A^T\mathbf{b} = \mathbf{0}$ and $P\mathbf{b} = \mathbf{0}$.

A typical $\mathbf{b}$ has a component $\mathbf{p}$ in the column space and a component $\mathbf{e}$ perpendicular to it, in $N(A^T)$. The projection keeps $\mathbf{p}$ and kills $\mathbf{e}$:

$$\mathbf{b} = \mathbf{p} + \mathbf{e}, \qquad \mathbf{p} = P\mathbf{b}, \qquad \mathbf{e} = \mathbf{b} - \mathbf{p} = (I - P)\mathbf{b}.$$

So $I - P$ is the projection onto $N(A^T)$, and it is again a projection matrix: it is symmetric, and

$$(I - P)^2 = I - 2P + P^2 = I - P.$$

16.2 Least squares

Return to the line $b = C + Dt$ through the data $(1, 1)$, $(2, 2)$, $(3, 2)$. The unsolvable system is $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} C \\ D \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

The columns of A are independent, so they form a basis for $C(A)$, but $\mathbf{b}$ is not in $C(A)$. The best line minimizes the total squared error

$$\|A\mathbf{x} - \mathbf{b}\|^2 = \|\mathbf{e}\|^2 = e_1^2 + e_2^2 + e_3^2,$$

where e_i is the vertical distance from the ith data point to the line. This is LEAST SQUARES, and fitting a line this way is LINEAR REGRESSION. Squaring makes a single bad point, an OUTLIER, very costly, so one far-off measurement can change the best line drastically; least squares is most reliable when there are no outliers.

There are two pictures of the same problem.

- In the tb-plane, $p_i = C + Dt_i$ is the height of the line at t_i, and $e_i = b_i - p_i$ is the vertical miss. The heights p_1, p_2, p_3 lie exactly on a line, so $A\mathbf{x} = \mathbf{p}$ is solvable.
- In $\mathbb{R}^3$, $\mathbf{p} = (p_1, p_2, p_3)$ is the projection of $\mathbf{b}$ onto the column space, and $\mathbf{e}$ lies in $N(A^T)$. The numbers C and D are the coefficients of the combination of columns that gives $\mathbf{p}$.

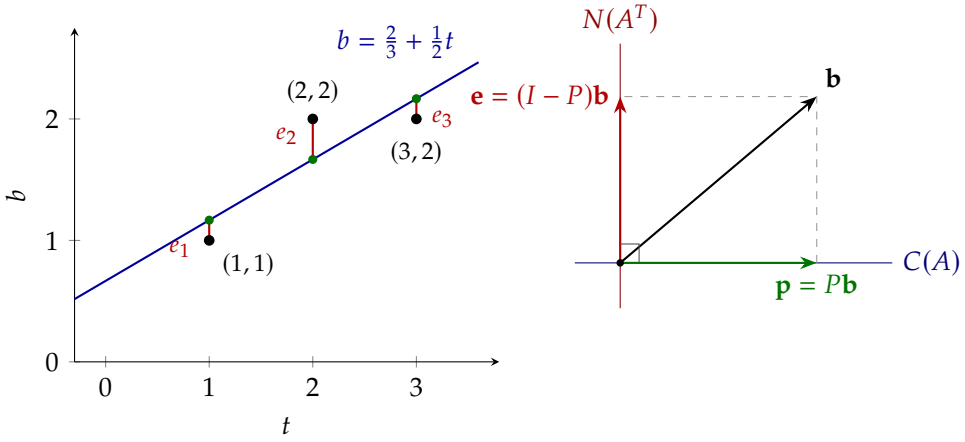


Figure 12: Left: the data points, the least-squares line $b = \frac{2}{3} + \frac{1}{2}t$, the fitted values p_i on the line, and the vertical errors e_i. Right: the same problem as a projection of $\mathbf{b} \in \mathbb{R}^3$, drawn schematically with $C(A)$ as a line; $\mathbf{b} = \mathbf{p} + \mathbf{e}$ with $\mathbf{p} \perp \mathbf{e}$.

16.3 Computing the best line

The normal equations $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$ are

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}, \qquad \text{that is,} \qquad \begin{aligned} 3\hat{C} + 6\hat{D} &= 5, \\ 6\hat{C} + 14\hat{D} &= 11. \end{aligned}$$

As expected, $A^T A$ is symmetric and invertible. Subtracting twice the first equation from the second gives $2\hat{D} = 1$, so $\hat{D} = \frac{1}{2}$, and back substitution gives $3\hat{C} = 5 - 3$, so $\hat{C} = \frac{2}{3}$. The best line is

$$\boxed{b = \tfrac{2}{3} + \tfrac{1}{2}t}.$$

Calculus gives the same equations. The error

$$E(C, D) = (C + D - 1)^2 + (C + 2D - 2)^2 + (C + 3D - 2)^2$$

is a function of two variables, and its partial derivatives are linear:

$$\tfrac{1}{2}\,\partial E/\partial C = 3C + 6D - 5, \qquad \tfrac{1}{2}\,\partial E/\partial D = 6C + 14D - 11.$$

Setting them to zero reproduces the normal equations.

Evaluating the line at $t = 1, 2, 3$ gives the projection, and subtracting from $\mathbf{b}$ gives the error:

i	1	2	3
b_i	1	2	2
p_i	7/6	5/3	13/6
e_i	−1/6	1/3	−1/6

$$\mathbf{b} = \mathbf{p} + \mathbf{e}: \quad \begin{bmatrix}1\\2\\2\end{bmatrix} = \begin{bmatrix}7/6\\5/3\\13/6\end{bmatrix} + \begin{bmatrix}-1/6\\2/6\\-1/6\end{bmatrix}.$$

Two checks confirm the geometry.

- $\mathbf{p} \perp \mathbf{e}$: $\mathbf{p}^T\mathbf{e} = \frac{-7+20-13}{36} = 0$.
- $\mathbf{e}$ is perpendicular to the whole column space, not just to $\mathbf{p}$: $(1,1,1)\cdot\mathbf{e} = \frac{-1+2-1}{6} = 0$ and $(1,2,3)\cdot\mathbf{e} = \frac{-1+4-3}{6} = 0$.

16.4 When is A^TA invertible?

Theorem 16.1 [Invertibility of A^TA]. *If A has independent columns, then A^TA is invertible.*

Proof. Suppose $A^TA\mathbf{x} = \mathbf{0}$; we show $\mathbf{x} = \mathbf{0}$. Take the dot product with $\mathbf{x}$:

$$\mathbf{x}^TA^TA\mathbf{x} = (A\mathbf{x})^T(A\mathbf{x}) = \|A\mathbf{x}\|^2 = 0,$$

so $A\mathbf{x} = \mathbf{0}$. Because the columns of A are independent, the only vector in $N(A)$ is $\mathbf{0}$, so $\mathbf{x} = \mathbf{0}$. A square matrix whose nullspace is $\{\mathbf{0}\}$ is invertible. □

So least squares works whenever the columns of A are independent. Columns are certainly independent when they are orthonormal: perpendicular unit vectors, such as the standard basis vectors $(1,0,0)$, $(0,1,0)$, $(0,0,1)$, or the pair

$$\begin{bmatrix}\cos\theta\\\sin\theta\end{bmatrix}, \qquad \begin{bmatrix}-\sin\theta\\\cos\theta\end{bmatrix}.$$

For a matrix with orthonormal columns, A^TA is the identity. Lecture 17 develops this case.

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Lecture 17: Orthogonal Matrices and Gram-Schmidt

An orthonormal basis, or a matrix with orthonormal columns, makes the formulas of this chapter trivial. The Gram–Schmidt process turns any basis into an orthonormal basis of the same space, and in matrix form it becomes the factorization $A = QR$.

17.1 Orthonormal vectors and matrices

Definition 17.1 [Orthonormal vectors]. Vectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$ are orthonormal if

$$\mathbf{q}_i^T \mathbf{q}_j = \begin{cases} 0, & i \neq j, \\ 1, & i = j. \end{cases}$$

They are perpendicular (ortho) and have unit length (normal).

Orthonormal vectors are independent: if $c_1\mathbf{q}_1 + \cdots + c_n\mathbf{q}_n = \mathbf{0}$, taking the dot product with $\mathbf{q}_i$ leaves $c_i = 0$.

Put orthonormal vectors into the columns of a matrix $Q = \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{bmatrix}$, which may be rectangular. The rows of Q^T are the $\mathbf{q}_i^T$, so the entries of Q^TQ are the dot products $\mathbf{q}_i^T\mathbf{q}_j$:

$$Q^TQ = \begin{bmatrix} \mathbf{q}_1^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{bmatrix} = I.$$

Matrices with orthonormal columns join the earlier families: triangular, diagonal, permutation, symmetric, reduced row echelon, and projection matrices.

Definition 17.2 [Orthogonal matrix]. A *square* matrix Q with orthonormal columns is an orthogonal matrix. Then $Q^TQ = I$ says

$$Q^T = Q^{-1}.$$

Example: Every permutation matrix is orthogonal; for instance

$$Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix},$$

and both are orthogonal with $Q^TQ = I$. Rotations are orthogonal:

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

The matrix $\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ has orthogonal columns of length $\sqrt{2}$, so it is not orthogonal, but dividing by the length fixes it:

$$Q = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad Q = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

The second repeats the 2×2 sign pattern in blocks, with a minus on the last block. Orthogonal matrices whose entries are all ± 1 before scaling are HADAMARD MATRICES; this block pattern produces larger ones, but which sizes admit a Hadamard matrix is not known in general.

Example: A rectangular matrix with orthonormal columns, and an orthogonal matrix extending it by a third column:

$$Q = \frac{1}{3}\begin{bmatrix} 1 & -2 \\ 2 & -1 \\ 2 & 2 \end{bmatrix}, \qquad \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}.$$

Each column $(1, 2, 2)$ has length 3, and the columns are perpendicular. Such examples are pleasant because no square roots appear; Gram–Schmidt usually produces them, since it divides by lengths.

17.2 Orthonormal columns simplify projections

Suppose Q has orthonormal columns. The projection onto its column space is

$$P = Q(Q^TQ)^{-1}Q^T = QQ^T,$$

since $Q^TQ = I$. If Q is square, its columns span the whole space and $P = QQ^T = I$. Otherwise $QQ^T \neq I$, but it has the two properties of a projection: it is symmetric, and $(QQ^T)(QQ^T) = Q(Q^TQ)Q^T = QQ^T$.

The normal equations $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$ become $Q^TQ\hat{\mathbf{x}} = Q^T\mathbf{b}$, so no system needs to be solved:

$$\boxed{\hat{\mathbf{x}} = Q^T\mathbf{b}}, \qquad \text{that is,} \qquad \hat{x}_i = \mathbf{q}_i^T\mathbf{b}.$$

With an orthonormal basis, the component of $\mathbf{b}$ along the ith basis vector is just a dot product.

17.3 The Gram–Schmidt process

Elimination aimed to make a matrix triangular; Gram–Schmidt aims to make its columns orthonormal. Start with two independent vectors $\mathbf{a}$ and $\mathbf{b}$. First find orthogonal vectors $\mathbf{A}$ and $\mathbf{B}$ spanning the same plane, then normalize:

$$\mathbf{q}_1 = \frac{\mathbf{A}}{\|\mathbf{A}\|}, \qquad \mathbf{q}_2 = \frac{\mathbf{B}}{\|\mathbf{B}\|}.$$

The first direction is kept: $\mathbf{A} = \mathbf{a}$. For the second, subtract from $\mathbf{b}$ its projection onto $\mathbf{A}$; what remains is the error vector $\mathbf{e}$ of Lecture 15:

$$\mathbf{B} = \mathbf{b} - \frac{\mathbf{A}^T\mathbf{b}}{\mathbf{A}^T\mathbf{A}}\,\mathbf{A}.$$

Multiplying by $\mathbf{A}^T$ confirms $\mathbf{A}^T\mathbf{B} = \mathbf{A}^T\mathbf{b} - \mathbf{A}^T\mathbf{b} = 0$. Because $\mathbf{a}$ and $\mathbf{b}$ are independent, $\mathbf{B} \neq \mathbf{0}$.

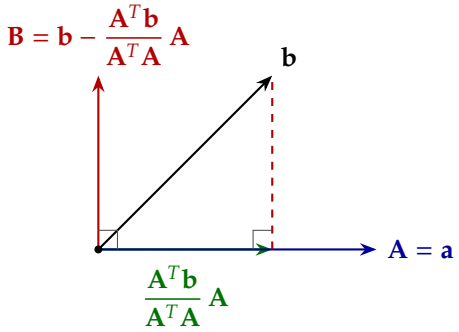


Figure 13: Gram–Schmidt for two vectors. Subtracting from $\mathbf{b}$ its projection onto $\mathbf{A} = \mathbf{a}$ leaves $\mathbf{B} \perp \mathbf{A}$. The dashed segment is the error vector, and $\mathbf{B}$ is that error drawn from the origin.

For a third independent vector $\mathbf{c}$, subtract its components in both earlier directions:

$$\mathbf{C} = \mathbf{c} - \frac{\mathbf{A}^T\mathbf{c}}{\mathbf{A}^T\mathbf{A}}\,\mathbf{A} - \frac{\mathbf{B}^T\mathbf{c}}{\mathbf{B}^T\mathbf{B}}\,\mathbf{B}, \qquad \mathbf{q}_3 = \frac{\mathbf{C}}{\|\mathbf{C}\|}.$$

Then $\mathbf{A}^T\mathbf{C} = \mathbf{A}^T\mathbf{c} - \mathbf{A}^T\mathbf{c} - 0 = 0$, and similarly $\mathbf{B}^T\mathbf{C} = 0$. Continuing in the same way, with the normalized vectors so that each projection coefficient is a dot product, gives the general process.

Algorithm 17.1: Gram–Schmidt

Input: Independent vectors $\mathbf{a}_1, \dots, \mathbf{a}_n \in \mathbb{R}^m$.
Output: Orthonormal $\mathbf{q}_1, \dots, \mathbf{q}_n$ with the same span.

```
for j ← 1 to n do
    v ← a_j − Σ_{i<j} (q_i^T a_j) q_i        // remove earlier components
    q_j ← v/‖v‖
return q_1, ..., q_n
```

Example: Let $\mathbf{a} = (1, 1, 1)$ and $\mathbf{b} = (1, 0, 2)$, which are not orthogonal. Then $\mathbf{A} = \mathbf{a}$, $\mathbf{A}^T\mathbf{b} = 3$, and

$\mathbf{A}^T\mathbf{A} = 3$, so

$$\mathbf{B} = \begin{bmatrix}1\\0\\2\end{bmatrix} - \frac{3}{3}\begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}0\\-1\\1\end{bmatrix}, \qquad \mathbf{A}^T\mathbf{B} = 0.$$

Dividing by the lengths $\sqrt{3}$ and $\sqrt{2}$,

$$Q = \begin{bmatrix}\mathbf{q}_1 & \mathbf{q}_2\end{bmatrix} = \begin{bmatrix}1/\sqrt{3} & 0\\ 1/\sqrt{3} & -1/\sqrt{2}\\ 1/\sqrt{3} & 1/\sqrt{2}\end{bmatrix}.$$

The column space of Q is the same plane as the column space of $\begin{bmatrix}\mathbf{a} & \mathbf{b}\end{bmatrix}$: $\mathbf{B}$ is a combination of $\mathbf{a}$ and $\mathbf{b}$, so each step stays in that plane.

17.4 The factorization $A = QR$

Elimination in matrix language was $A = LU$. Gram–Schmidt in matrix language is $A = QR$, where Q has orthonormal columns with the same column space as A, and R is upper triangular. For two columns,

$$\begin{bmatrix}\mathbf{a}_1 & \mathbf{a}_2\end{bmatrix} = \begin{bmatrix}\mathbf{q}_1 & \mathbf{q}_2\end{bmatrix}\begin{bmatrix}\mathbf{a}_1^T\mathbf{q}_1 & \mathbf{a}_2^T\mathbf{q}_1\\ \mathbf{a}_1^T\mathbf{q}_2 & \mathbf{a}_2^T\mathbf{q}_2\end{bmatrix}, \qquad \mathbf{a}_1^T\mathbf{q}_2 = 0.$$

In general $R = Q^TA$, with entries $r_{ij} = \mathbf{q}_i^T\mathbf{a}_j$, which is consistent with $Q^TQ = I$. It is upper triangular because each $\mathbf{q}_i$ was constructed perpendicular to all earlier vectors: $\mathbf{a}_j$ lies in the span of $\mathbf{q}_1, \ldots, \mathbf{q}_j$, so $\mathbf{q}_i^T\mathbf{a}_j = 0$ for $i > j$. In the example above,

$$\begin{bmatrix}1 & 1\\ 1 & 0\\ 1 & 2\end{bmatrix} = \begin{bmatrix}1/\sqrt{3} & 0\\ 1/\sqrt{3} & -1/\sqrt{2}\\ 1/\sqrt{3} & 1/\sqrt{2}\end{bmatrix}\begin{bmatrix}\sqrt{3} & \sqrt{3}\\ 0 & \sqrt{2}\end{bmatrix}.$$

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Lecture 18: Properties of Determinants

From here on the focus moves from rectangular to square matrices, and the next large topics are determinants and eigenvalues. The DETERMINANT is a number $\det A$, also written $|A|$, attached to every square matrix. It packs a great deal of information into one number; in particular, A is invertible exactly when $\det A \neq 0$. Rather than start from a big formula, we start from three defining properties and derive everything else from them.

18.1 The three defining properties

1. $\det I = 1$.
2. Exchanging two rows reverses the sign of the determinant.
3. The determinant is a linear function of each row separately, when the other rows are held fixed:
   (a) multiplying one row by t multiplies the determinant by t,
   $$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix};$$
   (b) if one row is a sum, the determinant splits accordingly,
   $$\begin{vmatrix} a + a' & b + b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}.$$

Linearity holds in one row at a time. It does *not* say $\det(A + B) = \det A + \det B$, which would change every row at once. Stated for the first row, property 3 extends to any row: exchange that row to the top, use linearity, and exchange back, with two sign changes that cancel.

Properties 1 and 2 already determine the determinant of every permutation matrix: it is $+1$ or -1 according to whether the permutation takes an even or an odd number of row exchanges. For example,
$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1, \qquad \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1.$$
The familiar 2×2 formula $ad - bc$ satisfies all three properties, and the properties will produce a formula for every size.

18.2 Consequences for elimination

The next properties follow from properties 1 to 3.

4. *If two rows are equal, then* $\det A = 0$. Exchanging the two equal rows leaves the matrix, and so its determinant, unchanged; but by property 2 it also reverses the sign. The only number equal to its own negative is 0. This matches the fact that a matrix with two equal rows has rank less than n.

5. *Subtracting ℓ times row i from row $k \neq i$ leaves the determinant unchanged.* In the 2×2 case,

$$\begin{vmatrix} a & b \\ c - \ell a & d - \ell b \end{vmatrix} \overset{3(b)}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -\ell a & -\ell b \end{vmatrix} \overset{3(a)}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \ell \begin{vmatrix} a & b \\ a & b \end{vmatrix} \overset{4}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

   For $n \times n$ the argument is the same: the extra determinant has row k equal to $-\ell$ times row i, so it is $-\ell$ times a determinant with two equal rows. Hence the elimination steps from A to an upper triangular U do not change the determinant, apart from the sign changes of any row exchanges.

6. *A row of zeros gives* $\det A = 0$. A zero row equals 0 times itself, so property 3(a) with $t = 0$ gives $\det A = 0 \cdot \det A = 0$.

7. *If U is triangular, then* $\det U = d_1 d_2 \cdots d_n$*, the product of its diagonal entries.* Suppose every $d_i \neq 0$. By property 5, elimination clears the entries off the diagonal without changing the determinant (for upper triangular U, work upward from the last row). This leaves the diagonal matrix with entries $d_1, \ldots, d_n$. Factoring d_i out of row i by property 3(a) leaves the identity, whose determinant is 1 by property 1. If some $d_i = 0$, elimination produces a row of zeros, so $\det U = 0$ by property 6, which again equals the product.

8. $\det A = 0$ *exactly when A is singular.* If A is singular, elimination produces a row of zeros, and $\det A = 0$. If A is invertible, elimination produces a full set of pivots $d_1, \ldots, d_n$, and

$$\det A = \pm\, d_1 d_2 \cdots d_n \neq 0,$$

   with the sign $-$ when an odd number of row exchanges was needed.

This is the practical formula: software computes a determinant by elimination, keeping track of the parity of the row exchanges, and multiplies the pivots. For 2×2 with $a \neq 0$,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longrightarrow \begin{bmatrix} a & b \\ 0 & d - \frac{c}{a} b \end{bmatrix}, \qquad \begin{vmatrix} a & b \\ c & d \end{vmatrix} = a\left(d - \frac{c}{a} b\right) = ad - bc.$$

If $a = 0$, a row exchange is needed first, and if that fails, the matrix is singular. The argument also shows that properties 1 to 3 determine the determinant completely: any function satisfying them must equal $\pm$ the product of the pivots.

18.3 Products and transposes

9. $\det(AB) = (\det A)(\det B)$.

   Proof. If B is singular, so is AB, and both sides are 0. Otherwise consider $D(A) = \det(AB)/\det B$ as a function of A. It satisfies property 1, since $D(I) = 1$. Exchanging two rows of A exchanges the same rows of AB, so property 2 holds. Row i of AB is (row i of A) B, which is linear in row i of A, so property 3 holds. Since the three properties determine the determinant, $D(A) = \det A$.

Although the determinant of a sum is not the sum of determinants, the determinant of a product is the product of determinants. Consequences:

- From $A^{-1}A = I$, $(\det A^{-1})(\det A) = \det I = 1$, so

$$\det A^{-1} = \frac{1}{\det A}.$$

  For instance, $\operatorname{diag}(2, 3)$ has determinant 6 and its inverse $\operatorname{diag}(\frac{1}{2}, \frac{1}{3})$ has determinant $\frac{1}{6}$. If A is singular, A^{-1} does not exist and $1/\det A$ makes no sense either.

- $\det A^2 = (\det A)^2$.

- $\det(2A) = 2^n \det A$ for $n \times n$ matrices, not $2 \det A$: a factor 2 comes out of each of the n rows by property 3(a). Like volume, doubling every side of a box in three dimensions multiplies its volume by $2^3 = 8$.

10. $\det A^T = \det A$. For 2×2,

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc.$$

    Proof. If A is singular, so is A^T, and both determinants are 0. Otherwise elimination gives $A = LU$, so $A^T = U^T L^T$, and by property 9 the claim becomes

$$|U^T|\,|L^T| = |L|\,|U|.$$

    The matrices L and L^T are triangular with 1s on the diagonal, so by property 7 both determinants are 1. The matrices U and U^T are triangular with the same diagonal, so $|U| = |U^T|$. If row exchanges are needed, $PA = LU$ and $A^T P^T = U^T L^T$; since $P^T P = I$ and $\det P = \pm 1$, property 9 gives $\det P^T = \det P$, and the same comparison goes through.

Property 10 turns every statement about rows into one about columns. A column of zeros gives determinant 0, exchanging two columns reverses the sign, and the determinant is linear in each column separately.

18.4 A loose end

Property 2 makes the determinant well defined only if no permutation can be reached both by an even and by an odd number of row exchanges. If seven exchanges and ten exchanges could produce the same matrix, the determinant would equal its own negative. The fact that rescues the definition is that every permutation is either even or odd: all ways of producing it by exchanges have the same parity. With that fact, properties 1 to 3 define the determinant, and properties 4 to 10 follow.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 18, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 19: Determinant Formulas and Cofactors

The three defining properties of the determinant (Lecture 18) lead to explicit formulas: the big formula, a sum of $n!$ signed products, and the cofactor formula, which expresses an $n \times n$ determinant through determinants of size $n - 1$.

19.1 The 2×2 and 3×3 cases

Recall the properties: $\det I = 1$; exchanging rows reverses the sign; the determinant is linear in each row separately. For a 2×2 matrix, split the first row as $(a, b) = (a, 0) + (0, b)$, then split the second row in each piece:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}.$$

The first and last pieces have a column of zeros, so they vanish. The second is diagonal and equals ad. The third becomes diagonal after one row exchange, so it equals $-bc$. Hence $ad - bc$.

For a 3×3 matrix, splitting each of the three rows into three pieces gives $3^3 = 27$ determinants. A piece survives only if it has exactly one entry from each row *and each column*; if some column is unused, it is a column of zeros. The six survivors are permutation matrices with their 1s replaced by entries of A. Factoring the entries out of their rows by linearity leaves a permutation matrix, whose determinant is $+1$ or -1:

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}.$$

For instance $a_{11}a_{23}a_{32}$ uses columns $(1, 3, 2)$, one exchange away from $(1, 2, 3)$, so it gets a minus sign; $a_{12}a_{23}a_{31}$ uses columns $(2, 3, 1)$, two exchanges away, so it gets a plus sign.

For 3×3 only, the plus terms run down and to the right (cyclically) and the minus terms down and to the left. This rule fails for 4×4: the product along the counter-diagonal, columns $(4, 3, 2, 1)$, takes two exchanges (rows $1 \leftrightarrow 4$ and $2 \leftrightarrow 3$), so it comes with a plus sign.

19.2 The big formula

The surviving terms number 2 for $n = 2$, 6 for $n = 3$, and $24 = 4!$ for $n = 4$. The entry in row 1 can come from any of n columns; then the entry in row 2 from any of the $n - 1$ remaining columns, and so on.

Theorem 19.1 [Big formula]. *For an $n \times n$ matrix A,*

$$\boxed{\det A = \sum_{n!\ terms} \pm a_{1\alpha} a_{2\beta} a_{3\gamma} \cdots a_{n\omega}},$$

where the sum runs over all permutations $(\alpha, \beta, \gamma, \ldots, \omega)$ of $(1, 2, \ldots, n)$. The sign is the determinant of the corresponding permutation matrix: $+$ if the column order can be restored to $(1, 2, \ldots, n)$ by an even number of exchanges, $-$ if by an odd number. Half the terms have each sign ($n \geq 2$).

The argument is the 3×3 one: split every row completely by linearity, discard pieces with a repeated column, and evaluate each survivor by factoring out its entries. As a check, for $A = I$ every term contains a zero entry except the one for the identity permutation $\alpha = 1, \beta = 2, \ldots, \omega = n$, which gives +1. The other properties could be verified from the formula as well, but $\det(AB) = \det A \det B$ would be very hard to see this way.

Example: In a matrix with many zeros most terms vanish. Consider

$$A = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}.$$

A nonzero term picks a 1 from each row and each column. Starting with a_{14} forces a_{23}, then a_{32}, then a_{41}: the permutation $(4, 3, 2, 1)$, two exchanges, contributes +1. Starting with a_{13} forces a_{22}, a_{31}, a_{44}: the permutation $(3, 2, 1, 4)$, one exchange, contributes -1. These are the only nonzero terms, so $\det A = 0$. Indeed A is singular: rows 1 and 3 add to $(1, 1, 1, 1)$, and so do rows 2 and 4, so row 1 $-$ row 2 + row 3 $-$ row 4 $= \mathbf{0}$.

19.3 Cofactors

Group the terms of the big formula by the entry they use from the first row. For 3×3,

$$\det A = a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(-a_{21}a_{33} + a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31}).$$

The factor multiplying a_{11} is the determinant of the 2×2 matrix left after deleting row 1 and column 1: once row 1 and column 1 are used, the remaining terms choose one entry from each remaining row and column. The factor multiplying a_{12} is *minus* the determinant left after deleting row 1 and column 2.

Definition 19.2 [Cofactor]. Let M_{ij} be the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j. The cofactor of a_{ij} is

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

It is $+\det M_{ij}$ when $i+j$ is even and $-\det M_{ij}$ when $i+j$ is odd. (Without the sign, $\det M_{ij}$ is called a MINOR.)

The signs follow a checkerboard: along row 1 they are $+,-,+,-,\ldots$, and along row 2 they start with $-$.

$$\begin{bmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{bmatrix}$$

Theorem 19.3 [COFACTOR FORMULA]. *For any row i,*

$$\boxed{\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}}.$$

Since $\det A^T = \det A$, the same expansion works along any column.

Proof. Take $i = 1$. The terms of the big formula containing a_{1j} pair the remaining rows $2,\ldots,n$ with the remaining columns, one each, so apart from signs they are the terms of $\det M_{1j}$. To fix the sign, move column j to the front by $j-1$ exchanges of neighboring columns; this keeps the order of the other columns, puts a_{1j} in the corner, and leaves M_{1j} in the lower right block. There, the terms containing a_{1j} add up to $a_{1j}\det M_{1j}$, and the $j-1$ exchanges contribute $(-1)^{j-1} = (-1)^{1+j}$. For row i, first move row i to the top by $i-1$ exchanges of neighboring rows, which keeps the order of the other rows and contributes $(-1)^{i-1}$. □

For 2×2 the cofactor of a is d and the cofactor of b is $-c$, giving $ad + b(-c)$. Expanding repeatedly reduces any determinant to 1×1 determinants, so cofactors could have served as the definition. There are now three formulas: the product of the pivots (the efficient one, which builds the complicated terms into the pivots), the big formula with its $n!$ terms spread out, and the cofactor formula in between, which builds determinants from smaller ones.

19.4 Tridiagonal matrices of ones

A TRIDIAGONAL MATRIX has nonzero entries only on the main diagonal and the two diagonals adjacent to it. Let A_n be the $n\times n$ tridiagonal matrix of 1s, for example

$$A_4 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

Then $|A_1| = 1$, and $|A_2| = 0$ because A_2 has two equal rows. For A_3, the big formula has only three nonzero terms: the diagonal product 1 and the two terms $a_{12}a_{21}a_{33}$ and $a_{11}a_{23}a_{32}$, each with a minus sign, so $|A_3| = 1 - 1 - 1 = -1$.

Proposition 19.4 [Recurrence]. *For $n \geq 3$, $|A_n| = |A_{n-1}| - |A_{n-2}|$.*

Proof. Expand along row 1, which has only $a_{11} = a_{12} = 1$. Deleting row 1 and column 1 leaves A_{n-1}, with sign +. Deleting row 1 and column 2 leaves a matrix whose first column is $(1, 0, \ldots, 0)$, with sign −; expanding that matrix along its first column leaves A_{n-2} with sign +. For $n = 4$,

$$|A_4| = 1\begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} - 1\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = |A_3| - |A_2| = -1. \qquad \square$$

The recurrence gives

$$|A_1| = 1,\ |A_2| = 0,\ |A_3| = -1,\ |A_4| = -1,\ |A_5| = 0,\ |A_6| = 1,\ |A_7| = 1,\ \ldots$$

Since each value depends only on the previous two and $(|A_7|, |A_8|) = (|A_1|, |A_2|)$, the sequence repeats with period 6; for example, $|A_{61}| = |A_1| = 1$.

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Lecture 20: Cramer's Rule, Inverse Matrix and Volume

The determinant packs a matrix into one number, and that number gives explicit formulas for things computed so far only by algorithms: the inverse, the solution of $A\mathbf{x} = \mathbf{b}$, and the volume of a box.

20.1 A formula for A^{-1}

For 2×2 matrices,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

The entries on the right are cofactors: $d = C_{11}$, $a = C_{22}$, and $-b = C_{21}$ is the cofactor of c (delete row 2 and column 1, leaving b, with sign $(-1)^{2+1}$), while $-c = C_{12}$. They appear transposed. The same holds for every n.

Theorem 20.1 [Inverse by cofactors]. *Let C be the* cofactor matrix *of A, with entries C_{ij}. Then*

$$AC^T = (\det A)\, I, \qquad \textit{so} \qquad \boxed{A^{-1} = \frac{1}{\det A}\, C^T} \quad \textit{whenever } \det A \neq 0.$$

Notice the transpose: the cofactors of row 1 of A form column 1 of A^{-1}. The formula also explains the degrees involved: $\det A$ is a sum of products of n entries, while each cofactor is a sum of products of $n - 1$ entries. Gauss-Jordan elimination computes the same matrix without revealing this algebraic structure.

Proof. Entry (i, k) of AC^T is row i of A times column k of C^T, that is, row i of A against the cofactors of row k:

$$(AC^T)_{ik} = \sum_{j=1}^{n} a_{ij} C_{kj}.$$

For $k = i$ this is the cofactor expansion of $\det A$ along row i (Lecture 19), so every diagonal entry equals $\det A$.

For $k \neq i$, let A_s be A with row k replaced by a copy of row i. The cofactors C_{kj} do not involve row k, so they are the same for A and A_s, and the cofactor expansion of $\det A_s$ along row k is exactly $\sum_j a_{ij} C_{kj}$. Since A_s has two equal rows, this sum is 0. For 2×2: row 1 against the cofactors of row 2 gives $a(-b) + b(a) = 0$, the determinant of $A_s = \left[\begin{smallmatrix} a & b \\ a & b \end{smallmatrix}\right]$. Hence $AC^T = (\det A)I$. □

The formula also shows how A^{-1} changes when an entry of A changes.

20.2 Cramer's rule

If A is invertible, the solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det A} C^T\mathbf{b}$. Each component of $C^T\mathbf{b}$ is a combination of cofactors weighted by the numbers b_i, hence itself a determinant.

Theorem 20.2 [Cramer's rule]. *If* $\det A \neq 0$*, the solution of* $A\mathbf{x} = \mathbf{b}$ *is*

$$x_j = \frac{\det B_j}{\det A}, \qquad j = 1, \ldots, n,$$

where B_j *is* A *with column* j *replaced by* $\mathbf{b}$*:*

$$B_1 = \begin{bmatrix} \mathbf{b} & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}, \qquad \ldots, \qquad B_n = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_{n-1} & \mathbf{b} \end{bmatrix}.$$

Proof. Expand $\det B_j$ by cofactors down column j. Deleting row i and column j of B_j leaves the same matrix as deleting them from A, so the cofactor of the entry b_i is C_{ij}. Thus $\det B_j = b_1 C_{1j} + \cdots + b_n C_{nj}$, which is component j of $C^T\mathbf{b}$. Dividing by $\det A$ gives x_j. □

Cramer's rule needs $n + 1$ determinants, which is hopelessly slow compared with elimination. Its value, like that of the cofactor inverse, is that it is a formula: it expresses the answer algebraically in terms of the entries.

20.3 $|\det A|$ is a volume

Theorem 20.3 [Determinant and volume]. *For an* $n \times n$ *matrix* A*,* $|\det A|$ *is the volume of the box (parallelepiped) whose edges from the origin are the rows of* A*. Since* $\det A^T = \det A$*, the box built on the columns has the same volume.*

The sign of $\det A$ records whether the edges form a right-handed or a left-handed system.

Proof. It suffices to check that volume satisfies the three defining properties of the determinant (Lecture 18), up to sign.

1. If $A = I$, the box is the unit cube, with volume $1 = \det I$. More generally, if $A = Q$ is orthogonal, the box is a rotated unit cube with volume 1; and indeed $Q^TQ = I$ gives $(\det Q)^2 = \det Q^T \det Q = 1$, so $|\det Q| = 1$.
2. Exchanging two rows does not change the box, and changes $\det A$ only by a sign.

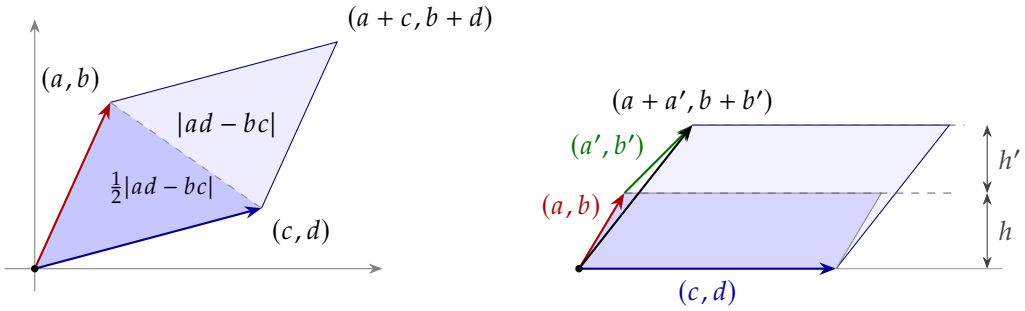
3(a). Multiplying one row by t stretches the box by $|t|$ in that edge, multiplying the volume by $|t|$.

3(b). Linearity in one row: in two dimensions,

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}.$$

The area of a parallelogram is its base times its height. Take the edge (c, d) as base; the height belonging to the other edge (a, b) is its component perpendicular to (c, d), which is linear in (a, b). So the heights for (a, b) and (a', b') add to the height for $(a + a', b + b')$ (Figure 14(b)), and the areas add. The same argument works for a box in n dimensions: volume is the base spanned by the other $n - 1$ edges times the height of the remaining edge.

A function with these properties agrees with $|\det A|$, which proves the claim. (Property 4 is also visible: if two edges are equal, the box is flat and has volume 0.) □



(a) area from coordinates

(b) heights add

Figure 14: (a) The parallelogram with edges (a, b) and (c, d) has area $|ad - bc|$; the shaded triangle is half of it. (b) Linearity in the first row, with the base (c, d) drawn horizontally: the heights h and h' of (a, b) and (a', b') add to the height of $(a + a', b + b')$. Schematic coordinates.

The practical point is that areas and volumes come straight from the coordinates of the corners, with no lengths, angles, or square roots. The parallelogram with edges (a, b) and (c, d) has area $|ad - bc|$ (base times height would involve $\sqrt{a^2 + b^2}$ and a messy height), and the triangle with vertices $(0, 0)$, (a, b), (c, d) is half of it. A triangle with arbitrary vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) has area

$$\frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right|.$$

Subtracting row 1 from rows 2 and 3 reduces this to the 2×2 determinant of the edges $(x_2 - x_1, y_2 - y_1)$ and $(x_3 - x_1, y_3 - y_1)$: the previous formula after moving vertex 1 to the origin.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 20, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 21: Eigenvalues and Eigenvectors

From here on, matrices are square and we look for special numbers and special vectors attached to them. A matrix acts on vectors like a function: in goes $\mathbf{x}$, out comes $A\mathbf{x}$. Most vectors come out pointing in a new direction; the eigenvectors are the ones that do not.

21.1 Definition and first examples

Definition 21.1 [Eigenvalues and eigenvectors]. A nonzero vector $\mathbf{x}$ is an eigenvector of A if $A\mathbf{x}$ is parallel to $\mathbf{x}$:

$$A\mathbf{x} = \lambda\mathbf{x}.$$

The number λ is the corresponding eigenvalue. It may be positive, negative, zero, or even complex.

The eigenvectors with $\lambda = 0$ satisfy $A\mathbf{x} = \mathbf{0}$: they are the nonzero vectors of the nullspace. So A is singular exactly when $\lambda = 0$ is an eigenvalue.

Some matrices reveal their eigenvectors geometrically, before any computation.

- *Projection onto a plane in $\mathbb{R}^3$.* A vector $\mathbf{b}$ outside the plane is not an eigenvector: $P\mathbf{b}$ points in a different direction. Every $\mathbf{x}$ in the plane has $P\mathbf{x} = \mathbf{x}$, so it is an eigenvector with $\lambda = 1$. Every $\mathbf{x}$ perpendicular to the plane has $P\mathbf{x} = \mathbf{0}$, an eigenvector with $\lambda = 0$. So the eigenvalues of a projection are 1 and 0, and here the eigenvectors (two independent ones in the plane, one perpendicular) span $\mathbb{R}^3$. Not every matrix has this property.
- *Permutation.* $B = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ swaps the two components. The vector $(1, 1)$ is unchanged, so $\lambda_1 = 1$; the vector $(-1, 1)$ is reversed, so $\lambda_2 = -1$. These eigenvectors are perpendicular, which will turn out to follow from $B = B^T$.

21.2 The characteristic equation

Both λ and $\mathbf{x}$ are unknown, and they are multiplied together, so elimination cannot be applied directly. Rewrite the equation as

$$(A - \lambda I)\mathbf{x} = \mathbf{0}.$$

A nonzero solution $\mathbf{x}$ exists only if $A - \lambda I$ is singular. This eliminates $\mathbf{x}$ and leaves an equation for λ alone.

Theorem 21.2 [Characteristic equation]. *λ is an eigenvalue of A if and only if*

$$\boxed{\det(A - \lambda I) = 0}.$$

For an $n \times n$ matrix this is a polynomial equation of degree n in λ, so A has n eigenvalues, counted with repetition. For each eigenvalue λ, the eigenvectors are the nonzero vectors in the nullspace of $A - \lambda I$, found by elimination.

The eigenvalues may be repeated, and repeated eigenvalues are the source of all trouble (see the last subsection). Each eigenvalue comes with a whole line (or larger subspace) of eigenvectors; we choose a basis, for a line a single vector.

Example: Let $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$. Then

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2),$$

so $\lambda_1 = 4$ and $\lambda_2 = 2$. For $\lambda_1 = 4$, $A - 4I = \left[\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ is singular with nullspace spanned by $\mathbf{x}_1 = (1, 1)$. For $\lambda_2 = 2$, $A - 2I = \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right]$ has nullspace spanned by $\mathbf{x}_2 = (-1, 1)$. Again the matrix is symmetric, the eigenvalues are real, and the eigenvectors are perpendicular (Figure 15).

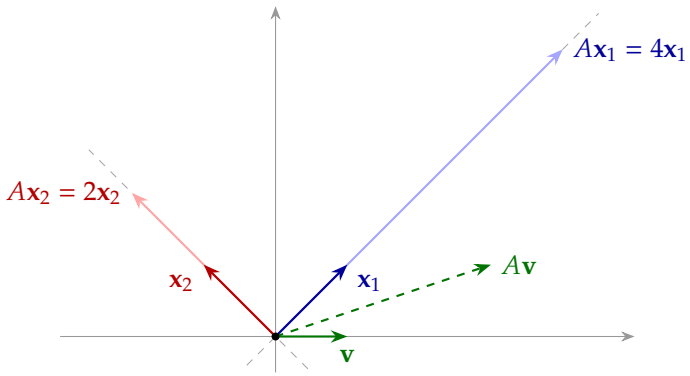


Figure 15: The matrix $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$ stretches its eigenvectors along their own lines, $\mathbf{x}_1$ by $\lambda_1 = 4$ and $\mathbf{x}_2$ by $\lambda_2 = 2$. A vector such as $\mathbf{v} = (1, 0)$ is turned to $A\mathbf{v} = (3, 1)$.

21.3 Trace and determinant

In the example, the coefficient 6 is the trace $3 + 3$ and the constant 8 is $\det A$. For every 2×2 matrix,

$$\det(A - \lambda I) = \lambda^2 - (\text{trace } A)\,\lambda + \det A,$$

and comparing with $(\lambda - \lambda_1)(\lambda - \lambda_2)$ gives $\lambda_1 + \lambda_2 = \text{trace } A$ and $\lambda_1 \lambda_2 = \det A$. The same two facts hold for all n.

Theorem 21.3 [Trace and determinant]. *For an $n \times n$ matrix with eigenvalues $\lambda_1, \ldots, \lambda_n$,*

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = a_{11} + a_{22} + \cdots + a_{nn} = \operatorname{trace} A, \qquad \lambda_1 \lambda_2 \cdots \lambda_n = \det A.$$

For a 2×2 matrix, one eigenvalue and the trace give the other: for the permutation B above, $\lambda_1 = 1$ and trace 0 force $\lambda_2 = -1$.

21.4 Shifts, sums, and products

The matrix $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$ is $B + 3I$, and its eigenvalues $4, 2$ are those of B plus 3, with the same eigenvectors. This holds for every shift: if $B\mathbf{x} = \lambda\mathbf{x}$, then

$$(B + cI)\mathbf{x} = \lambda\mathbf{x} + c\mathbf{x} = (\lambda + c)\mathbf{x}.$$

Note. For general sums this reasoning fails. If $A\mathbf{x} = \lambda\mathbf{x}$ and B has eigenvalue α, adding $B\mathbf{x} = \alpha\mathbf{x}$ would give $(A + B)\mathbf{x} = (\lambda + \alpha)\mathbf{x}$, but nothing says B has the same eigenvector $\mathbf{x}$; usually its eigenvector is some other $\mathbf{y}$. So the eigenvalues of $A + B$ are generally not sums of eigenvalues of A and B, and the eigenvalues of AB are generally not products. This works when A and B share their eigenvectors, as when $B = cI$.

21.5 Complex eigenvalues

The rotation by 90°, $Q = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$, turns every nonzero vector, so no real vector can be an eigenvector. The trace says $\lambda_1 + \lambda_2 = 0$ and the determinant says $\lambda_1 \lambda_2 = 1$, which no pair of real numbers satisfies. Indeed

$$\det(Q - \lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1, \qquad \lambda_1 = i, \quad \lambda_2 = -i.$$

A real matrix can have complex eigenvalues, and when it does they come in conjugate pairs: if $a + ib$ is an eigenvalue, so is $a - ib$. Symmetric matrices have only real eigenvalues. At the opposite extreme are antisymmetric matrices, $A^T = -A$, like Q: their eigenvalues are purely imaginary, $\lambda = ib$.

21.6 Triangular matrices and repeated eigenvalues

For a triangular matrix the eigenvalues are the diagonal entries, because $\det(A - \lambda I)$ is the product of the diagonal entries of the triangular matrix $A - \lambda I$. For

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}, \qquad \det(A - \lambda I) = \begin{vmatrix} 3 - \lambda & 1 \\ 0 & 3 - \lambda \end{vmatrix} = (3 - \lambda)^2,$$

so $\lambda_1 = \lambda_2 = 3$. The eigenvectors solve

$$(A - 3I)\mathbf{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{x} = \mathbf{0},$$

whose nullspace is the line through $\mathbf{x}_1 = (1, 0)$. There is no second independent eigenvector. This 2×2 matrix has only one line of eigenvectors: the repeated eigenvalue has produced a shortage of eigenvectors.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 21, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 22: Diagonalization and Powers of A

Once the eigenvalues and eigenvectors are known, they diagonalize the matrix. The factorization $A = S\Lambda S^{-1}$ joins $A = LU$ and $A = QR$, and it is the one that makes powers of A, and so difference equations, easy.

22.1 Diagonalizing a matrix

Suppose A has n linearly independent eigenvectors $\mathbf{x}_1, \ldots, \mathbf{x}_n$. Put them in the columns of the eigenvector matrix S. Multiplying a column at a time,

$$AS = A\begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\lambda_1\mathbf{x}_1 & \cdots & \lambda_n\mathbf{x}_n\end{bmatrix} = \begin{bmatrix}\mathbf{x}_1 & \cdots & \mathbf{x}_n\end{bmatrix}\begin{bmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{bmatrix} = S\Lambda.$$

The last step uses the rule that multiplying S on the right by a diagonal matrix scales its columns. Here Λ is the diagonal eigenvalue matrix.

Theorem 22.1 [Diagonalization]. *If A has n independent eigenvectors, then S is invertible and*

$$\boxed{S^{-1}AS = \Lambda}, \qquad \textit{equivalently} \qquad \boxed{A = S\Lambda S^{-1}}.$$

To keep the order straight: A multiplies its eigenvectors, so AS is the product that appears, and S^{-1} then makes it diagonal. The independence assumption is exactly what makes S^{-1} exist.

22.2 Powers of A

If $A\mathbf{x} = \lambda\mathbf{x}$, then $A^2\mathbf{x} = \lambda A\mathbf{x} = \lambda^2\mathbf{x}$: the eigenvalues of A^2 are the squares λ^2, with the same eigenvectors. In matrix form the inner $S^{-1}S$ cancels:

$$A^2 = S\Lambda S^{-1}S\Lambda S^{-1} = S\Lambda^2 S^{-1}, \qquad A^k = S\Lambda^k S^{-1}.$$

So A^k has eigenvalues λ_i^k and the same eigenvectors. Powers of LU show nothing of this kind; the eigenvalues carry the information about powers that the pivots do not.

Theorem 22.2 [Stable powers]. *Suppose A has n independent eigenvectors. Then $A^k \to 0$ as $k \to \infty$ if and only if $|\lambda_i| < 1$ for every eigenvalue.*

Proof. $A^k = S\Lambda^k S^{-1}$ with S fixed, so $A^k \to 0$ exactly when $\Lambda^k = S^{-1}A^k S \to 0$. The diagonal entries of Λ^k are λ_i^k, which tend to 0 exactly when $|\lambda_i| < 1$. The absolute value matters because eigenvalues may be negative or complex. □

22.3 Which matrices are diagonalizable?

Theorem 22.3 [Distinct eigenvalues**].** *If the n eigenvalues of A are all different, then the eigenvectors are independent and A is diagonalizable.*

A random matrix has distinct eigenvalues, so most matrices are diagonalizable. With a repeated eigenvalue one has to look more closely; there may or may not be enough eigenvectors.

- The $n \times n$ identity has $\lambda = 1$ repeated n times, yet every vector is an eigenvector. Any n independent vectors form S, and $S^{-1}IS = I = \Lambda$: a diagonal matrix is already its own Λ.
- $A = \left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ has $\det(A - \lambda I) = (2-\lambda)^2$, so $\lambda = 2, 2$. But $A - 2I = \left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$ has a one-dimensional nullspace, spanned by $(1, 0)$. There is no second independent eigenvector, and A cannot be diagonalized.

The number of times λ is a root of $\det(A - \lambda I)$ is its algebraic multiplicity; the dimension of the nullspace of $A - \lambda I$, the number of independent eigenvectors, is its geometric multiplicity. In the last example the algebraic multiplicity is 2 and the geometric multiplicity is 1. When the eigenvalues are distinct, both multiplicities are 1 for every eigenvalue. Everything that follows assumes n independent eigenvectors.

22.4 Difference equations $\mathbf{u}_{k+1} = A\mathbf{u}_k$

Start from a given $\mathbf{u}_0$ and multiply by A at every step. This first-order difference equation (a system, since $\mathbf{u}_k$ is a vector) has the compact solution $\mathbf{u}_k = A^k\mathbf{u}_0$. To actually compute it, expand $\mathbf{u}_0$ in eigenvectors:

$$\mathbf{u}_0 = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_n\mathbf{x}_n = S\mathbf{c}.$$

Each piece is a pure eigenvector and goes its own way, picking up a factor λ_i at each step:

$$A\mathbf{u}_0 = c_1\lambda_1\mathbf{x}_1 + \cdots + c_n\lambda_n\mathbf{x}_n, \qquad \boxed{\mathbf{u}_k = A^k\mathbf{u}_0 = c_1\lambda_1^k\mathbf{x}_1 + \cdots + c_n\lambda_n^k\mathbf{x}_n = S\Lambda^k\mathbf{c}}.$$

This is $A^k\mathbf{u}_0 = S\Lambda^k S^{-1}\mathbf{u}_0$ with $\mathbf{c} = S^{-1}\mathbf{u}_0$. So the recipe is: find the eigenvalues and eigenvectors, solve $S\mathbf{c} = \mathbf{u}_0$ for the coefficients, and attach λ_i^k to each term. The eigenvalues alone already say whether $\mathbf{u}_k$ grows, decays, or settles down.

22.5 Fibonacci numbers

The Fibonacci numbers $0, 1, 1, 2, 3, 5, 8, 13, \ldots$ satisfy $F_{k+2} = F_{k+1} + F_k$ with $F_0 = 0$, $F_1 = 1$. How fast do they grow, and what is F_{100}? The recurrence is a single second-order equation. To make it a first-order system, use the unknown $\mathbf{u}_k = (F_{k+1}, F_k)$ and add the trivial equation $F_{k+1} = F_{k+1}$:

$$\begin{aligned} F_{k+2} &= F_{k+1} + F_k \\ F_{k+1} &= F_{k+1} \end{aligned} \qquad \Longleftrightarrow \qquad \mathbf{u}_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{u}_k.$$

This A is symmetric, so its eigenvalues are real (and its eigenvectors perpendicular). They add to the trace 1 and multiply to the determinant -1:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1 = 0, \qquad \lambda_1 = \frac{1+\sqrt{5}}{2} \approx 1.618, \quad \lambda_2 = \frac{1-\sqrt{5}}{2} \approx -0.618.$$

The equation $\lambda^2 = \lambda + 1$ mirrors the recurrence itself. The eigenvalues are distinct, so A is diagonalizable.

The growth is controlled by λ_1, the only eigenvalue larger than 1 in absolute value. In $\mathbf{u}_k = c_1\lambda_1^k\mathbf{x}_1 + c_2\lambda_2^k\mathbf{x}_2$ the second term dies out since $|\lambda_2| < 1$, so

$$F_{100} \approx c_1\left(\frac{1+\sqrt{5}}{2}\right)^{100},$$

and each Fibonacci number is about 1.618 times the previous one.

For the exact formula: since $\lambda^2 - \lambda - 1 = 0$, the vector $(\lambda, 1)$ is in the nullspace of $A - \lambda I = \left[\begin{smallmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{smallmatrix}\right]$ (the first row gives $\lambda - \lambda^2 + 1 = 0$, the second $\lambda - \lambda = 0$). So $\mathbf{x}_1 = (\lambda_1, 1)$ and $\mathbf{x}_2 = (\lambda_2, 1)$. The initial vector is $\mathbf{u}_0 = (F_1, F_0) = (1, 0) = c_1\mathbf{x}_1 + c_2\mathbf{x}_2$, so $c_1 + c_2 = 0$ and $c_1(\lambda_1 - \lambda_2) = 1$, giving $c_1 = -c_2 = 1/\sqrt{5}$. Reading off the second component of $\mathbf{u}_k$,

$$\boxed{F_k = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^k - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^k}.$$

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 22, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 23: Differential Equations and exp(At)

A system of first-order linear differential equations with constant coefficients, $d\mathbf{u}/dt = A\mathbf{u}$, is solved by exponentials. The linear algebra is in finding the exponents and the vectors that multiply them: they are the eigenvalues and eigenvectors of A. Everything runs parallel to the powers A^k of Lecture 22, with $e^{\lambda t}$ in place of λ^k.

23.1 An example

Consider

$$\frac{du_1}{dt} = -u_1 + 2u_2, \qquad \frac{du_2}{dt} = u_1 - 2u_2, \qquad \text{that is} \qquad \frac{d\mathbf{u}}{dt} = A\mathbf{u}, \quad A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}, \quad \mathbf{u}(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Everything starts in u_1; since $du_2/dt > 0$ at the start, it flows from u_1 into u_2.

Eigenvalues. A is singular (column 2 is -2 times column 1), so $\lambda_1 = 0$; the trace is -3, so $\lambda_2 = -3$. Indeed $\det(A - \lambda I) = \lambda^2 + 3\lambda = \lambda(\lambda + 3)$.

Eigenvectors. For $\lambda_1 = 0$, $\mathbf{x}_1 = (2, 1)$ spans the nullspace of A. For $\lambda_2 = -3$, $A + 3I = \left[\begin{smallmatrix} 2 & 2 \\ 1 & 1 \end{smallmatrix}\right]$ gives $\mathbf{x}_2 = (1, -1)$, and $A\mathbf{x}_2 = -3\mathbf{x}_2$.

Pure solutions. Each eigenpair gives a solution $e^{\lambda t}\mathbf{x}$: if $\mathbf{u} = e^{\lambda_1 t}\mathbf{x}_1$ then

$$\frac{d\mathbf{u}}{dt} = \lambda_1 e^{\lambda_1 t}\mathbf{x}_1 = e^{\lambda_1 t} A\mathbf{x}_1 = A\mathbf{u}.$$

These are the analogues of the pure powers $\lambda^k\mathbf{x}$ for $\mathbf{u}_{k+1} = A\mathbf{u}_k$. The general solution combines them:

$$\mathbf{u}(t) = c_1 e^{\lambda_1 t}\mathbf{x}_1 + c_2 e^{\lambda_2 t}\mathbf{x}_2 = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Initial condition. At $t = 0$,

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \text{i.e.} \quad S\mathbf{c} = \mathbf{u}(0), \quad S = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix},$$

the eigenvector matrix again. So $c_1 = c_2 = \frac{1}{3}$ and

$$\mathbf{u}(t) = \frac{1}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{1}{3}e^{-3t}\begin{bmatrix} 1 \\ -1 \end{bmatrix} \longrightarrow \mathbf{u}(\infty) = \begin{bmatrix} 2/3 \\ 1/3 \end{bmatrix}.$$

The e^{-3t} part decays and the $e^{0t} = 1$ part is a STEADY STATE: an eigenvalue 0 gives a constant solution (Figure 16). Choosing $\mathbf{c}$ at the start fixes how much of each pure exponential is present forever after.

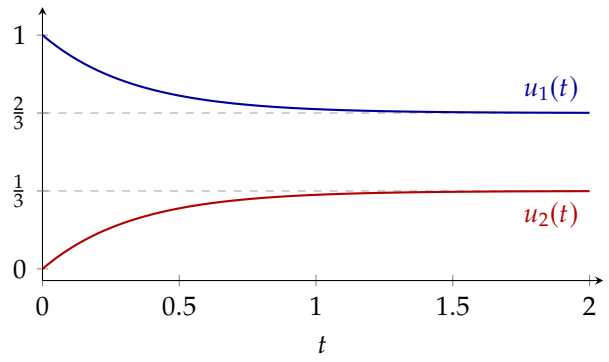


Figure 16: The solution $u_1(t) = \frac{2}{3} + \frac{1}{3}e^{-3t}$, $u_2(t) = \frac{1}{3} - \frac{1}{3}e^{-3t}$ moves from $(1, 0)$ to the steady state $(\frac{2}{3}, \frac{1}{3})$; the total $u_1 + u_2 = 1$ is conserved.

23.2 Stability

Write $\lambda = a + ib$. Then $|e^{\lambda t}| = e^{at}$, since $e^{ibt} = \cos bt + i \sin bt$ has absolute value 1: the imaginary part only makes the solution oscillate, and the real part decides its size. For A with n independent eigenvectors, the eigenvalues give three cases.

1. *Stability*, $\mathbf{u}(t) \to \mathbf{0}$ for every start: $\operatorname{Re}\lambda < 0$ for all eigenvalues.
2. *Steady state*: one eigenvalue is 0 and all others have $\operatorname{Re}\lambda < 0$.
3. *Blow-up* (for starts with a component along the offending eigenvector): $\operatorname{Re}\lambda > 0$ for some eigenvalue.

Replacing A by $-A$ flips every eigenvalue: the example above would have e^{3t} and blow up.

Proposition 23.1 [TWO BY TWO STABILITY]. *A real 2×2 matrix A has both eigenvalues with negative real part if and only if* $\operatorname{trace} A < 0$ *and* $\det A > 0$.

Proof. If both real parts are negative, then $\operatorname{trace} A = \lambda_1 + \lambda_2 < 0$ (for a complex pair the imaginary parts cancel). The product $\det A = \lambda_1\lambda_2$ is positive: two negative numbers, or $(a + ib)(a - ib) = a^2 + b^2$. Conversely, if the eigenvalues are real, $\det A > 0$ makes them of the same sign and $\operatorname{trace} A < 0$ makes that sign negative; if they are a complex pair $a \pm ib$, then $\operatorname{trace} A = 2a < 0$. □

A negative trace alone is not enough: $\left[\begin{smallmatrix} -2 & 0 \\ 0 & 1 \end{smallmatrix}\right]$ has trace -1 but eigenvalue 1, and e^t blows up. Its determinant -2 fails the test.

23.3 Uncoupling with S

The matrix A couples the unknowns; the eigenvectors uncouple them. Set $\mathbf{u} = S\mathbf{v}$. Then

$$S\frac{d\mathbf{v}}{dt} = AS\mathbf{v} \quad \Longrightarrow \quad \frac{d\mathbf{v}}{dt} = S^{-1}AS\,\mathbf{v} = \Lambda\mathbf{v},$$

which is n separate equations $dv_i/dt = \lambda_i v_i$ with solutions $v_i(t) = e^{\lambda_i t}v_i(0)$. In matrix form,

$$\mathbf{v}(t) = e^{\Lambda t}\mathbf{v}(0), \qquad \boxed{\mathbf{u}(t) = Se^{\Lambda t}S^{-1}\mathbf{u}(0) = e^{At}\mathbf{u}(0)}.$$

It remains to say what the exponential of a matrix is, and why $e^{At} = Se^{\Lambda t}S^{-1}$.

23.4 The matrix exponential

Definition 23.2 [Matrix exponential]. Following the power series $e^x = \sum_{n\geq 0} x^n/n!$,

$$e^{At} = I + At + \frac{(At)^2}{2} + \frac{(At)^3}{6} + \cdots = \sum_{n=0}^{\infty}\frac{(At)^n}{n!}.$$

The factorials make this series converge for every A and every t. The other basic series, the geometric series $1/(1-x) = \sum x^n$, also extends to matrices,

$$(I - At)^{-1} = I + At + (At)^2 + (At)^3 + \cdots,$$

but converges only when the eigenvalues of At are smaller than 1 in absolute value. For small t, $(I - At)^{-1} \approx I + At$.

Theorem 23.3 [Exponential by diagonalization]. *If $A = S\Lambda S^{-1}$ has n independent eigenvectors, then*

$$e^{At} = Se^{\Lambda t}S^{-1}, \qquad e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t}\end{bmatrix}.$$

Proof. Use $(At)^n = S\Lambda^n S^{-1}t^n$ in each term and write $I = SS^{-1}$:

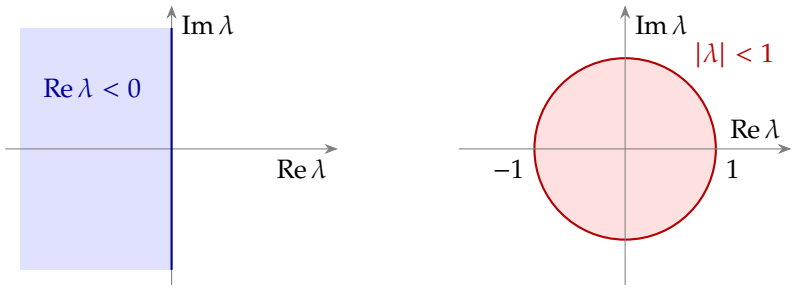
$$e^{At} = SS^{-1} + S\Lambda S^{-1}t + \frac{S\Lambda^2 S^{-1}}{2}t^2 + \cdots = S\Big(I + \Lambda t + \frac{(\Lambda t)^2}{2} + \cdots\Big)S^{-1} = Se^{\Lambda t}S^{-1}.$$

For diagonal Λ every term of the series is diagonal, and the diagonal entries are the scalar series for $e^{\lambda_i t}$. □

The definition of e^{At} needs no assumption; the formula $Se^{\Lambda t}S^{-1}$ needs n independent eigenvectors.

(A matrix without them can still be made triangular, though not diagonal.)

Since S and S^{-1} do not move, $e^{At} \to 0$ exactly when every $e^{\lambda_i t} \to 0$, that is, when every $\operatorname{Re} \lambda_i < 0$: the eigenvalues lie in the left half-plane. For powers, $A^k \to 0$ when every $|\lambda_i| < 1$: the eigenvalues lie inside the unit circle (Figure 17).



(a) $e^{At} \to 0$ (b) $A^k \to 0$

Figure 17: Stability regions in the complex plane: eigenvalues with $\operatorname{Re} \lambda < 0$ make $e^{At} \to 0$; eigenvalues with $|\lambda| < 1$ make $A^k \to 0$.

23.5 Second-order equations

As with Fibonacci, a second-order equation becomes a first-order system. For $y'' + by' + ky = 0$, take $\mathbf{u} = (y', y)$ and add the trivial equation $y' = y'$:

$$\mathbf{u}' = \begin{bmatrix} y'' \\ y' \end{bmatrix} = \begin{bmatrix} -b & -k \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y' \\ y \end{bmatrix}.$$

The first row is the differential equation and the second row is trivial. The eigenvalues come straight from the equation: $\det(A - \lambda I) = \lambda^2 + b\lambda + k$. In general a kth-order equation gives a $k \times k$ matrix with the coefficients of the equation (with signs reversed) in the first row, 1s just below the diagonal, and zeros elsewhere.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 23, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 24: Markov Matrices; Fourier Series

Two applications: Markov matrices, where an eigenvalue $\lambda = 1$ gives the steady state of $\mathbf{u}_{k+1} = A\mathbf{u}_k$, and Fourier series, which are expansions in an orthogonal basis of a space of functions.

24.1 Eigenvalues of A^T

Proposition 24.1 [Same eigenvalues]. *A and A^T have the same eigenvalues.*

Proof. $(A - \lambda I)^T = A^T - \lambda I$, and a matrix and its transpose have the same determinant (property 10, Lecture 18). So $\det(A - \lambda I) = \det(A^T - \lambda I)$, and the two characteristic equations have the same roots. □

The eigenvectors of A and A^T are usually different: the nullspace of $A - \lambda I$ is not the nullspace of its transpose.

24.2 Markov matrices

Definition 24.2 [Markov matrix]. A square matrix is a Markov matrix if all its entries are nonnegative and each column adds to 1. For example,

$$A = \begin{bmatrix} .1 & .01 & .3 \\ .2 & .99 & .3 \\ .7 & 0 & .4 \end{bmatrix}.$$

Both properties come from probability: entries are probabilities, and each column accounts for everything. Both survive multiplication, so every power A^k is again a Markov matrix. For differential equations a steady state came from $\lambda = 0$; for powers it comes from $\lambda = 1$.

Theorem 24.3 [Eigenvalues of a Markov matrix]. *Let A be a Markov matrix. Then*

1. *$\lambda = 1$ is an eigenvalue of A;*
2. *every other eigenvalue has $|\lambda| \leq 1$, and typically $|\lambda| < 1$;*
3. *the eigenvector $\mathbf{x}_1$ for $\lambda = 1$ can be chosen with no negative components.*

Proof of (1). Each column of $A - I$ adds to 0, since 1 was subtracted from one entry of each column. So the sum of the rows of $A - I$ is the zero row: the rows are dependent and $A - I$ is singular. In other words $(1, 1, \ldots, 1)$ is in the nullspace of $(A - I)^T$, so 1 is an eigenvalue of A^T, and by Proposition 24.1 also of A. Its eigenvector $\mathbf{x}_1$ for A is in the nullspace of $A - I$. □

Parts (2) and (3) are stated without proof. For the 3×3 example,

$$A - I = \begin{bmatrix} -.9 & .01 & .3 \\ .2 & -.01 & .3 \\ .7 & 0 & -.6 \end{bmatrix}, \qquad \mathbf{x}_1 = \begin{bmatrix} .6 \\ 33 \\ .7 \end{bmatrix}:$$

the last row forces $x_1 : x_3 = .6 : .7$, and then the first row requires $-.54 + .01x_2 + .21 = 0$, so $x_2 = 33$. The components are indeed positive.

If A has n independent eigenvectors, then $\mathbf{u}_k = A^k\mathbf{u}_0 = c_1\lambda_1^k\mathbf{x}_1 + \cdots + c_n\lambda_n^k\mathbf{x}_n$ (Lecture 22). With $\lambda_1 = 1$ and $|\lambda_i| < 1$ for the others, every term but the first dies out:

$$\mathbf{u}_k \to c_1\mathbf{x}_1,$$

the steady state, which is the $\mathbf{x}_1$ component of $\mathbf{u}_0$.

24.3 Example: populations of two states

Let $\mathbf{u}_k = (u_{\text{Cal}}, u_{\text{Mass}})$ be the populations of California and Massachusetts in year k. Each year 90% of the people in California stay and 10% move to Massachusetts, while 80% of the people in Massachusetts stay and 20% move to California (Figure 18):

$$\begin{bmatrix} u_{\text{Cal}} \\ u_{\text{Mass}} \end{bmatrix}_{k+1} = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix} \begin{bmatrix} u_{\text{Cal}} \\ u_{\text{Mass}} \end{bmatrix}_k.$$

Column 1 accounts for everyone in California at time k, and column 2 for everyone in Massachusetts, so the total population is conserved. The model assumes the same probabilities act every year.

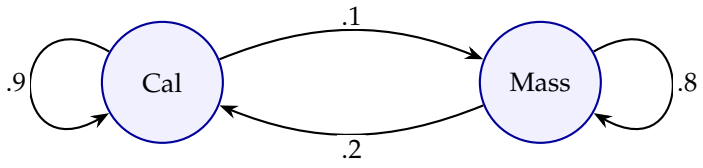


Figure 18: The two-state Markov chain: the entry in row i, column j of A is the fraction moving from state j to state i in one step.

Start with $\mathbf{u}_0 = (0, 1000)$. After one step $\mathbf{u}_1 = A\mathbf{u}_0 = (200, 800)$; then California keeps growing and Massachusetts shrinking, always with total 1000.

Eigenvalues. $\lambda_1 = 1$, and the trace 1.7 gives $\lambda_2 = .7$ (consistent with $\det A = .72 - .02 = .7$).

Eigenvectors. $A - I = \begin{bmatrix} -.1 & .2 \\ .1 & -.2 \end{bmatrix}$ gives $\mathbf{x}_1 = (2, 1)$, which is positive; $A - .7I = \begin{bmatrix} .2 & .2 \\ .1 & .1 \end{bmatrix}$ gives $\mathbf{x}_2 = (-1, 1)$.

Solution. $\mathbf{u}_k = c_1 1^k (2, 1) + c_2 (.7)^k (-1, 1)$, and $k = 0$ gives $(0, 1000) = c_1(2, 1) + c_2(-1, 1)$, so $c_1 = \frac{1000}{3}$ and $c_2 = \frac{2000}{3}$:

$$\mathbf{u}_k = \frac{1000}{3}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \frac{2000}{3}(.7)^k \begin{bmatrix} -1 \\ 1 \end{bmatrix} \longrightarrow \mathbf{u}_\infty = \begin{bmatrix} 2000/3 \\ 1000/3 \end{bmatrix}.$$

In the limit $\frac{2}{3}$ of the people are in California and $\frac{1}{3}$ in Massachusetts. The steady state is $\mathbf{x}_1$ scaled to the total population, so it is the same from any starting vector with that total; the $(.7)^k$ term describes the approach to it.

Note. Some applications use row vectors multiplied from the left. Then the matrix is the transpose of ours and its *rows* add to 1; all the calculations transpose accordingly.

24.4 Expansion in an orthonormal basis

Let $\mathbf{q}_1, \ldots, \mathbf{q}_n$ be an orthonormal basis of $\mathbb{R}^n$. Every $\mathbf{v}$ is a combination

$$\mathbf{v} = x_1\mathbf{q}_1 + x_2\mathbf{q}_2 + \cdots + x_n\mathbf{q}_n.$$

Taking the inner product with $\mathbf{q}_i$ kills every term except one, since $\mathbf{q}_i^T\mathbf{q}_j = 0$ for $j \neq i$ and $\mathbf{q}_i^T\mathbf{q}_i = 1$:

$$\boxed{x_i = \mathbf{q}_i^T\mathbf{v}}.$$

In matrix form the expansion is $Q\mathbf{x} = \mathbf{v}$ with the $\mathbf{q}_i$ as the columns of Q, so $\mathbf{x} = Q^{-1}\mathbf{v} = Q^T\mathbf{v}$, whose ith component is again $\mathbf{q}_i^T\mathbf{v}$. No normal equations are needed; orthogonality does all the work, and it is the property Fourier series are built on.

24.5 Fourier series

A Fourier series expands a function in sines and cosines:

$$f(x) = a_0 + a_1\cos x + b_1\sin x + a_2\cos 2x + b_2\sin 2x + \cdots.$$

The setting is now an infinite-dimensional vector space whose vectors are functions, with basis $1, \cos x, \sin x, \cos 2x, \sin 2x, \ldots$. These functions have period 2π, so $f(x) = f(x + 2\pi)$ and everything lives on $[0, 2\pi]$.

The inner product of vectors, $\mathbf{v}^T\mathbf{w} = v_1w_1 + \cdots + v_nw_n$, multiplies components and adds. Functions

have a continuum of values $f(x)$, so the analogue multiplies values and integrates:

$$f^T g = \int_0^{2\pi} f(x)g(x)\,dx.$$

With this inner product the basis functions are orthogonal. For example

$$\int_0^{2\pi} \sin x \cos x\,dx = \tfrac{1}{2}\sin^2 x\Big|_0^{2\pi} = 0,$$

and the other pairs follow in the same way with trigonometric identities. The basis is orthogonal but not normalized: for instance $\int_0^{2\pi} \cos^2 x\,dx = \pi$.

Fourier coefficients. To find a_1, take the inner product of both sides with $\cos x$, that is, multiply by $\cos x$ and integrate. By orthogonality only the a_1 term survives:

$$\int_0^{2\pi} f(x)\cos x\,dx = a_1 \int_0^{2\pi} \cos^2 x\,dx = \pi a_1, \qquad \boxed{a_1 = \frac{1}{\pi}\int_0^{2\pi} f(x)\cos x\,dx}.$$

The same technique gives every coefficient. In particular, the inner product with the constant 1 (whose squared length is 2π) gives the average value of f,

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} f(x)\,dx.$$

This is exactly the expansion $x_i = \mathbf{q}_i^T\mathbf{v}$, divided by the squared length of each basis function.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 24, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 25: Symmetric Matrices and Positive Definiteness

Special matrices have special eigenvalues and eigenvectors: Markov matrices have $\lambda = 1$. Symmetric matrices, $A = A^T$, are the most important class: their eigenvalues are real and their eigenvectors can be chosen perpendicular. Positive definite matrices are the symmetric matrices with positive eigenvalues, and they tie together pivots, determinants, and eigenvalues.

25.1 The spectral theorem

Theorem 25.1 [Spectral theorem]. *Let A be a real symmetric matrix. Then*

1. *all eigenvalues of A are real;*
2. *there is a complete set of eigenvectors that can be chosen orthonormal.*

Consequently

$$\boxed{A = Q\Lambda Q^{-1} = Q\Lambda Q^T},$$

with Q orthogonal (orthonormal eigenvectors in its columns) and Λ the real diagonal eigenvalue matrix. Conversely, every matrix $Q\Lambda Q^T$ is symmetric, since its transpose is $(Q^T)^T\Lambda^T Q^T = Q\Lambda Q^T$.

"Can be chosen" matters when an eigenvalue is repeated. For the identity every vector is an eigenvector, and the statement is that perpendicular ones can be picked. In the typical case the eigenvalues are distinct, each has a line of eigenvectors, and these lines are automatically perpendicular. Normalizing the eigenvectors to unit length turns the eigenvector matrix S of $A = S\Lambda S^{-1}$ into an orthogonal Q, with $Q^{-1} = Q^T$. The name comes from the spectrum, the set of eigenvalues; in mechanics the same result is the principal axis theorem: along the right axes, directions do not couple.

25.2 Why the eigenvalues are real

Complex conjugation, written with a bar, changes $a + ib$ to $a - ib$, and it respects products. Conjugating $A\mathbf{x} = \lambda\mathbf{x}$ gives $\bar{A}\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$, and for real A this is

$$A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}.$$

So for any real matrix, if λ is an eigenvalue with eigenvector $\mathbf{x}$, then $\bar{\lambda}$ is an eigenvalue with eigenvector $\bar{\mathbf{x}}$: complex eigenvalues of real matrices come in conjugate pairs. Symmetry has not been used yet.

Proof of real eigenvalues. Transpose $A\bar{\mathbf{x}} = \bar{\lambda}\bar{\mathbf{x}}$ to get $\bar{\mathbf{x}}^T A^T = \bar{\mathbf{x}}^T \bar{\lambda}$. Now use $A^T = A$ and multiply on the right by $\mathbf{x}$:

$$\bar{\mathbf{x}}^T A\mathbf{x} = \bar{\lambda}\, \bar{\mathbf{x}}^T \mathbf{x}.$$

On the other hand, multiplying $A\mathbf{x} = \lambda\mathbf{x}$ on the left by $\bar{\mathbf{x}}^T$ gives

$$\bar{\mathbf{x}}^T A\mathbf{x} = \lambda\, \bar{\mathbf{x}}^T \mathbf{x}.$$

The left sides agree, so $\lambda\, \bar{\mathbf{x}}^T \mathbf{x} = \bar{\lambda}\, \bar{\mathbf{x}}^T \mathbf{x}$. Finally

$$\bar{\mathbf{x}}^T \mathbf{x} = \bar{x}_1 x_1 + \cdots + \bar{x}_n x_n = |x_1|^2 + \cdots + |x_n|^2 > 0$$

for $\mathbf{x} \neq \mathbf{0}$, because $(a - ib)(a + ib) = a^2 + b^2$. Cancelling it gives $\lambda = \bar{\lambda}$, so λ is real. □

The quantity $\bar{\mathbf{x}}^T \mathbf{x}$ is the right squared length of a complex vector: multiply each component by its conjugate. For complex matrices the proof goes through with A^T replaced by $\bar{A}^T$, so the "good" complex matrices, with real eigenvalues and perpendicular eigenvectors, are those with $\bar{A}^T = A$. For real matrices this is just symmetry.

25.3 A symmetric matrix as a sum of projections

Multiplying out $A = Q\Lambda Q^T$ as columns times rows,

$$A = \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^T + \cdots + \lambda_n \mathbf{q}_n \mathbf{q}_n^T.$$

Since $\mathbf{q}_k$ is a unit vector, $\mathbf{q}_k \mathbf{q}_k^T$ is the projection matrix onto the line through $\mathbf{q}_k$ (Lecture 15, with $\mathbf{q}_k^T \mathbf{q}_k = 1$): it is symmetric and squares to itself. So every symmetric matrix is a combination of projections onto mutually perpendicular lines, with the eigenvalues as coefficients.

25.4 Signs of eigenvalues from pivots

Once the eigenvalues are known to be real, the next question is their signs, which decide stability for differential equations. Computing the eigenvalues of a large matrix by finding the roots of $\det(A - \lambda I)$ is a very bad (unstable) method, and other numerical methods are used. The pivots, however, are easy and reliable to compute, and for symmetric matrices they reveal the signs.

Theorem 25.2 [Pivots and eigenvalues]. *For a symmetric matrix, the number of positive pivots equals the number of positive eigenvalues.*

The pivots are not the eigenvalues; only their products agree, since with no row exchanges both

equal $\det A$. Shifting helps locate the eigenvalues: $A - bI$ has eigenvalues $\lambda_i - b$, so the signs of its pivots tell how many eigenvalues of A lie above and below any number b.

25.5 Positive definite matrices

Definition 25.3 [Positive definite]. A symmetric matrix is positive definite if all its eigenvalues are positive.

By Theorem 25.2 this holds exactly when all the pivots are positive, which is the practical test.

Example: $A = \left[\begin{smallmatrix} 5 & 2 \\ 2 & 3 \end{smallmatrix}\right]$ is symmetric with $\det A = 11$. Its pivots are 5 and $11/5$ (the product of the pivots is the determinant), both positive, so A is positive definite. Indeed

$$\det(A - \lambda I) = \lambda^2 - 8\lambda + 11 = 0, \qquad \lambda = 4 \pm \sqrt{5} > 0,$$

real and positive, but much less simple than the pivots.

The determinant of a positive definite matrix is positive, being the product of positive eigenvalues. But $\det A > 0$ alone is not enough: $\left[\begin{smallmatrix} -1 & 0 \\ 0 & -3 \end{smallmatrix}\right]$ has determinant 3, yet its pivots and eigenvalues are -1 and -3. The test must look at all the upper-left determinants.

Theorem 25.4 [Determinant test]. *A symmetric $n \times n$ matrix A is positive definite if and only if all of its* leading subdeterminants, *the determinants of the upper-left $k \times k$ submatrices for $k = 1, \ldots, n$, are positive.*

For $\left[\begin{smallmatrix} 5 & 2 \\ 2 & 3 \end{smallmatrix}\right]$ these are 5 and 11, so the test passes; for $\left[\begin{smallmatrix} -1 & 0 \\ 0 & -3 \end{smallmatrix}\right]$ the first is -1, so it fails although the full determinant is positive. For symmetric matrices, then, the three subjects of the course meet. Positive eigenvalues, positive pivots, and positive leading subdeterminants go together: any one of them implies the others.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 25, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 26: Complex Matrices; Fast Fourier Transform (FFT)

Even a real matrix can have complex eigenvalues and eigenvectors, so lengths, inner products, symmetry, and orthogonality must be redefined for complex vectors. The most important complex matrix is the Fourier matrix F_n. Multiplying by a full $n \times n$ matrix normally takes n^2 multiplications; the fast Fourier transform (FFT) multiplies by F_n in about $\frac{1}{2} n \log_2 n$.

26.1 Complex vectors

A vector $\mathbf{z} = (z_1, \ldots, z_n)$ with complex entries lies in $\mathbb{C}^n$. The old formula for length fails: $\mathbf{z}^T\mathbf{z} = z_1^2 + \cdots + z_n^2$ need not be positive. For $\mathbf{z} = (1, i)$,

$$\mathbf{z}^T\mathbf{z} = \begin{bmatrix} 1 & i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = 1 + i^2 = 0,$$

but this nonzero vector should not have length 0. The fix is to conjugate the first factor, since $\overline{z}_k z_k = |z_k|^2$.

Definition 26.1 [Length and inner product in $\mathbb{C}^n$]. The conjugate transpose of $\mathbf{z}$ is $\mathbf{z}^H = \overline{\mathbf{z}}^T$ (read "$\mathbf{z}$ Hermitian"). The length of $\mathbf{z}$ is given by

$$\|\mathbf{z}\|^2 = \mathbf{z}^H\mathbf{z} = |z_1|^2 + |z_2|^2 + \cdots + |z_n|^2,$$

and the inner product of $\mathbf{y}$ and $\mathbf{x}$ is

$$\mathbf{y}^H\mathbf{x} = \overline{y}_1 x_1 + \overline{y}_2 x_2 + \cdots + \overline{y}_n x_n.$$

For $\mathbf{z} = (1, i)$ this gives $\mathbf{z}^H\mathbf{z} = \begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = 2$, so $\|\mathbf{z}\| = \sqrt{2}$. The inner product is usually a complex number, but the inner product of a vector with itself is its length squared. The rule is: whenever we transpose, we also conjugate.

26.2 Hermitian and unitary matrices

Definition 26.2 [Hermitian matrix]. A complex matrix A is Hermitian if $A^H = \overline{A}^T = A$. This replaces symmetry $A^T = A$.

Flipping across the diagonal and conjugating leaves the diagonal in place, so the diagonal entries of

a Hermitian matrix are real, and the entry opposite a_{jk} is $\overline{a_{jk}}$. For example,

$$A = \begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix} = A^H.$$

Proposition 26.3 [Hermitian matrices]. *A Hermitian matrix has real eigenvalues and perpendicular eigenvectors (for distinct eigenvalues).*

Proof. If $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{x}^H A\mathbf{x} = \lambda\, \mathbf{x}^H\mathbf{x}$. The number $\mathbf{x}^H A\mathbf{x}$ equals its own conjugate, since $(\mathbf{x}^H A\mathbf{x})^H = \mathbf{x}^H A^H\mathbf{x} = \mathbf{x}^H A\mathbf{x}$, so it is real; and $\mathbf{x}^H\mathbf{x} > 0$. Hence λ is real. If also $A\mathbf{y} = \mu\mathbf{y}$ with $\mu \neq \lambda$, then $\lambda\, \mathbf{y}^H\mathbf{x} = \mathbf{y}^H A\mathbf{x} = (A\mathbf{y})^H\mathbf{x} = \mu\, \mathbf{y}^H\mathbf{x}$, using μ real, so $\mathbf{y}^H\mathbf{x} = 0$. □

Complex vectors $\mathbf{q}_1, \ldots, \mathbf{q}_n$ are orthonormal when

$$\mathbf{q}_j^H \mathbf{q}_k = \begin{cases} 0, & j \neq k, \\ 1, & j = k. \end{cases}$$

Placed in the columns of Q, this says $Q^H Q = I$.

Definition 26.4 [Unitary matrix]. A square complex matrix with orthonormal columns, $Q^H Q = I$, is unitary. Unitary is to orthogonal as Hermitian is to symmetric, and $Q^{-1} = Q^H$.

26.3 The Fourier matrix

A Fourier series writes a periodic function as a combination of frequencies, $f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \cdots$. For finite data the decomposition is the discrete Fourier transform, which is multiplication by the Fourier matrix. Rows and columns are numbered from 0 to $n-1$.

Definition 26.5 [Fourier matrix]. Let $w = e^{2\pi i/n}$. The $n \times n$ Fourier matrix has entries $(F_n)_{jk} = w^{jk}$ for $j, k = 0, 1, \ldots, n-1$:

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & w & w^2 & \cdots & w^{n-1} \\ 1 & w^2 & w^4 & \cdots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \cdots & w^{(n-1)^2} \end{bmatrix}.$$

F_n is symmetric, $F_n = F_n^T$ (but not Hermitian), and no entry is zero. The number $w = \cos(2\pi/n) + i\sin(2\pi/n)$ lies on the unit circle at angle $2\pi/n$, and $w^n = 1$. Its powers are best computed in polar form: $w^{jk} = e^{2\pi ijk/n}$ is on the unit circle at jk times that angle. The powers $1, w, \ldots, w^{n-1}$ are the n roots of $z^n = 1$.

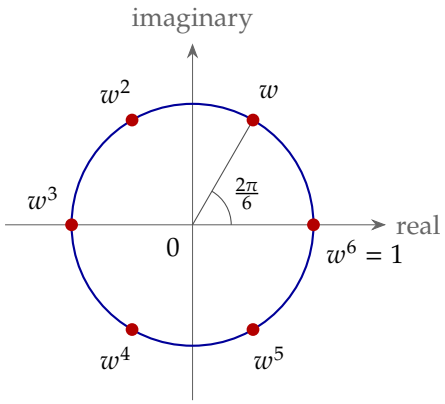


Figure 19: For $n = 6$, $w = e^{2\pi i/6}$ is one sixth of the way around the unit circle, at $60°$. Its powers $w^2, \dots, w^6 = 1$ are the six sixth roots of 1.

Example: For $n = 4$, $w = e^{2\pi i/4} = i$, with powers i, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. The exponent of entry (j, k) is jk:

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & i^2 & i^3 \\ 1 & i^2 & i^4 & i^6 \\ 1 & i^3 & i^6 & i^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}.$$

Without conjugation, columns 1 and 3 would seem to have inner product $1 + 1 + 1 + 1 = 4$. With it, $\overline{(1, i, -1, -i)} = (1, -i, -1, i)$ and

$$\begin{bmatrix} 1 & -i & -1 & i \end{bmatrix} \begin{bmatrix} 1 \\ -i \\ -1 \\ i \end{bmatrix} = 1 - 1 + 1 - 1 = 0.$$

Proposition 26.6 [Orthogonal columns]. *The columns of F_n are orthogonal, each of length $\sqrt{n}$. Hence $\frac{1}{\sqrt{n}} F_n$ is unitary and*

$$\overline{F}_n F_n = nI, \qquad F_n^{-1} = \tfrac{1}{n} \overline{F}_n.$$

Proof. Column k is $(1, w^k, w^{2k}, \dots, w^{(n-1)k})$, and $\overline{w} = w^{-1}$ on the unit circle. The inner product of columns j and k is

$$\sum_{m=0}^{n-1} \overline{w}^{mj} w^{mk} = \sum_{m=0}^{n-1} \left(w^{k-j}\right)^m.$$

For $j = k$ every term is 1 and the sum is n. For $j \neq k$ the ratio $r = w^{k-j} \neq 1$ satisfies $r^n = 1$, so the geometric sum is $(1 - r^n)/(1 - r) = 0$. Since F_n is symmetric, $F_n^H = \overline{F}_n$. □

For $n = 4$ the columns have length 2, so $\frac{1}{4} F_4^H F_4 = I$. Since the inverse is again a Fourier-type matrix, any fast method for F_n also inverts the transform quickly.

Example: A single impulse at time zero contains all frequencies equally:

$$F_4 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \qquad F_4 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 0 \\ 0 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Multiplying by F_4 again almost returns the impulse; dividing by $n = 4$ inverts the transform exactly.

26.4 The fast Fourier transform

The FFT rests on one fact about the roots: with $w_n = e^{2\pi i/n}$, squaring doubles the angle, so $w_{2n}^2 = w_n$. Thus F_{2n} (built from w_{2n}) is connected to F_n (built from w_n), half its size.

Theorem 26.7 [FFT FACTORIZATION**].**

$$F_{2n} = \begin{bmatrix} I & D \\ I & -D \end{bmatrix} \begin{bmatrix} F_n & 0 \\ 0 & F_n \end{bmatrix} P,$$

where I is $n \times n$, $D = \operatorname{diag}(1, w_{2n}, w_{2n}^2, \ldots, w_{2n}^{n-1})$, and P is the $2n \times 2n$ permutation that puts the even components $x_0, x_2, \ldots, x_{2n-2}$ first and the odd components $x_1, x_3, \ldots, x_{2n-1}$ after them.

Proof. Let $\mathbf{y} = F_{2n}\mathbf{x}$, and let $\mathbf{x}' = (x_0, x_2, \ldots)$ and $\mathbf{x}'' = (x_1, x_3, \ldots)$ be the even and odd parts. For $0 \le j < n$, split the sum over even and odd indices and use $w_{2n}^{2jk} = w_n^{jk}$:

$$y_j = \sum_{k=0}^{n-1} w_n^{jk} x_{2k} + w_{2n}^{j} \sum_{k=0}^{n-1} w_n^{jk} x_{2k+1} = (F_n \mathbf{x}')_j + w_{2n}^{j} (F_n \mathbf{x}'')_j.$$

For the index $j + n$, $w_n^{(j+n)k} = w_n^{jk}$ because $w_n^n = 1$, and $w_{2n}^{j+n} = -w_{2n}^{j}$ because $w_{2n}^n = -1$. So $y_{j+n} = (F_n \mathbf{x}')_j - w_{2n}^{j} (F_n \mathbf{x}'')_j$. These two formulas are the three factors read from right to left. □

Multiplying by F_{2n} directly costs $(2n)^2 = 4n^2$. The factorization costs two half-size transforms, $2n^2$, plus about n multiplications by D; the identity blocks and the even-odd permutation cost essentially nothing. Now apply the same idea to each F_n, and so on recursively. For $n = 64$, the even-odd permutation is followed by splitting the evens into even-evens $(x_0, x_4, x_8, \ldots)$ and even-odds $(x_2, x_6, \ldots)$, and similarly for the odds, so the four quarters are transformed by F_{16}.

Algorithm 26.1: FFT

Input: $\mathbf{x} = (x_0, \ldots, x_{N-1})$ with N a power of 2.
Output: $\mathbf{y} = F_N \mathbf{x}$.

```
if N = 1 then return x
y' ← FFT(x_0, x_2, ..., x_{N-2})                    // size N/2
y'' ← FFT(x_1, x_3, ..., x_{N-1})
for j ← 0 to N/2 − 1 do
    y_j ← y'_j + w_N^j y''_j
    y_{j+N/2} ← y'_j − w_N^j y''_j
return y
```

Each level of the recursion costs $N/2$ multiplications by powers of w, and there are $\log_2 N$ levels before the transforms have size 1. So the cost $M(N)$ satisfies $M(N) = 2M(N/2) + N/2$, $M(1) = 0$, and

$$\boxed{M(N) = \tfrac{1}{2} N \log_2 N} \qquad \text{instead of } N^2.$$

For $N = 64$ there are six fix-up steps of 32 each. For $N = 1024 = 2^{10}$, direct multiplication needs over a million multiplications, while the FFT needs $\frac{1}{2} \cdot 1024 \cdot 10 = 5 \cdot 1024$, about 200 times fewer. This speed rests on the special structure of F_n, a matrix with orthogonal columns.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 26, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 27: Positive Definite Matrices and Minima

Positive definite matrices bring the course together: pivots, determinants, eigenvalues, and stability all give tests for them. The new quantity is the quadratic form $\mathbf{x}^T A\mathbf{x}$. It explains what positive definiteness means (a minimum) and leads to the geometry of ellipses and ellipsoids. Throughout, A is real and symmetric.

27.1 Four tests

Definition 27.1 [Positive definite]. A symmetric matrix A is positive definite if $\mathbf{x}^T A\mathbf{x} > 0$ for every real vector $\mathbf{x} \neq \mathbf{0}$. It is positive semidefinite if $\mathbf{x}^T A\mathbf{x} \geq 0$ for all $\mathbf{x}$.

For a 2×2 symmetric matrix the quadratic form is

$$\mathbf{x}^T A\mathbf{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = ax_1^2 + 2bx_1x_2 + cx_2^2.$$

It is purely of degree two, with no linear or constant terms. Each of the following is a complete test for $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ to be positive definite:

1. eigenvalues: $\lambda_1 > 0$ and $\lambda_2 > 0$;
2. determinants: $a > 0$ and $ac - b^2 > 0$;
3. pivots: $a > 0$ and $(ac - b^2)/a > 0$;
4. energy: $\mathbf{x}^T A\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$ (the definition).

The determinant test uses the leading (upper-left) submatrices; the second pivot is the determinant divided by the first pivot. For $n \times n$ matrices the tests are the same: all n eigenvalues positive, all n leading determinants positive, all n pivots positive.

Example: Consider $A = \begin{bmatrix} 2 & 6 \\ 6 & c \end{bmatrix}$. The first pivot is 2, and the determinant test requires $2c - 36 > 0$, that is, $c > 18$.

- $c = 18$ is the borderline. Then A is singular and positive semidefinite: its eigenvalues are 0 and 20 (the trace), and it has only one pivot. Its form is $2x_1^2 + 12x_1x_2 + 18x_2^2 = 2(x_1 + 3x_2)^2$, which vanishes at $(x_1, x_2) = (3, -1)$.
- $c = 7$ fails badly: the determinant is $14 - 36 = -22$, so the pivots are 2 and -11, and one

eigenvalue is negative. The form $f(x, y) = 2x^2 + 12xy + 7y^2$ is negative at $(1, -1)$, where it equals $2 - 12 + 7 = -3$.

- $c = 20$ passes: the determinant is $40 - 36 = 4$ and the trace is 22. The eigenvalues have positive product and positive sum, so both are positive.

27.2 Graphs, minima, and completing the square

The graph of $f(x, y) = \mathbf{x}^T A\mathbf{x}$ passes through the origin, where the first derivatives f_x and f_y are zero. For $c = 7$ the graph goes up along the axes but down along the direction $x = -y$: the origin is a SADDLE POINT, a maximum in some directions and a minimum in others. For $c = 20$ the graph is a bowl and the origin is a minimum. The squares $2x^2$ and $20y^2$ overwhelm the cross term $12xy$, which can have either sign.

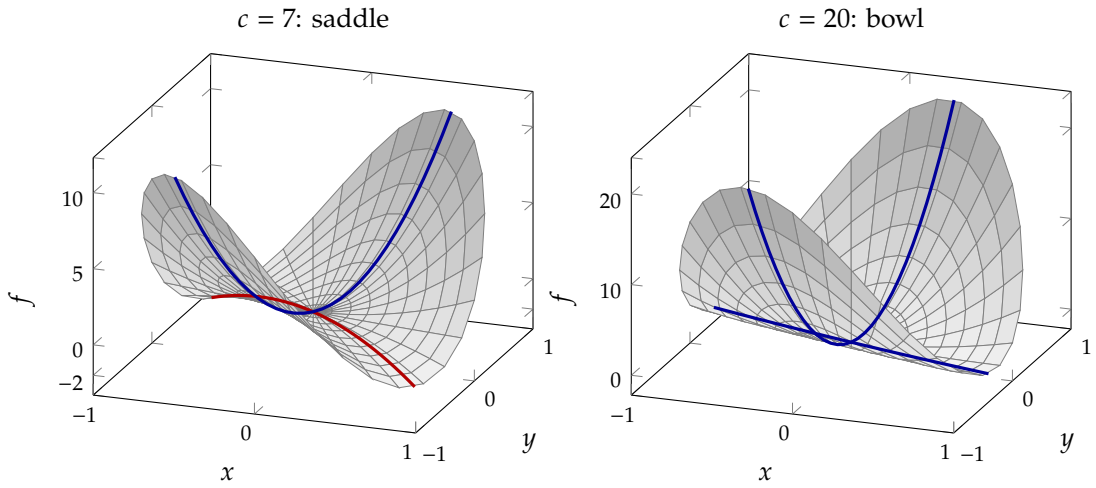


Figure 20: Graphs of $f(x, y) = 2x^2 + 12xy + cy^2$ over the unit disk; the thick curves follow the eigenvector directions of A. Left, $c = 7$: a saddle point, with f rising along $(2, 3)$ (eigenvalue 11) and falling along $(3, -2)$ (eigenvalue -2). Right, $c = 20$: a bowl with a minimum at the origin, rising slowly along the eigenvector of the small eigenvalue $11 - \sqrt{117}$; its horizontal cross-sections are ellipses.

To prove that $2x^2 + 12xy + 20y^2$ is positive, write it as a sum of squares by completing the square:

$$2x^2 + 12xy + 20y^2 = 2(x + 3y)^2 + 2y^2.$$

Since $2(x + 3y)^2 = 2x^2 + 12xy + 18y^2$, the borderline 18 appears again. With $c = 7$ the same step gives $2(x + 3y)^2 - 11y^2$, which is negative at $(x, y) = (-3, 1)$. Cutting the bowl at height 1 gives the curve $2x^2 + 12xy + 20y^2 = 1$, an ellipse; cutting the saddle gives a hyperbola.

Proposition 27.2 [COMPLETING THE SQUARE IS ELIMINATION]. *If $a \neq 0$, then*

$$ax^2 + 2bxy + cy^2 = a\left(x + \frac{b}{a}\,y\right)^2 + \frac{ac - b^2}{a}\,y^2.$$

The coefficients outside the squares are the pivots of A, and the number b/a inside is the multiplier ℓ_{21} from elimination. So the form is positive for all $(x, y) \neq (0, 0)$ exactly when both pivots are positive.

Proof. Expanding the right side gives $ax^2 + 2bxy + \frac{b^2}{a}y^2 + \frac{ac-b^2}{a}y^2$, which is the left side. If both outside coefficients are positive, the sum of squares is zero only when $y = 0$ and then $x = 0$. If $a < 0$, the point $(1, 0)$ gives $a < 0$; if $(ac - b^2)/a \leq 0$, the point $(-b/a, 1)$ gives $(ac - b^2)/a \leq 0$. □

For the example, subtracting 3 times row 1 from row 2 gives

$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 6 \\ 0 & 2 \end{bmatrix} = U, \qquad L = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}:$$

the pivots $2, 2$ multiply the squares, and the multiplier 3 sits inside $(x + 3y)^2$. Positive pivots give a sum of squares with positive weights, so the graph goes up and the origin is a minimum.

Note. In calculus, a minimum of $f(x)$ needs $f' = 0$ and $f'' > 0$. For $f(x_1, \ldots, x_n)$, a minimum needs all first derivatives zero and the matrix of second derivatives positive definite. In two variables this is the Hessian

$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix},$$

which is symmetric because $f_{xy} = f_{yx}$. The pure second derivatives must be positive and large enough to overcome the cross derivative: the familiar test $f_{xx}f_{yy} > f_{xy}^2$ is the determinant test. For $f = \mathbf{x}^T A\mathbf{x}$ the Hessian is $2A$.

27.3 An $n \times n$ example and ellipsoids

Example: Take

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}.$$

The leading determinants are

$$\det[2] = 2, \qquad \det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 3, \qquad \det A = 4,$$

all positive, so A is positive definite. The pivots are the ratios 2, $\frac{3}{2}$, $\frac{4}{3}$, since products of pivots are the leading determinants. The eigenvalues are $2 - \sqrt{2}$, 2, $2 + \sqrt{2}$, all positive; as checks, their sum 6 is the trace and their product $(2 - \sqrt{2}) \cdot 2 \cdot (2 + \sqrt{2}) = 4$ is the determinant. The form is

$$f(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3,$$

and completing squares writes it with the pivots outside:

$$f(\mathbf{x}) = 2\left(x_1 - \tfrac{1}{2}x_2\right)^2 + \tfrac{3}{2}\left(x_2 - \tfrac{2}{3}x_3\right)^2 + \tfrac{4}{3}x_3^2.$$

So $f > 0$ except at $\mathbf{x} = \mathbf{0}$, and its graph, in four dimensions, is a bowl (a paraboloid) with minimum at the origin.

Cutting the bowl at height 1 gives the surface

$$2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 = 1,$$

an ellipsoid. It has a major axis, a middle axis, and a minor axis. A sphere would come from $A = I$ (all eigenvalues equal), and a rugby ball from a repeated eigenvalue; here the three eigenvalues are different.

Theorem 27.3 [Principal axes]. *Write $A = Q\Lambda Q^T$ with orthonormal eigenvectors $\mathbf{q}_i$ in Q. The axes of the ellipsoid $\mathbf{x}^T A\mathbf{x} = 1$ point along the eigenvectors $\mathbf{q}_i$, and the half-length of the axis along $\mathbf{q}_i$ is $1/\sqrt{\lambda_i}$.*

Proof. Set $\mathbf{y} = Q^T\mathbf{x}$, the coordinates of $\mathbf{x}$ along the eigenvectors. Then $\mathbf{x}^T A\mathbf{x} = \mathbf{y}^T\Lambda\mathbf{y} = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2 = 1$, an ellipsoid whose axes are the y_i directions, meeting the y_i axis at $y_i = \pm 1/\sqrt{\lambda_i}$. □

In the example, the major axis comes from the smallest eigenvalue $2 - \sqrt{2}$ and the minor axis from $2 + \sqrt{2}$.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 27, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 28: Similar Matrices and Jordan Form

The lecture first shows where positive definite matrices come from, above all A^TA. It then groups square matrices into families of similar matrices, which share their eigenvalues, and describes the best representative of each family: the Jordan form.

28.1 A^TA is positive definite

Recall that a symmetric A is positive definite if $\mathbf{x}^TA\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$. Each test from Lecture 27 suits a different question.

- If A is symmetric positive definite, so is A^{-1}: its eigenvalues $1/\lambda_i$ are positive. (Little is known about the pivots of A^{-1}.)
- If A and B are positive definite, so is $A + B$. The eigenvalues and pivots of $A + B$ are not known, but $\mathbf{x}^T(A + B)\mathbf{x} = \mathbf{x}^TA\mathbf{x} + \mathbf{x}^TB\mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$.

Proposition 28.1 [A^TA]. *Let A be any $m \times n$ matrix. Then A^TA is square, symmetric, and positive semidefinite. It is positive definite exactly when A has rank n (independent columns).*

Proof. Move the parentheses:

$$\mathbf{x}^T(A^TA)\mathbf{x} = (A\mathbf{x})^T(A\mathbf{x}) = \|A\mathbf{x}\|^2 \geq 0.$$

This is zero only when $A\mathbf{x} = \mathbf{0}$. If the columns are independent, $N(A) = \{\mathbf{0}\}$, so $\mathbf{x}^TA^TA\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$. Otherwise a nonzero $\mathbf{x} \in N(A)$ gives 0. □

This is the matrix of least squares: when A has independent columns, A^TA is invertible and even positive definite. Positive definite matrices are also good for computation: elimination never needs row exchanges, since zeros or unsuitably small numbers never appear in the pivot positions.

28.2 Similar matrices

Definition 28.2 [Similar matrices]. Square $n \times n$ matrices A and B are similar if

$$B = M^{-1}AM$$

for some invertible matrix M.

Similarity sorts matrices into families, each matrix similar to every other in its family. If A has n independent eigenvectors, the eigenvector matrix S gives $S^{-1}AS = \Lambda$: A is similar to the diagonal matrix Λ, the nicest member of its family. Any other invertible M produces another member $M^{-1}AM$, usually not diagonal.

Example: $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1, so A is similar to $\Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$. With $M = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$,

$$B = M^{-1}AM = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 9 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} -2 & -15 \\ 1 & 6 \end{bmatrix}.$$

B has trace 4 and determinant $-12 + 15 = 3$, so its eigenvalues are again 3 and 1.

Theorem 28.3 [Similar matrices share eigenvalues]. *If $B = M^{-1}AM$, then A and B have the same eigenvalues. If $A\mathbf{x} = \lambda\mathbf{x}$, then $M^{-1}\mathbf{x}$ is an eigenvector of B for the same λ. Similar matrices also have the same number of independent eigenvectors.*

Proof. Insert $MM^{-1} = I$ and multiply on the left by M^{-1}:

$$AMM^{-1}\mathbf{x} = \lambda\mathbf{x} \quad \Longrightarrow \quad M^{-1}AM\,(M^{-1}\mathbf{x}) = \lambda\, M^{-1}\mathbf{x},$$

that is, $B(M^{-1}\mathbf{x}) = \lambda(M^{-1}\mathbf{x})$, with $M^{-1}\mathbf{x} \neq \mathbf{0}$. The invertible map $\mathbf{x} \mapsto M^{-1}\mathbf{x}$ takes independent eigenvectors of A to independent eigenvectors of B, and $A = MBM^{-1}$ gives the reverse. □

The eigenvectors change; only the eigenvalues are shared. Diagonalization is the special case where the eigenvectors become the standard basis vectors $(1, 0)$ and $(0, 1)$. The family of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ consists of all 2×2 matrices with eigenvalues 3 and 1: each is diagonalizable, since its eigenvalues are distinct, and so similar to the same Λ. Other members are $\begin{bmatrix} 3 & 7 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 1 & 7 \\ 0 & 3 \end{bmatrix}$.

28.3 Repeated eigenvalues

If $\lambda_1 = \lambda_2$, there may not be a full set of eigenvectors. Take $\lambda_1 = \lambda_2 = 4$. The matrices with eigenvalues 4, 4 split into two families.

- $4I = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$ is alone in its family, since $M^{-1}(4I)M = 4M^{-1}M = 4I$ for every invertible M.
- Every other matrix with eigenvalues 4, 4 is similar to $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$. None of them is diagonalizable, since a diagonalizable one would be similar to $4I$; each has only one independent eigenvector.

Members are found by choosing trace 8 and determinant 16:

$$\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}, \quad \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \quad \begin{bmatrix} 4 & 0 \\ 17 & 4 \end{bmatrix}, \quad \begin{bmatrix} a & b \\ (8a - a^2 - 16)/b & 8 - a \end{bmatrix} \ (b \neq 0).$$

The best member $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$ of the large family is as close to diagonal as possible: it is the Jordan form.

28.4 Jordan form

Equal eigenvalues and an equal number of eigenvectors do not guarantee similarity. Consider

$$A = \left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 \end{array}\right], \qquad C = \left[\begin{array}{cc|cc} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

Both have four zero eigenvalues and rank 2, so each nullspace, the space of eigenvectors, has dimension $4 - 2 = 2$: two eigenvectors and two missing. Changing the $(1, 3)$ entry of A from 0 to 7 keeps the rank at 2, and the result is similar to A, but A, with a 1 above the diagonal for each missing eigenvector, is its nicer representative. Yet C is not similar to A: A breaks into blocks of sizes 3 and 1, while C breaks into two blocks of size 2. (One way to see it: $C^2 = 0$ but $A^2 \neq 0$, and $B = M^{-1}AM$ would give $B^2 = M^{-1}A^2M \neq 0$.)

Definition 28.4 [Jordan block]. A Jordan block J_i has a repeated eigenvalue λ_i on the diagonal, 1's just above the diagonal, and zeros elsewhere:

$$J_i = \begin{bmatrix} \lambda_i & 1 & & \\ & \lambda_i & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{bmatrix}.$$

It has only one independent eigenvector.

Theorem 28.5 [Jordan]. *Every square matrix A is similar to a* Jordan matrix

$$J = \begin{bmatrix} J_1 & & \\ & \ddots & \\ & & J_d \end{bmatrix}$$

with Jordan blocks on the diagonal. The number d of blocks is the number of independent eigenvectors of A, one per block. Two matrices whose Jordan blocks have different sizes are not similar.

So J has the eigenvalues on its diagonal, possibly some 1's just above, and zeros elsewhere.

- If A has n distinct eigenvalues, it is diagonalizable and $J = \Lambda$, with $d = n$ blocks of size 1.
- If A has repeated eigenvalues and missing eigenvectors, J has $n - d$ ones above the diagonal.

The Jordan form was once the climax of linear algebra, but it is not numerically useful: it depends on eigenvalues being exactly equal and on the exact rank, and the slightest change in the entries can change both. Computing J for a matrix with missing eigenvectors is not covered; the diagonalizable case is the important one.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 28, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 29: Singular Value Decomposition

The singular value decomposition (SVD) is the final and best factorization of a matrix:

$$A = U\Sigma V^T,$$

with U orthogonal, Σ diagonal, and V orthogonal. Here A can be *any* matrix. For a symmetric positive definite matrix, $A = Q\Lambda Q^T$ is already an SVD with $U = V = Q$. For a general square matrix, the diagonalization $A = S\Lambda S^{-1}$ is not of this form, since the eigenvector matrix S is usually not orthogonal. The SVD needs two, usually different, orthogonal matrices.

29.1 Orthonormal bases that A keeps orthogonal

Think of A as taking a vector $\mathbf{v}_1$ in its row space to $A\mathbf{v}_1$ in its column space. The goal is an orthonormal basis $\mathbf{v}_1, \ldots, \mathbf{v}_r$ of the row space that A carries to orthogonal vectors:

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i, \qquad i = 1, \ldots, r,$$

with $\mathbf{u}_1, \ldots, \mathbf{u}_r$ an orthonormal basis of the column space and stretching factors $\sigma_i > 0$, the singular values. Gram–Schmidt easily gives *some* orthonormal basis of the row space, but in general A does not take it to an orthogonal set; the special basis is the point. In matrix form,

$$A \begin{bmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_r \end{bmatrix} = \begin{bmatrix} \sigma_1\mathbf{u}_1 & \cdots & \sigma_r\mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}.$$

The nullspaces cause no trouble. Complete the $\mathbf{v}$'s with an orthonormal basis $\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$ of $N(A)$ and the $\mathbf{u}$'s with an orthonormal basis $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$ of $N(A^T)$. Since $A\mathbf{v}_i = \mathbf{0}$ for $i > r$, the equation becomes $AV = U\Sigma$, where V is $n \times n$, U is $m \times m$, and the $m \times n$ matrix Σ has $\sigma_1, \ldots, \sigma_r$ on its diagonal and zeros elsewhere. Multiplying on the right by $V^{-1} = V^T$,

$$\boxed{A = U\Sigma V^T}.$$

29.2 Computing U, Σ, and V

To find V without U, multiply A by its transpose $A^T = V\Sigma^T U^T$. The U's cancel because $U^T U = I$:

$$A^T A = V\Sigma^T U^T U\Sigma V^T = V\,\Sigma^T\Sigma\,V^T = V\begin{bmatrix}\sigma_1^2 & & \\ & \ddots & \\ & & \sigma_n^2\end{bmatrix}V^T,$$

with $\sigma_i = 0$ for $i > r$. This is the factorization $Q\Lambda Q^T$ of the symmetric positive semidefinite matrix $A^T A$. So the columns of V are orthonormal eigenvectors of $A^T A$, its eigenvalues are the σ_i^2, and σ_i is the positive square root. In the other order, $AA^T = U\Sigma\Sigma^T U^T$, so the columns of U are eigenvectors of AA^T, with the same nonzero eigenvalues σ_i^2 (the nonzero eigenvalues of AB and BA agree).

An eigenvector is determined only up to sign, so U and V cannot be chosen independently. Once $\mathbf{v}_1, \ldots, \mathbf{v}_r$ are fixed, take

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\sigma_i}, \qquad i \le r.$$

These are orthonormal: $\mathbf{u}_i^T\mathbf{u}_j = \mathbf{v}_i^T A^T A\mathbf{v}_j/(\sigma_i\sigma_j) = \sigma_j^2\,\mathbf{v}_i^T\mathbf{v}_j/(\sigma_i\sigma_j)$, which is 1 for $i = j$ and 0 otherwise. This construction proves that every matrix has an SVD, and AA^T serves as a check.

Example: Let $A = \begin{bmatrix}4 & 4\\ -3 & 3\end{bmatrix}$, invertible, so the row and column spaces are both $\mathbb{R}^2$. Then

$$A^T A = \begin{bmatrix}4 & -3\\ 4 & 3\end{bmatrix}\begin{bmatrix}4 & 4\\ -3 & 3\end{bmatrix} = \begin{bmatrix}25 & 7\\ 7 & 25\end{bmatrix},$$

with eigenvectors $(1,1)$ and $(1,-1)$ and eigenvalues $\sigma_1^2 = 32$, $\sigma_2^2 = 18$. Normalizing, $\mathbf{v}_1 = (1/\sqrt{2}, 1/\sqrt{2})$, $\mathbf{v}_2 = (1/\sqrt{2}, -1/\sqrt{2})$, and $\sigma_1 = \sqrt{32} = 4\sqrt{2}$, $\sigma_2 = \sqrt{18} = 3\sqrt{2}$. The other product happens to be diagonal,

$$AA^T = \begin{bmatrix}4 & 4\\ -3 & 3\end{bmatrix}\begin{bmatrix}4 & -3\\ 4 & 3\end{bmatrix} = \begin{bmatrix}32 & 0\\ 0 & 18\end{bmatrix},$$

again with eigenvalues 32 and 18, and eigenvectors along $(1,0)$ and $(0,1)$. The signs must match the $\mathbf{v}$'s: $A\mathbf{v}_1 = (4\sqrt{2}, 0)$ gives $\mathbf{u}_1 = (1,0)$, but $A\mathbf{v}_2 = (0, -3\sqrt{2})$ gives $\mathbf{u}_2 = (0,-1)$. Choosing $\mathbf{u}_2 = (0,1)$ instead would produce the second row with the wrong signs. So

$$\begin{bmatrix}4 & 4\\ -3 & 3\end{bmatrix} = \underbrace{\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}}_{U}\underbrace{\begin{bmatrix}4\sqrt{2} & 0\\ 0 & 3\sqrt{2}\end{bmatrix}}_{\Sigma}\underbrace{\begin{bmatrix}1/\sqrt{2} & 1/\sqrt{2}\\ 1/\sqrt{2} & -1/\sqrt{2}\end{bmatrix}}_{V^T}.$$

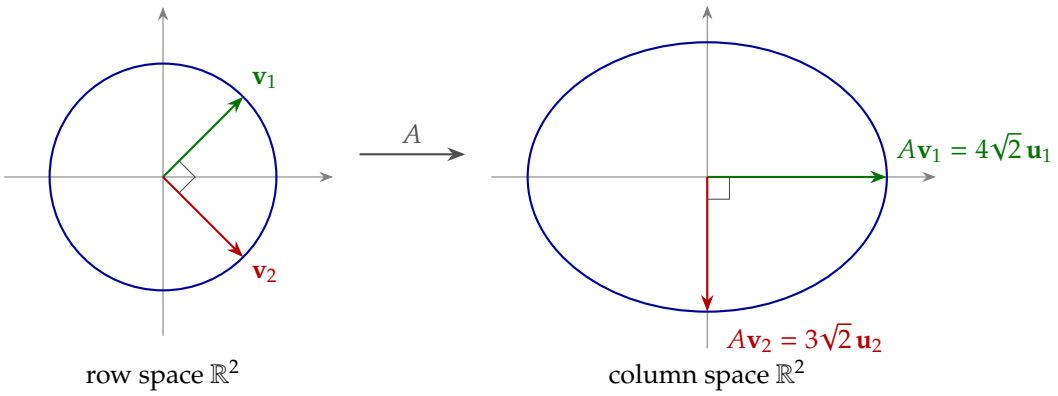


Figure 21: The SVD of the first example. The orthonormal vectors $\mathbf{v}_1, \mathbf{v}_2$ go to the orthogonal vectors $A\mathbf{v}_1 = 4\sqrt{2}\,\mathbf{u}_1$ and $A\mathbf{v}_2 = 3\sqrt{2}\,\mathbf{u}_2$; the unit circle goes to an ellipse with semi-axes σ_1 and σ_2. The two panels are drawn at different scales.

Example: Let $A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$, of rank 1. Its row space is the line through $(4, 3)$, its column space is the line through $(4, 8)$, or $(1, 2)$, and the nullspace and left nullspace are the perpendicular lines. The unit vectors in the row and column spaces are

$$\mathbf{v}_1 = \begin{bmatrix} .8 \\ .6 \end{bmatrix}, \qquad \mathbf{u}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

For σ_1, find the nonzero eigenvalue of

$$A^T A = \begin{bmatrix} 4 & 8 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 80 & 60 \\ 60 & 45 \end{bmatrix}.$$

It has rank 1, so one eigenvalue is 0 and the other is the trace, $\sigma_1^2 = 125$. Completing with $\mathbf{v}_2 = (.6, -.8)$ in the nullspace and $\mathbf{u}_2 = (2, -1)/\sqrt{5}$ in the left nullspace,

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \underbrace{\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} .8 & .6 \\ .6 & -.8 \end{bmatrix}}_{V^T}.$$

The $\sqrt{5}$ cancels into $\sqrt{125}$ to leave 5, and $5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} .8 & .6 \end{bmatrix}$ reproduces A. The nullspace vectors $\mathbf{v}_2$ and $\mathbf{u}_2$ are multiplied by the zero in Σ.

29.3 The right bases for the four subspaces

The SVD chooses orthonormal bases for the four fundamental subspaces of an $m \times n$ matrix of rank r:

Basis	Subspace	Dimension
$\mathbf{v}_1, \ldots, \mathbf{v}_r$	row space $C(A^T)$	r
$\mathbf{u}_1, \ldots, \mathbf{u}_r$	column space $C(A)$	r
$\mathbf{v}_{r+1}, \ldots, \mathbf{v}_n$	nullspace $N(A)$	$n - r$
$\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$	left nullspace $N(A^T)$	$m - r$

Gram–Schmidt could produce orthonormal bases; these are the right ones because they also make the matrix diagonal: $A\mathbf{v}_i = \sigma_i \mathbf{u}_i$, with no coupling between different i. Finding them requires the eigenvalues and eigenvectors of A^TA and AA^T, so the SVD brings together positive definite matrices and the four subspaces.

Source. Gilbert Strang, MIT 18.06SC, *Linear Algebra*, Lecture 29, Fall 2011. Adapted from MIT OpenCourseWare, CC BY-NC-SA 4.0.

Lecture 30: Linear Transformations and their Matrices

A linear transformation makes sense without coordinates: it describes what happens to the whole space at once. To compute, we choose bases, and then every linear transformation is described by a matrix. Matrix multiplication comes from composing transformations.

30.1 Linear transformations without coordinates

A transformation (or mapping) T assigns an output $T(\mathbf{v})$ to each input $\mathbf{v}$, like a function. For example, projection onto a fixed line through the origin takes every vector $\mathbf{v}$ in the plane to the vector $T(\mathbf{v})$ on that line; here $T : \mathbb{R}^2 \to \mathbb{R}^2$.

Definition 30.1 [Linear transformation]. A transformation T is linear if

$$T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w}) \qquad \text{and} \qquad T(c\mathbf{v}) = c\,T(\mathbf{v})$$

for all vectors $\mathbf{v}, \mathbf{w}$ and scalars c. Equivalently, $T(c\mathbf{v} + d\mathbf{w}) = c\,T(\mathbf{v}) + d\,T(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w}$ and all c, d.

Every linear T has $T(\mathbf{0}) = \mathbf{0}$: take $\mathbf{v} = \mathbf{0}$ and $c = 3$ to get $T(\mathbf{0}) = 3\,T(\mathbf{0})$. Projection is linear: doubling $\mathbf{v}$ doubles its projection, reversing $\mathbf{v}$ reverses it, and projecting a sum gives the sum of the projections. Rotation of the plane by 45° is also linear, and this can be seen without any coordinates.

Example: Two non-examples.

- Shifting the whole plane by a fixed $\mathbf{v}_0 \neq \mathbf{0}$, $T(\mathbf{v}) = \mathbf{v} + \mathbf{v}_0$, is not linear: $T(2\mathbf{v}) = 2\mathbf{v} + \mathbf{v}_0 \neq 2T(\mathbf{v})$, and $T(\mathbf{0}) = \mathbf{v}_0 \neq \mathbf{0}$.
- The length $T(\mathbf{v}) = \|\mathbf{v}\|$, from $\mathbb{R}^3$ to $\mathbb{R}^1$, is not linear. It does satisfy $T(\mathbf{0}) = 0$ and $T(2\mathbf{v}) = 2T(\mathbf{v})$, but $T(-2\mathbf{v}) = 2\|\mathbf{v}\| \neq -2T(\mathbf{v})$.

From now on T always denotes a linear transformation. The geometric description shows the big picture: we can see at once what a rotation does to a whole figure, such as a house in the plane, whereas a matrix alone would make the image harder to guess.

30.2 Transformations from matrices

Any matrix A gives a linear transformation $T(\mathbf{v}) = A\mathbf{v}$, because $A(\mathbf{v}+\mathbf{w}) = A\mathbf{v}+A\mathbf{w}$ and $A(c\mathbf{v}) = c\,A\mathbf{v}$. A 2×3 matrix gives a linear $T : \mathbb{R}^3 \to \mathbb{R}^2$, and every 2×3 matrix gives such an example. Linear transformations are an abstract description of matrix multiplication.

Example: $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ keeps the x component of $\mathbf{v}$ and reverses the sign of the y component: it reflects the plane across the x axis, and the house lands upside down.

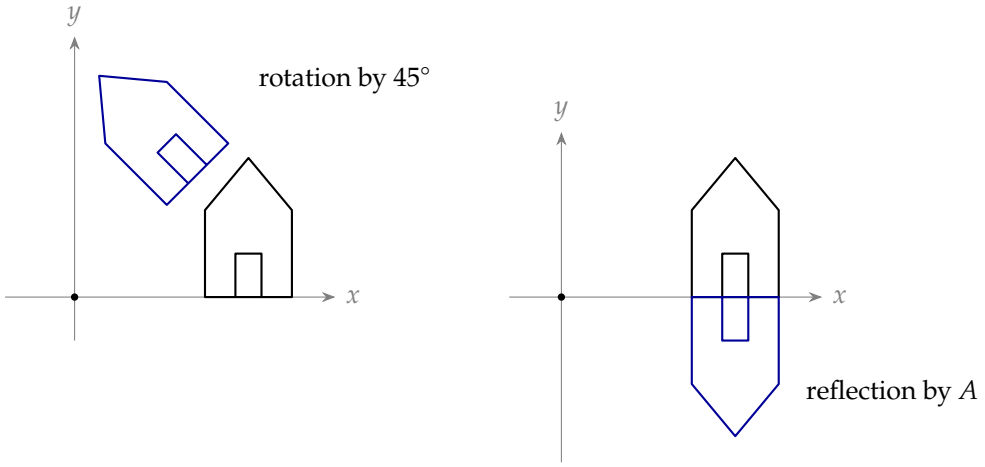


Figure 22: A house (black) and its image (blue) under two linear transformations of the plane: rotation by 45° counterclockwise, and reflection across the x axis by $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. Each transformation fixes the origin.

30.3 A basis determines the transformation

How much must we know to find $T(\mathbf{v})$ for every $\mathbf{v}$? Knowing $T(\mathbf{v}_1)$ gives $T(c\mathbf{v}_1)$ for all c. Knowing also $T(\mathbf{v}_2)$ for an independent $\mathbf{v}_2$ gives T on the whole plane of combinations $c\mathbf{v}_1 + d\mathbf{v}_2$. (If $\mathbf{v}_2 = 6\mathbf{v}_1$, then $T(\mathbf{v}_2) = 6T(\mathbf{v}_1)$ adds nothing new.) In general:

Proposition 30.2 [Determined by a basis]. *If $\mathbf{v}_1, \ldots, \mathbf{v}_n$ is a basis of the input space, then $T(\mathbf{v}_1), \ldots, T(\mathbf{v}_n)$ determine T: for $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$,*

$$T(\mathbf{v}) = c_1T(\mathbf{v}_1) + c_2T(\mathbf{v}_2) + \cdots + c_nT(\mathbf{v}_n).$$

Every $\mathbf{v}$ is a combination of the basis vectors in exactly one way. The numbers $c_1, \ldots, c_n$ are the coordinates of $\mathbf{v}$ in that basis. Coordinates come from a basis, and changing the basis changes the coordinates. Writing $\mathbf{v} = (3, 2, 4)$ silently uses the standard basis, 3 times the first coordinate vector plus 2 times the second plus 4 times the third; another basis, such as eigenvectors of a matrix, gives other coordinates for the same vector.

30.4 The matrix of a linear transformation

Let $T : \mathbb{R}^n \to \mathbb{R}^m$. Choose an input basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ of $\mathbb{R}^n$ and an output basis $\mathbf{w}_1, \ldots, \mathbf{w}_m$ of $\mathbb{R}^m$. We want a matrix A that multiplies the input coordinates of $\mathbf{v}$ to give the output coordinates of $T(\mathbf{v})$.

Theorem 30.3 [The matrix of T]. *Write each $T(\mathbf{v}_j)$ in the output basis,*

$$T(\mathbf{v}_j) = a_{1j}\mathbf{w}_1 + a_{2j}\mathbf{w}_2 + \cdots + a_{mj}\mathbf{w}_m,$$

and put these coefficients in column j of the $m \times n$ matrix $A = (a_{ij})$. If $\mathbf{v}$ has input coordinates $\mathbf{c}$, then $T(\mathbf{v})$ has output coordinates $A\mathbf{c}$.

Proof. For $\mathbf{v} = \mathbf{v}_j$ the coordinates are $\mathbf{c} = \mathbf{e}_j$ (a 1 in place j, zeros elsewhere), and $A\mathbf{e}_j$ is column j of A, the output coordinates of $T(\mathbf{v}_j)$ by construction. Both $\mathbf{c} \mapsto A\mathbf{c}$ and the coordinates of $T(\mathbf{v})$ are linear in $\mathbf{c}$, and they agree on every basis vector, so they agree for every $\mathbf{v}$. □

Example: Let T project the plane onto a line, using the same basis for inputs and outputs: $\mathbf{v}_1$ a unit vector on the line and $\mathbf{v}_2$ a unit vector perpendicular to it. Then $T(\mathbf{v}_1) = \mathbf{v}_1$ and $T(\mathbf{v}_2) = \mathbf{0}$, so $T(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = c_1\mathbf{v}_1$ and

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad A \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ 0 \end{bmatrix}.$$

This basis consists of eigenvectors of the projection, and the matrix is the diagonal eigenvalue matrix Λ, with eigenvalues 1 and 0. In the standard basis $\mathbf{v}_1 = \mathbf{w}_1 = (1, 0)$, $\mathbf{v}_2 = \mathbf{w}_2 = (0, 1)$, projection onto the 45° line through $\mathbf{a} = (1, 1)$ has the familiar matrix

$$P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T\mathbf{a}} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}.$$

Indeed $P(1, 0) = P(0, 1) = (\frac{1}{2}, \frac{1}{2})$, the projections of the basis vectors. It is still symmetric with $P^2 = P$, but not diagonal: the handiest basis is not the best one.

Example: Differentiation $T = d/dx$ is linear (which is why we can differentiate combinations of a few known functions):

$$T(c_1 + c_2x + c_3x^2) = c_2 + 2c_3x.$$

The input space of quadratics has basis $\mathbf{v}_1 = 1$, $\mathbf{v}_2 = x$, $\mathbf{v}_3 = x^2$; the output space has basis $\mathbf{w}_1 = 1$, $\mathbf{w}_2 = x$. Then

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \qquad A \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}.$$

So every linear transformation has a matrix once bases are chosen. If T is invertible, its inverse has the matrix A^{-1}. If $T_1 : \mathbf{v} \mapsto A_1\mathbf{v}$ is followed by $T_2 : \mathbf{w} \mapsto A_2\mathbf{w}$, the composite has matrix A_2A_1. This is where the rule for matrix multiplication comes from.

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Lecture 31: Change of Basis; Image Compression

A good choice of basis makes computations easier. Image and video compression is exactly a change of basis: the data can be stored and transmitted efficiently only after it is rewritten in a better basis. The linear algebra is the rule for how coordinates and matrices change when the basis changes.

31.1 Compression is a change of basis

A black-and-white image of 512×512 pixels records a gray level for each pixel, typically an integer from 0 to 255 (eight bits). So an image is a vector $\mathbf{x} \in \mathbb{R}^n$ with $n = 512^2$; a color image needs three times as many numbers. This is far too much to transmit, so the image is compressed. *Lossless* compression only removes redundancy; here we consider *lossy* compression.

Storing every pixel uses the standard basis $(1, 0, \ldots, 0)$, $(0, 1, 0, \ldots, 0)$, and so on. That basis ignores the fact that neighboring pixels tend to have nearly the same gray level. A blank blackboard, with all pixels equal, needs n numbers in the standard basis, but just one coefficient if the vector of all ones is a basis vector. Other useful basis vectors are the checkerboard $(1, -1, 1, -1, \ldots)$ and half ones, half minus ones (bright on the left, dark on the right). Which basis to choose is a large practical decision, and the best basis for one kind of image may not be best for another.

The JPEG standard breaks the image into 8×8 blocks of 64 pixels and changes basis within each block, using the Fourier basis: the columns of the Fourier matrix (Lecture 26), here

$$\begin{bmatrix}1\\1\\1\\1\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\\omega\\\omega^2\\\omega^3\\\omega^4\\\omega^5\\\omega^6\\\omega^7\end{bmatrix}, \begin{bmatrix}1\\\omega^2\\\omega^4\\\omega^6\\\omega^8\\\omega^{10}\\\omega^{12}\\\omega^{14}\end{bmatrix}, \ldots, \begin{bmatrix}1\\\omega^7\\\omega^{14}\\\omega^{21}\\\omega^{28}\\\omega^{35}\\\omega^{42}\\\omega^{49}\end{bmatrix}, \qquad \omega = e^{2\pi i/8}.$$

JPEG actually uses cosines, the real parts of ω^{jk}. The vector of all ones is the zero-frequency (DC) vector; the checkerboard, column $k = 4$ (counting from 0) since $\omega^4 = -1$, is the highest frequency. The steps are

$$\text{signal } \mathbf{x} \xrightarrow{\text{lossless}} 64 \text{ coefficients } \mathbf{c} \xrightarrow{\text{lossy}} \hat{\mathbf{c}} \text{ (many zeros)} \longrightarrow \hat{\mathbf{x}} = \sum_i \hat{c}_i \mathbf{w}_i.$$

The first step only changes basis. The compression step, for example THRESHOLDING, sets to zero

every coefficient below a threshold chosen so that the eye can hardly see the difference. A smooth image has a large coefficient for the all-ones vector and very little of the high frequencies. Keeping 3 of the 64 coefficients would be compression by about 21 to 1. For video, consecutive frames are highly correlated, so one predicts each frame from the previous one and encodes and compresses only the correction.

31.2 The Haar wavelet basis

The main competitor to the Fourier basis is a wavelet basis (the JPEG2000 standard uses wavelets, smoother than the ones below). The Haar wavelet basis of $\mathbb{R}^8$ consists of the columns of

$$W = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & -1 & 0 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 & -1 \end{bmatrix}:$$

all ones; four ones and four minus ones; two ones, two minus ones, and zeros, in each half; and four vectors with one 1, one -1, and six zeros. Except for the all-ones vector, the nonzero entries are half 1's and half -1's, and half or even three quarters of the entries are 0. The columns are orthogonal, with lengths $\sqrt{8}, \sqrt{8}, 2, 2$, and $\sqrt{2}$ for the last four, so dividing by these lengths makes them orthonormal. For instance, the fastest Fourier vector $(1, -1, 1, -1, 1, -1, 1, -1)$ is the sum of the last four wavelets.

Given the pixel values $\mathbf{x} = (p_1, \ldots, p_8)$ in the standard basis, the transform step finds the coefficients in

$$\mathbf{x} = c_1\mathbf{w}_1 + \cdots + c_8\mathbf{w}_8 = W\mathbf{c}, \qquad \text{so} \qquad \mathbf{c} = W^{-1}\mathbf{x}.$$

A good basis for compression therefore needs two properties.

- *Speed.* Multiplying by W and by W^{-1} must be fast, as with the FFT for the Fourier basis and the fast wavelet transform. With orthonormal columns, $W^{-1} = W^T$, so the inverse is as fast as W itself.
- *Good compression.* A few basis vectors should come close to the signal, so that most coefficients c_i are small and can safely be set to zero.

The standard basis is the fastest of all (no change at all) but compresses badly: discarding 90 percent of the pixel values destroys the image, while discarding the small coefficients $c_5, \ldots, c_8$ in a wavelet or Fourier basis only removes small blips.

31.3 Change of basis for vectors and matrices

If the columns of W are the new basis vectors, a vector with old coordinates $\mathbf{x}$ has new coordinates $\mathbf{c}$ given by

$$\mathbf{x} = W\mathbf{c}.$$

Now let T be a linear transformation of $\mathbb{R}^n$ (for instance, a rotation or a projection of $\mathbb{R}^8$). As in Lecture 30, its matrix A in a basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ comes from writing each output in that basis,

$$T(\mathbf{v}_j) = a_{1j}\mathbf{v}_1 + a_{2j}\mathbf{v}_2 + \cdots + a_{nj}\mathbf{v}_n,$$

so column j of A holds the coefficients of $T(\mathbf{v}_j)$; then $T(c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n) = c_1T(\mathbf{v}_1) + \cdots + c_nT(\mathbf{v}_n)$ gives T everywhere.

Theorem 31.1 [Change of basis]. *Let T have matrix A in the basis $\mathbf{v}_1, \ldots, \mathbf{v}_n$ and matrix B in the basis $\mathbf{w}_1, \ldots, \mathbf{w}_n$. Then A and B are similar:*

$$B = M^{-1}AM,$$

where column j of the change of basis matrix M holds the coordinates of $\mathbf{w}_j$ in the basis of $\mathbf{v}$'s.

Proof. Take a vector with coordinates $\mathbf{c}$ in the $\mathbf{w}$ basis. Its coordinates in the $\mathbf{v}$ basis are $M\mathbf{c}$. Applying T in the $\mathbf{v}$ basis gives the output coordinates $AM\mathbf{c}$, and converting back to the $\mathbf{w}$ basis gives $M^{-1}AM\mathbf{c}$. This must equal $B\mathbf{c}$ for every $\mathbf{c}$, so $B = M^{-1}AM$. □

When the old basis is the standard one, M is the matrix W above. So coordinates change by $\mathbf{x} = W\mathbf{c}$ and matrices change by $B = M^{-1}AM$.

Example: The best basis is a basis of eigenvectors, $T(\mathbf{v}_i) = \lambda_i\mathbf{v}_i$. The first output $\lambda_1\mathbf{v}_1$ is already written in the basis: λ_1 times the first basis vector and 0 times the others. So column 1 is $(\lambda_1, 0, \ldots, 0)$, and in general $A = \Lambda$, the diagonal matrix of eigenvalues. For images this basis would be ideal, but finding the eigenvectors costs far more than using a Fourier or wavelet basis, which is cheap and nearly as good.

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Lecture 32: Left and Right Inverses; Pseudoinverse

For an $m \times n$ matrix of rank r, the four subspaces decide what kind of inverse exists. Full rank gives a two-sided inverse, full column rank a left inverse, and full row rank a right inverse. Every matrix has a pseudoinverse, which inverts A between its row space and column space.

32.1 Two-sided, left, and right inverses

Definition 32.1 [Two-sided inverse]. A two-sided inverse of A is a matrix A^{-1} with $AA^{-1} = I = A^{-1}A$. It exists exactly when $r = m = n$: A is square with full rank.

Full column rank, $r = n$. The columns are independent and $N(A) = \{\mathbf{0}\}$, so $A\mathbf{x} = \mathbf{b}$ has either exactly one solution or none. The $n \times n$ symmetric matrix A^TA is invertible (Lecture 28), so

$$\big((A^TA)^{-1}A^T\big)A = (A^TA)^{-1}(A^TA) = I.$$

Definition 32.2 [Left inverse]. For A of full column rank, $A^{-1}_{\text{left}} = (A^TA)^{-1}A^T$ is a left inverse: $A^{-1}_{\text{left}}A = I_n$. There can be other left inverses, but this is the preferred one; it is the matrix of least squares.

In the other order, AA^{-1}_{left} is $m \times m$ and equals I only if $m = n$. In fact

$$AA^{-1}_{\text{left}} = A(A^TA)^{-1}A^T = P,$$

the projection onto the column space of A: it tries to be the identity, and is the identity on $C(A)$ and zero on $N(A^T)$. A rectangular matrix cannot have a two-sided inverse, because A or A^T then has a nonzero nullspace.

Full row rank, $r = m$. The rows are independent and $N(A^T) = \{\mathbf{0}\}$. Elimination never produces a zero row, so $A\mathbf{x} = \mathbf{b}$ always has a solution; with $n - m$ free variables, $N(A)$ has dimension $n - m$, and if $n > m$ there are infinitely many solutions. Now AA^T is the invertible $m \times m$ matrix.

Definition 32.3 [Right inverse]. For A of full row rank, $A^{-1}_{\text{right}} = A^T(AA^T)^{-1}$ is a right inverse: $AA^{-1}_{\text{right}} = AA^T(AA^T)^{-1} = I_m$.

In the other order, $A^{-1}_{\text{right}}A = A^T(AA^T)^{-1}A$ is the projection of $\mathbb{R}^n$ onto the row space of A. The four cases are summarized below.

Rank	Nullspaces	Solutions of $A\mathbf{x} = \mathbf{b}$	Inverse
$r = m = n$	$N(A) = N(A^T) = \{\mathbf{0}\}$	exactly 1	two-sided A^{-1}
$r = n < m$	$N(A) = \{\mathbf{0}\}$	0 or 1	left $(A^TA)^{-1}A^T$
$r = m < n$	$N(A^T) = \{\mathbf{0}\}$	infinitely many	right $A^T(AA^T)^{-1}$
$r < m, r < n$	both nonzero	0 or infinitely many	pseudoinverse A^+

32.2 From row space to column space, A is invertible

Nonzero nullspaces are what prevent inversion: if $A\mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \neq \mathbf{0}$, no matrix can bring $\mathbf{0}$ back to $\mathbf{x}$. Every $A\mathbf{x}$ lies in the column space, and both the row space and the column space have dimension r. Almost every vector has a row space component and a nullspace component, and A kills the nullspace component; but on the row space alone, A loses nothing.

Proposition 32.4 [One-to-one on the row space]. *If $\mathbf{x} \neq \mathbf{y}$ are both in the row space $C(A^T)$, then $A\mathbf{x} \neq A\mathbf{y}$. So $\mathbf{x} \mapsto A\mathbf{x}$ is an invertible map from the row space onto the column space.*

Proof. Suppose $A\mathbf{x} = A\mathbf{y}$. Then $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$, so $\mathbf{x} - \mathbf{y}$ is in the nullspace. It is also in the row space, since the row space is a subspace. The only vector in both is $\mathbf{0}$, because they are orthogonal complements. So $\mathbf{x} = \mathbf{y}$, a contradiction. The map is one-to-one between two spaces of dimension r, hence onto. □

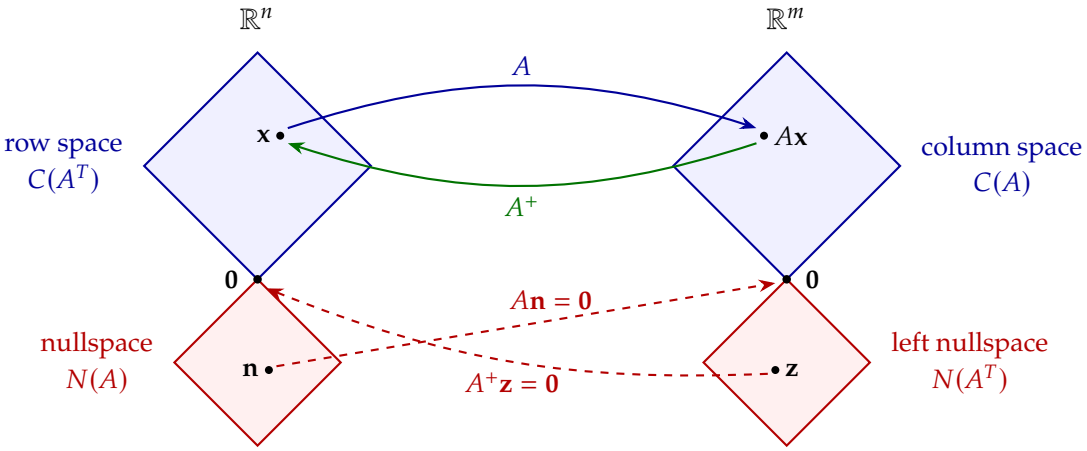


Figure 23: A takes the row space one-to-one onto the column space and sends $N(A)$ to $\mathbf{0}$. The pseudoinverse A^+ reverses A between the column space and the row space and sends $N(A^T)$ to $\mathbf{0}$.

32.3 The pseudoinverse

Definition 32.5 [Pseudoinverse]. The pseudoinverse A^+ of A is the $n \times m$ matrix that inverts A from the column space back to the row space, $A^+(A\mathbf{x}) = \mathbf{x}$ for every $\mathbf{x}$ in the row space, and whose

nullspace is $N(A^T)$.

Statisticians need A^+ in least squares (linear regression) when the columns of A may be dependent, for example after repeating an experiment, so that A^TA is singular.

One way to find A^+ starts from the SVD $A = U\Sigma V^T$ (Lecture 29). The orthogonal factors are easy to invert, $U^{-1} = U^T$ and $(V^T)^{-1} = V$, so all the difficulty sits in the $m \times n$ diagonal matrix Σ, which has $\sigma_1, \ldots, \sigma_r$ on its diagonal and zeros elsewhere. It is the simplest example of a matrix of rank r with both nullspaces. Its pseudoinverse is the $n \times m$ matrix

$$\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & & \\ & \ddots & & \\ & & 1/\sigma_r & \\ & & & 0 \end{bmatrix},$$

with $1/\sigma_1, \ldots, 1/\sigma_r$ in the first r diagonal positions and zeros elsewhere. If $r = m = n$, then $\Sigma^+ = \Sigma^{-1}$. In general

$$\Sigma\Sigma^+ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} (m \times m), \qquad \Sigma^+\Sigma = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} (n \times n):$$

neither is the identity, but they are projections onto the column space and the row space of Σ. Then

$$\boxed{A^+ = V\Sigma^+U^T}.$$

Indeed $A^+\mathbf{u}_i = \mathbf{v}_i/\sigma_i$ for $i \le r$, undoing $A\mathbf{v}_i = \sigma_i\mathbf{u}_i$, and $A^+\mathbf{u}_i = \mathbf{0}$ for the left nullspace vectors $\mathbf{u}_{r+1}, \ldots, \mathbf{u}_m$. Likewise $AA^+ = U\Sigma\Sigma^+U^T$ is the projection onto $C(A)$ and $A^+A = V\Sigma^+\Sigma V^T$ is the projection onto $C(A^T)$: the pseudoinverse brings vectors into the two good spaces and wipes out the nullspaces. Extra nonzero entries in the lower right corner of Σ^+ would multiply only zeros and give the same products $\Sigma\Sigma^+$ and $\Sigma^+\Sigma$, but Σ^+ as defined is the smallest choice, with no extra entries.

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